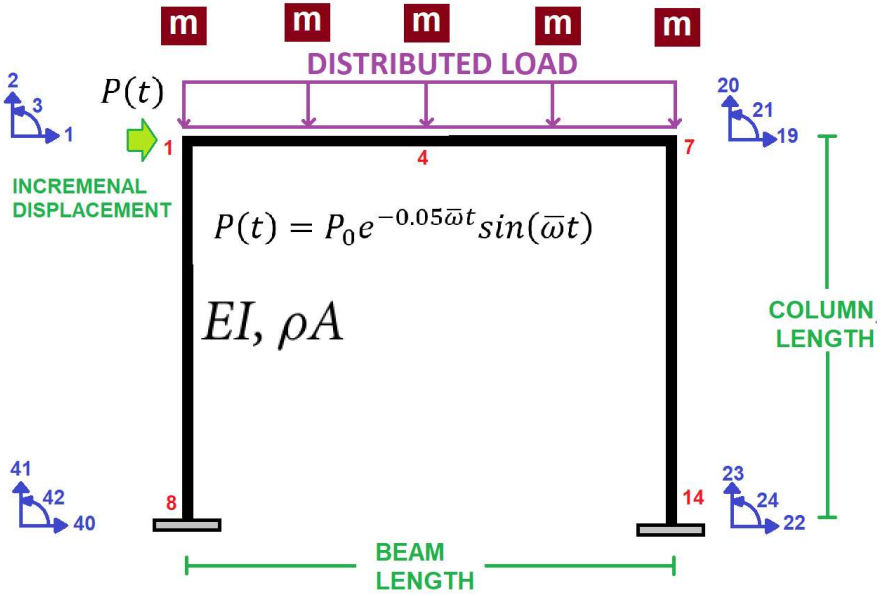
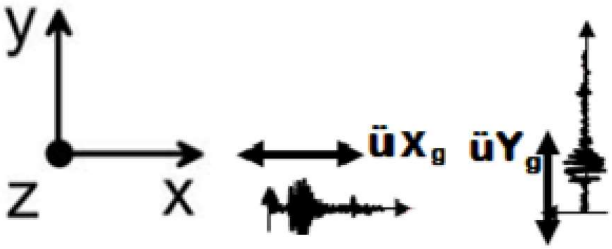
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A CONCRETE FRAME: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

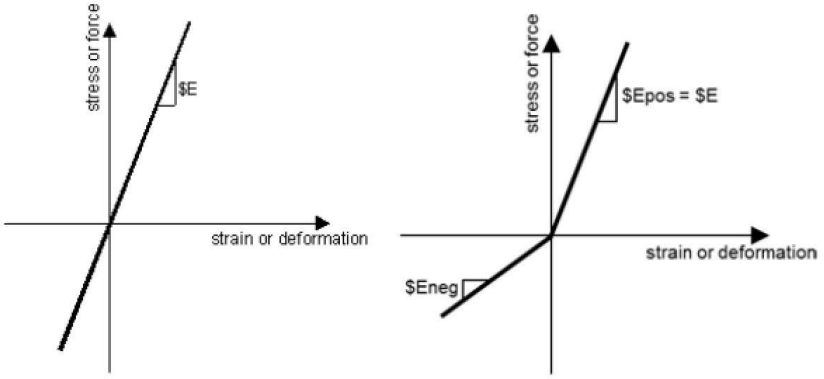




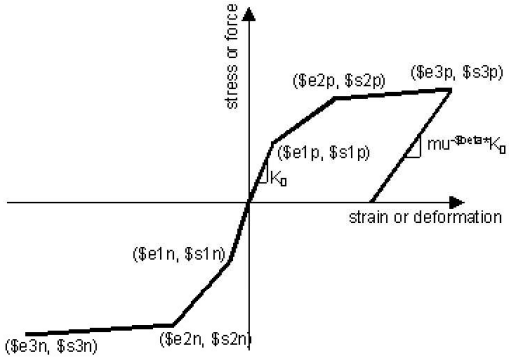
$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

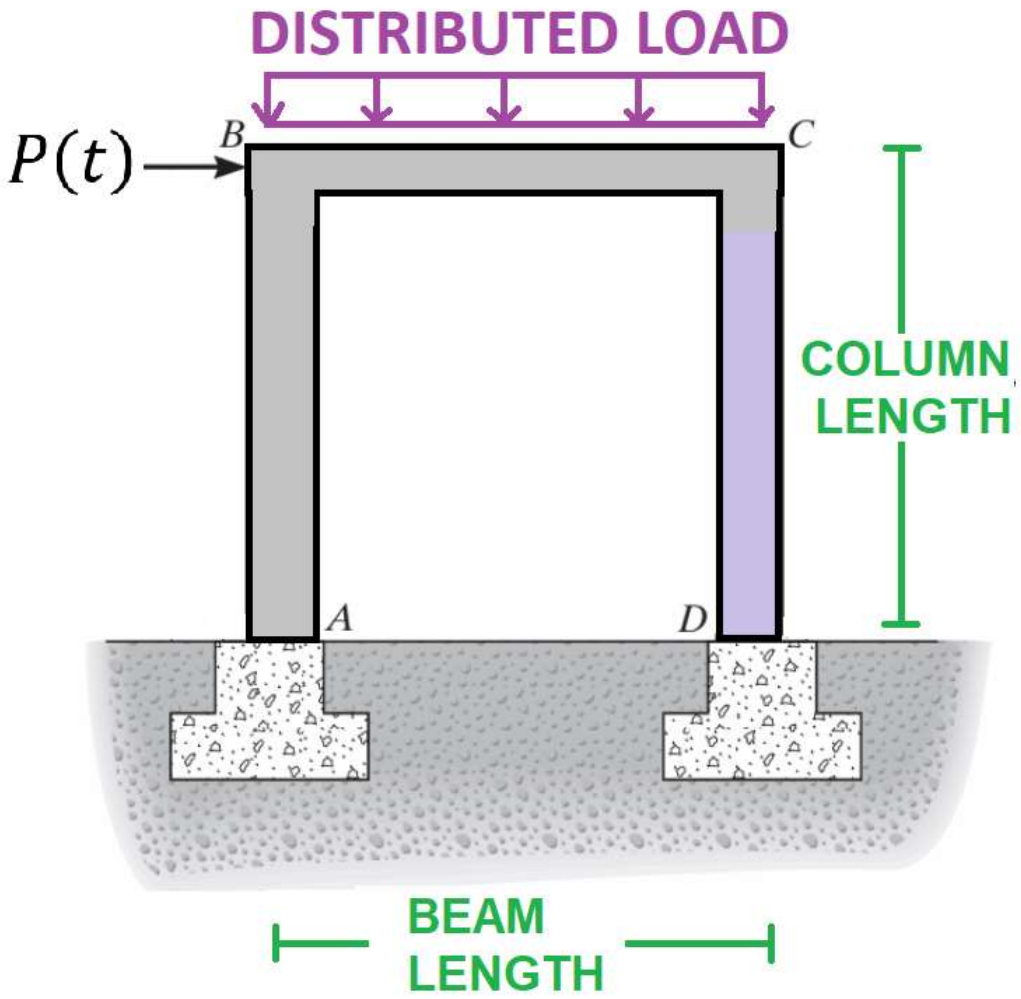


ELEMENT ELASTIC MATERIAL

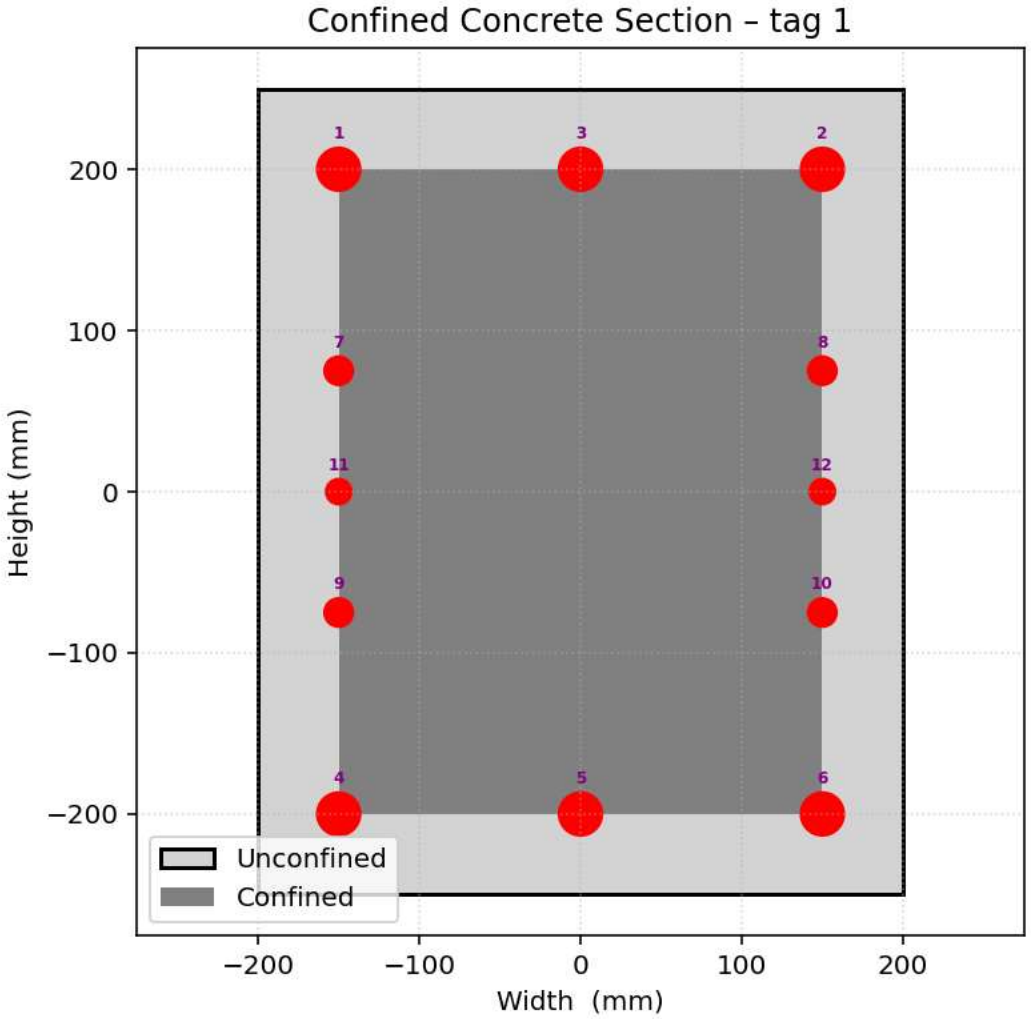


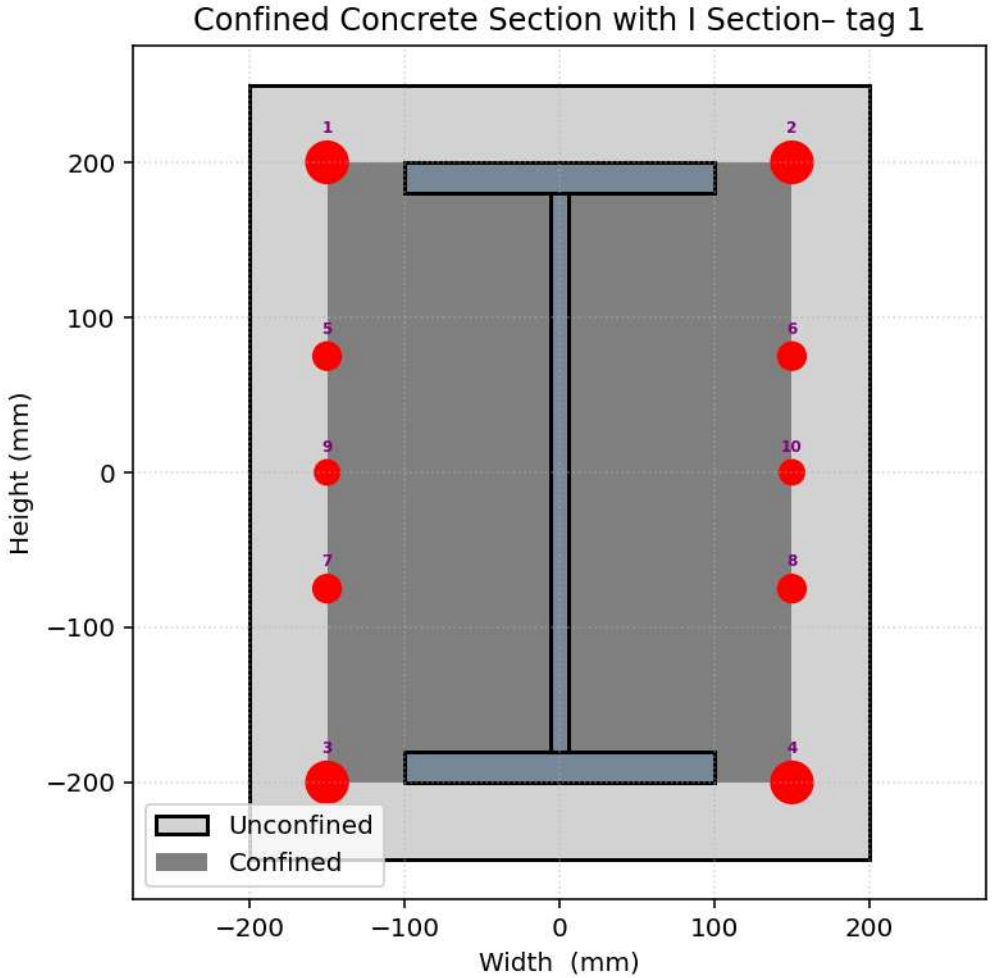
ELEMENT INELASTIC MATERIAL

$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$



$$\rho A \frac{\partial^2 v(x,t)}{\partial t^2} + EI \frac{\partial^4 v(x,t)}{\partial x^4} = P(t)$$


Confined Concrete Section – tag 1
Height (mm)
Width (mm)
Unconfined
Confined
1
2
3
4
5
6
7
8
9
10
11
12


Confined Concrete Section with I Section– tag 1
Height (mm)
Width (mm)
Unconfined
Confined
1
2
3
4
5
6
7
8
9
10

Spyder (Python 3.12)

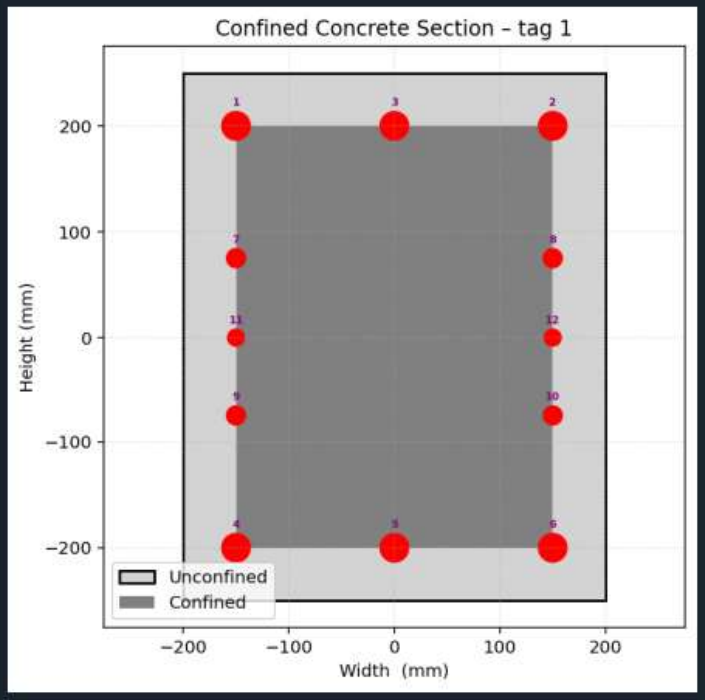
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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CONCRETE_FRAME_8_ANA\P(t)_CONCRETE_FRAME_8_ANA.py

P(t)_CONCRETE_FRAME_8_ANA.py | CONCRETE_SECTION_FUN_NEW.py

```
1  ##########################################################################################
2  #                  >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3  #         COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A CONCRETE FRAME : AN OPENSEES
4  #FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-H
5  #----------------------------------------------------------------------------------------
6  #                                          P(t) = P0 sin(wt)
7  #                                    P(t) = P0 exp(-0.05wt) sin(wt)
8  #----------------------------------------------------------------------------------------
9  #         ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND E
10 #                                    OF ENERGY DISSIPATION CAPACITY INDEX
11 #----------------------------------------------------------------------------------------
12 #                          THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
13 #                                    EMAIL: salar.d.ghashghaei@gmail.com
14 ##########################################################################################
15 """
16 Nonlinear Seismic Performance Assessment of a concrete frame:
17 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and
18
19 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a Beam E
20  for performance-based earthquake engineering. The 2D model incorporates both material non
21  (elastic-perfectly plastic or hysteretic steel with strain hardening)
22  and geometric nonlinearity to capture P-delta effects
23  and large displacements—essential for collapse assessment.
24
25 Eight analysis protocols are implemented:
26 (1) [PERIOD] : Structural Period
27 (2) [STATIC] : Gravity load analysis establishing dead load state
28 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
29  and plastic mechanism identification
30 (4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
31  pinching behavior, and energy dissipation degradation
32 (5) [STATIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Static Analysis of External time-dependent
33 (6) [DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Dynamic Analysis of External time-dependen
34 (7) [FREE-VIBRATION] : Free-vibration with initial conditions extracting damping ratios
```



IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 130, Col 58 | UTF-8 | CRLF | RW | Mem 41%

Spyder (Python 3.12)

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P(t)_CONCRETE_FRAME_8_ANA.py | CONCRETE_SECTION_FUN_NEW.py

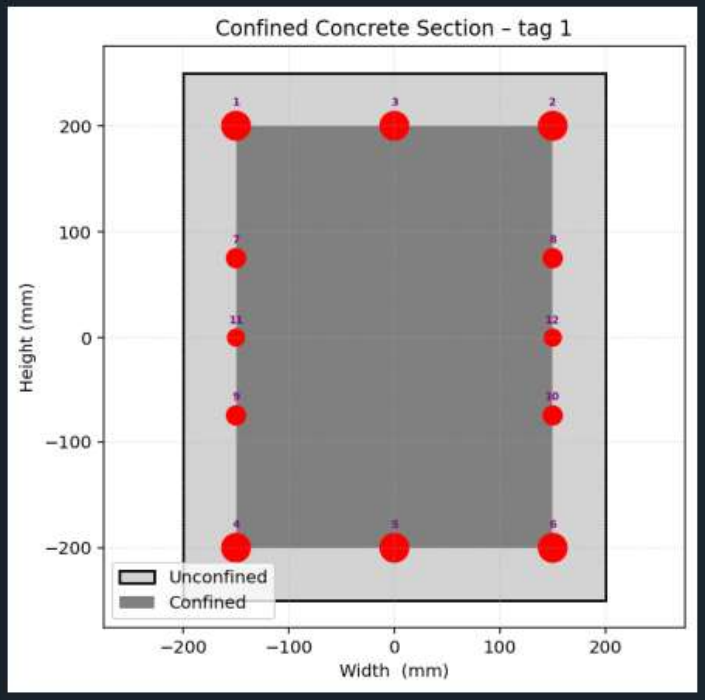
```python
def CONFINED_CONCRETE_SECTION(secTag, HSec, BSec, cover,
                              STEEL_TYPE, fc, Kc,
                              CONCRETE_DENSITY, PLOT):

    """
    Create a rectangular confined-concrete fiber section (OpenSees) and,
    if PLOT=True, draw a simple 2-D sketch of the section.

    Parameters
    ----------
    secTag          : int   - section identifier
    HSec, BSec      : float - height and width  (mm)
    cover           : float - concrete cover (mm)
    As              : float - area of one reinforcement bar (mm²)
    STEEL_TYPE      : str   - 'ELASTIC' or 'INELASTIC'
    fc              : float - unconfined concrete compressive strength (MPa, negative)
    Kfc             : float - ratio of confined to unconfined concrete strength
    CONCRETE_DENSITY: float - ρc (kg/m³) - will be converted to N·s²/mm³
    COL             : bool  - True → column (no middle layer of steel)
    PLOT            : bool  - draw the section if True

    THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
    """
    import numpy as np
    import openseespy.opensees as ops
    import matplotlib.pyplot as plt
    import matplotlib.pyplot as plt, matplotlib.patches as patches

    # Define materials for nonlinear elements
    # Define parameters (units: mm, N)
    # ------------------------------------------
    # CONCRETE                  tag   f'c        ec0   f'cu        ecu
    # Cover concrete (unconfined)
    fcU = -fc                   # [N/mm²] Concrete Compressive Strength
```



IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 120, Col 58 | UTF-8 | CRLF | RW | Mem 41%

STATIC ANALYSIS

Spyder (Python 3.12)

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P(t)_CONCRETE_FRAME_8_ANA.py | CONCRETE_SECTION_FUN_NEW.py

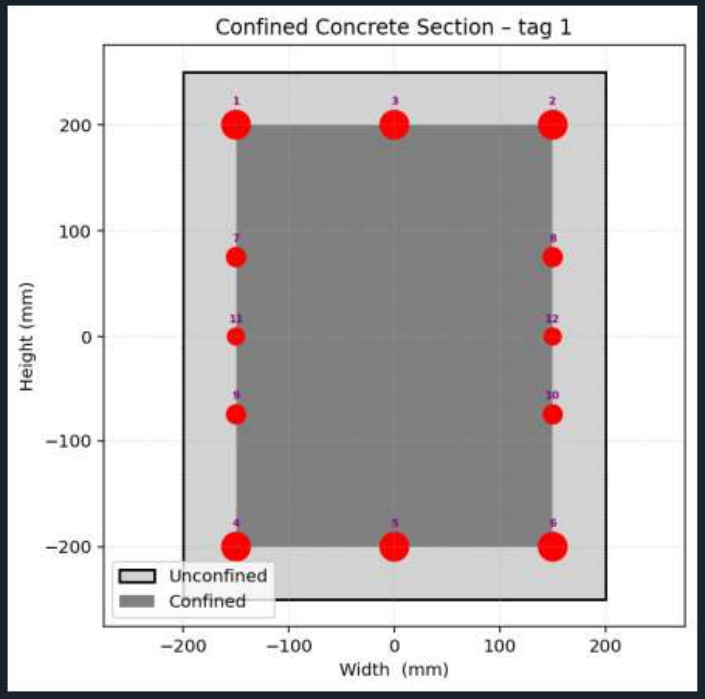
```
#%%-----------------------------------------------------
# STATIC ANALYSIS
#ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'STATIC'


DATA = CONCRETE_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(reactionX, reactionY, disp_mid,
 ele_axialforce, ele_shearforce, ele_momentforce,
 node_displacementsX, node_displacementsY,
 dispX, dispY) = DATA

S01.PLOT_2D_FRAME(deformed_scale=1.0)
#%%-----------------------------------------------------
# PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'PUSHOVER'

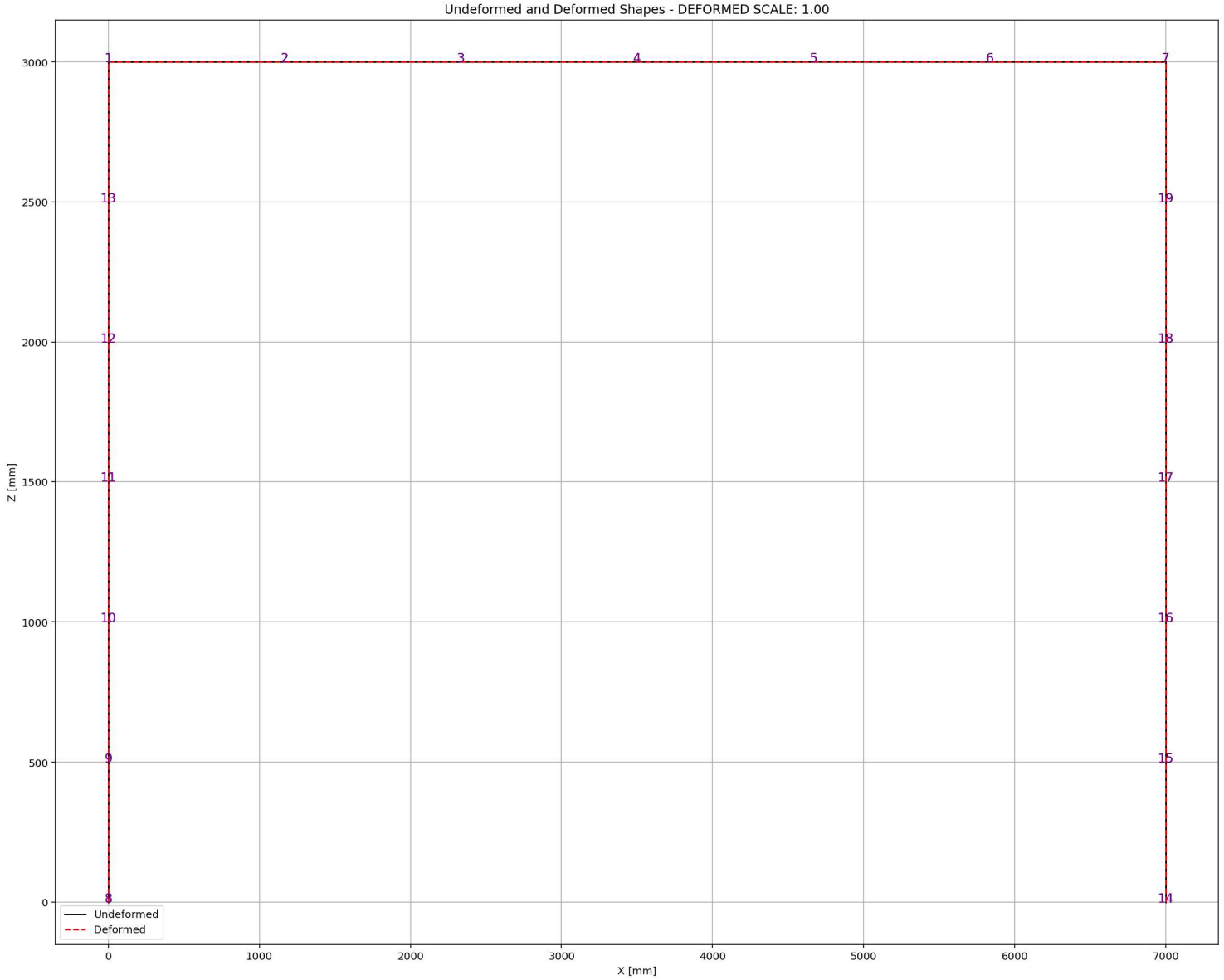

DATA = CONCRETE_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
(reactionX_PUSH, reactionY_PUSH, disp_mid_PUSH,
 ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
 node_displacementsX_PUSH, node_displacementsY_PUSH,
  dispX_PUSH, dispY_PUSH,
 PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
 STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINt_PUSH, CURVATURE_PUSH, NA_PUSH, CD_PUSH)


XDATA = disp_mid_PUSH
YDATA = reactionX_PUSH
```



IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 130, Col 58 | UTF-8 | CRLF | RW | Mem 41%


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
Undeformed
Deformed

PUSHOVER ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

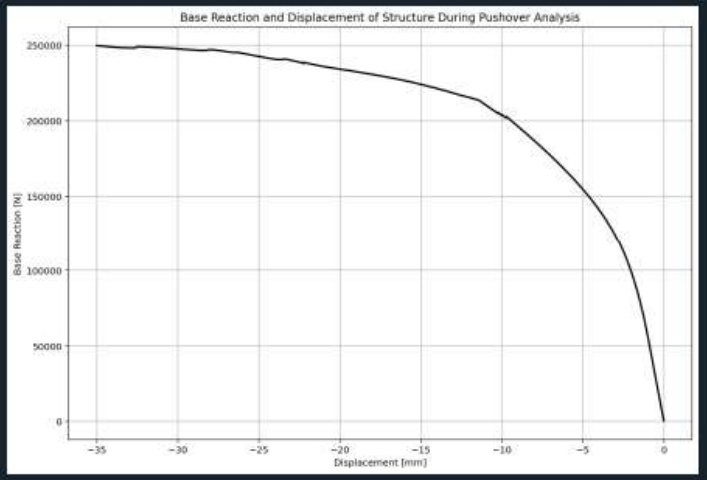
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P(t)_CONCRETE_FRAME_8_ANA.py | CONCRETE_SECTION_FUN_NEW.py

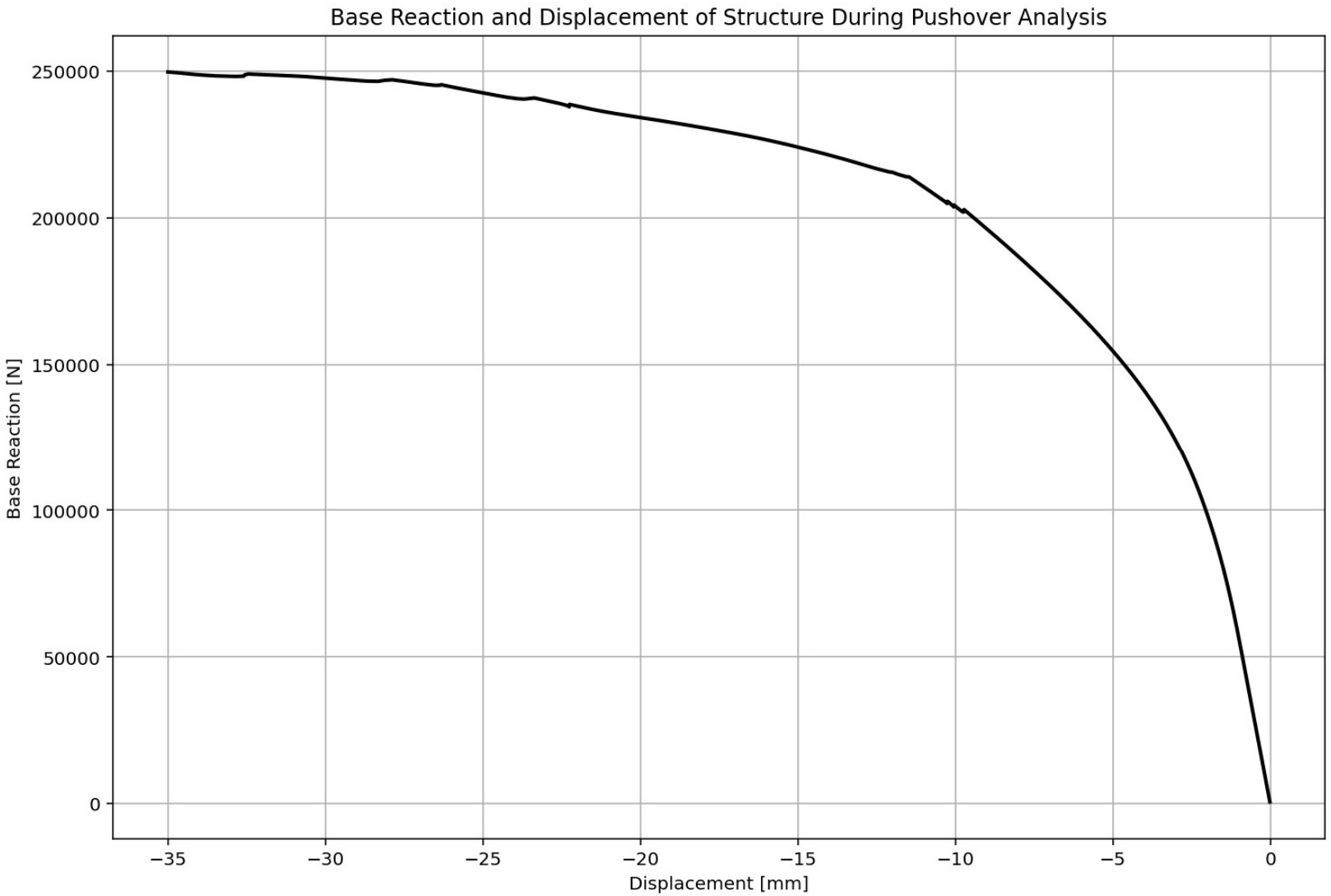
```
1253  # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1254  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1255  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIES
1256  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1257  ANAL_TYPE = 'PUSHOVER'
1258
1259
1260  DATA = CONCRETE_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1261  (reactionX_PUSH, reactionY_PUSH, disp_mid_PUSH,
1262   ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
1263   node_displacementsX_PUSH, node_displacementsY_PUSH,
1264    dispX_PUSH, dispY_PUSH,
1265   PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
1266   STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINt_PUSH, CURVATURE_PUSH, NA_PUSH, CD_PUSH)
1267
1268
1269  XDATA = disp_mid_PUSH
1270  YDATA = reactionX_PUSH
1271  XLABEL = 'Displacement [mm]'
1272  YLABEL = 'Base Reaction [N]'
1273  TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
1274  COLOR = 'black'
1275  SEMILOGY = False
1276  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1277
1278  DATA = S07.BILNEAR_CURVE(np.abs(disp_mid_PUSH), np.abs(reactionY_PUSH), SLOPE_NODE=10)
1279  (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor)
1280
1281  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1282  plt.figure(0, figsize=(12, 8))
1283  plt.plot(disp_mid_PUSH, PERIOD_MIN_PUSH, linewidth=3)
1284  plt.plot(disp_mid_PUSH, PERIOD_MAX_PUSH, linewidth=3)
1285  plt.title('Period of Structure During Pushover Analysis')
1286  plt.ylabel('Structural Period [s]')
```

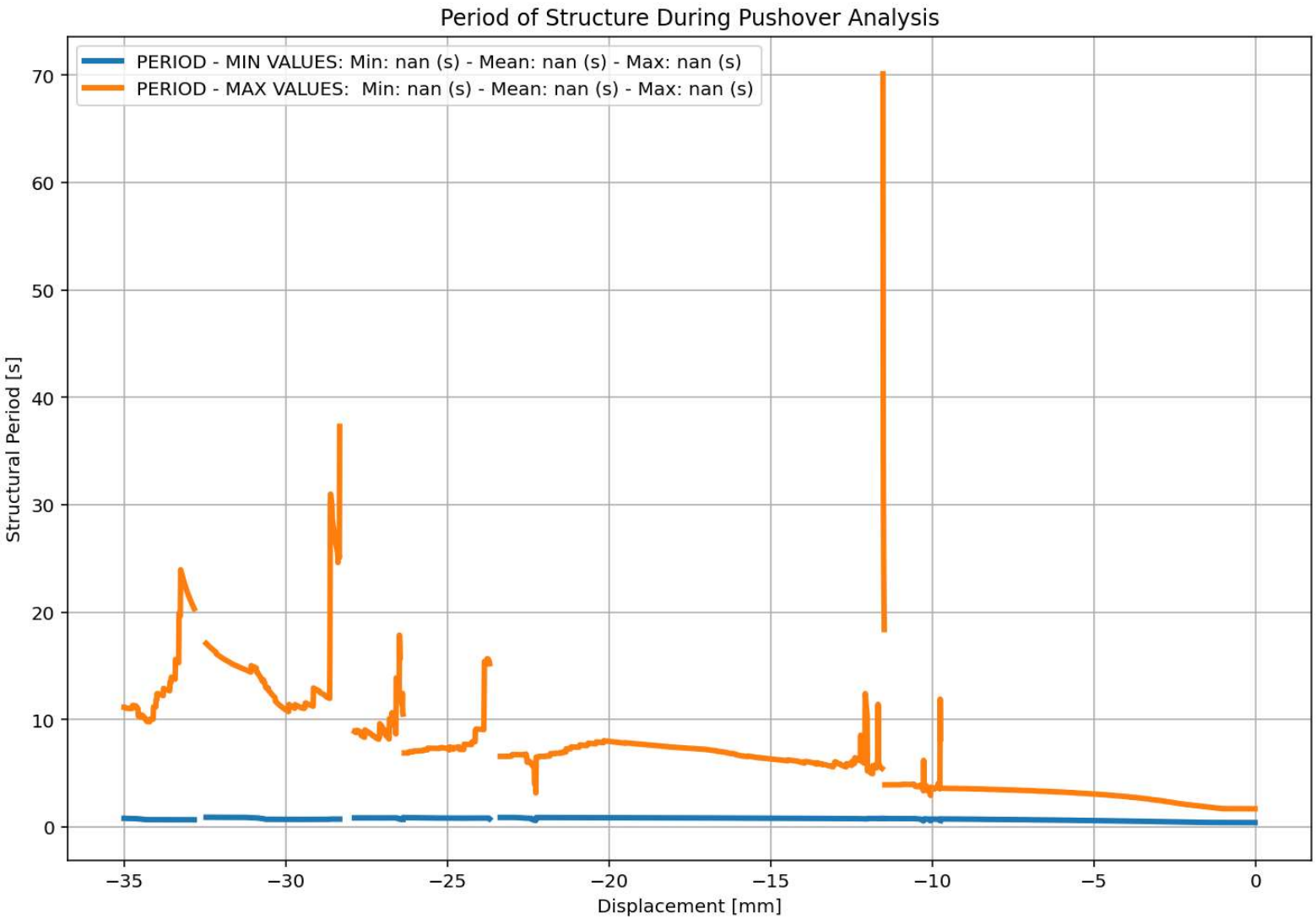
30 %

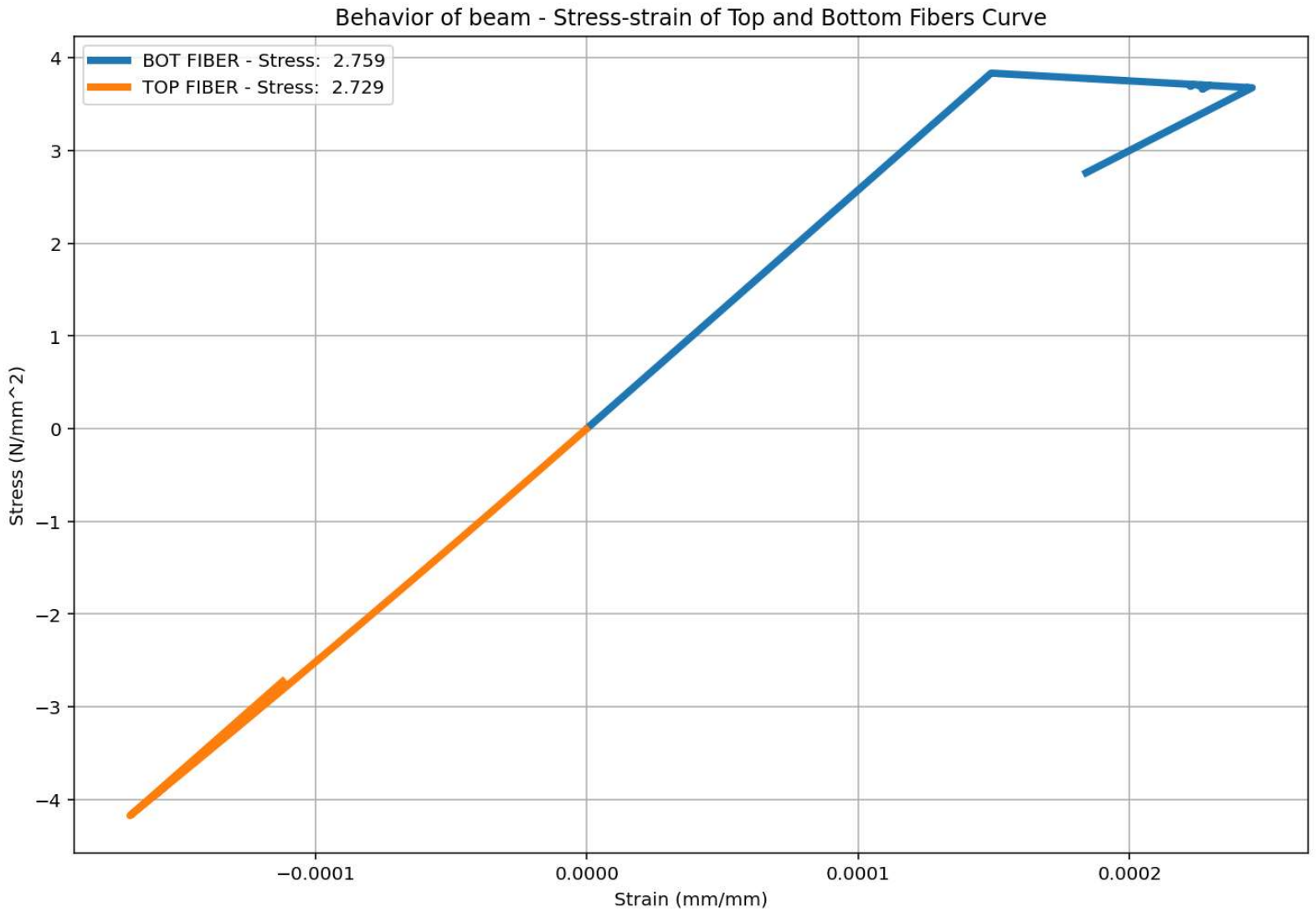


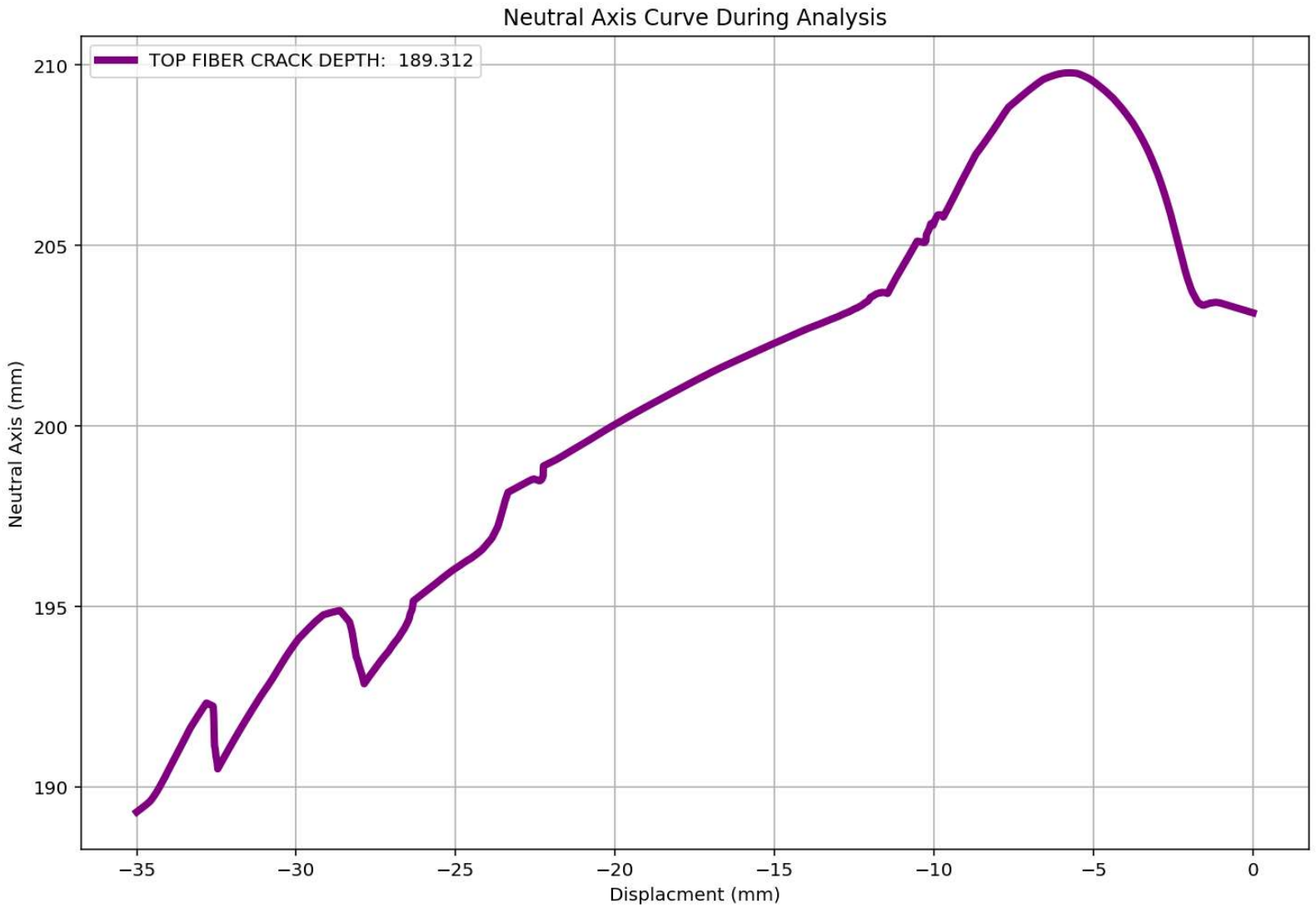
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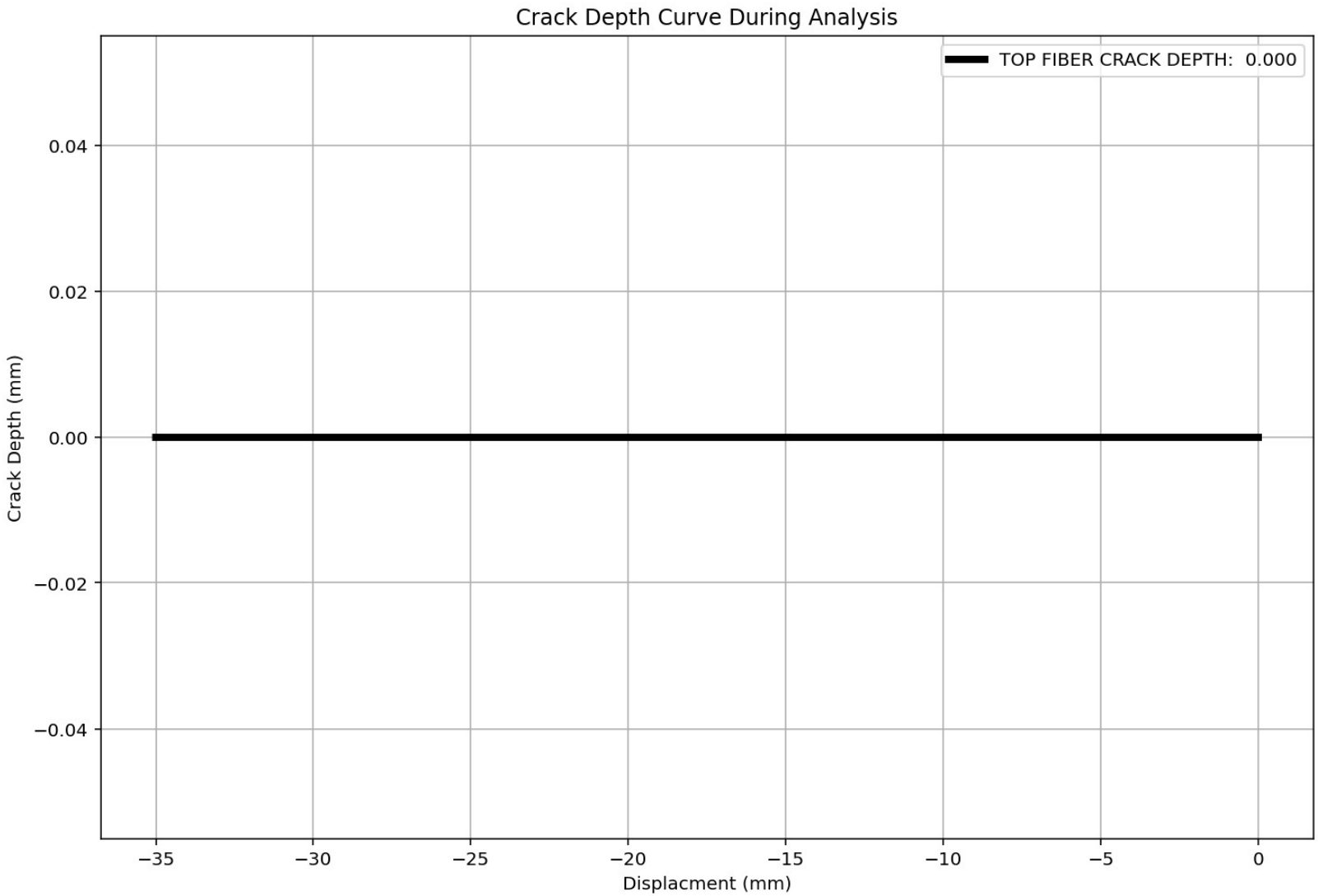
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 130, Col 58 | UTF-8 | CRLF | RW | Mem 42%

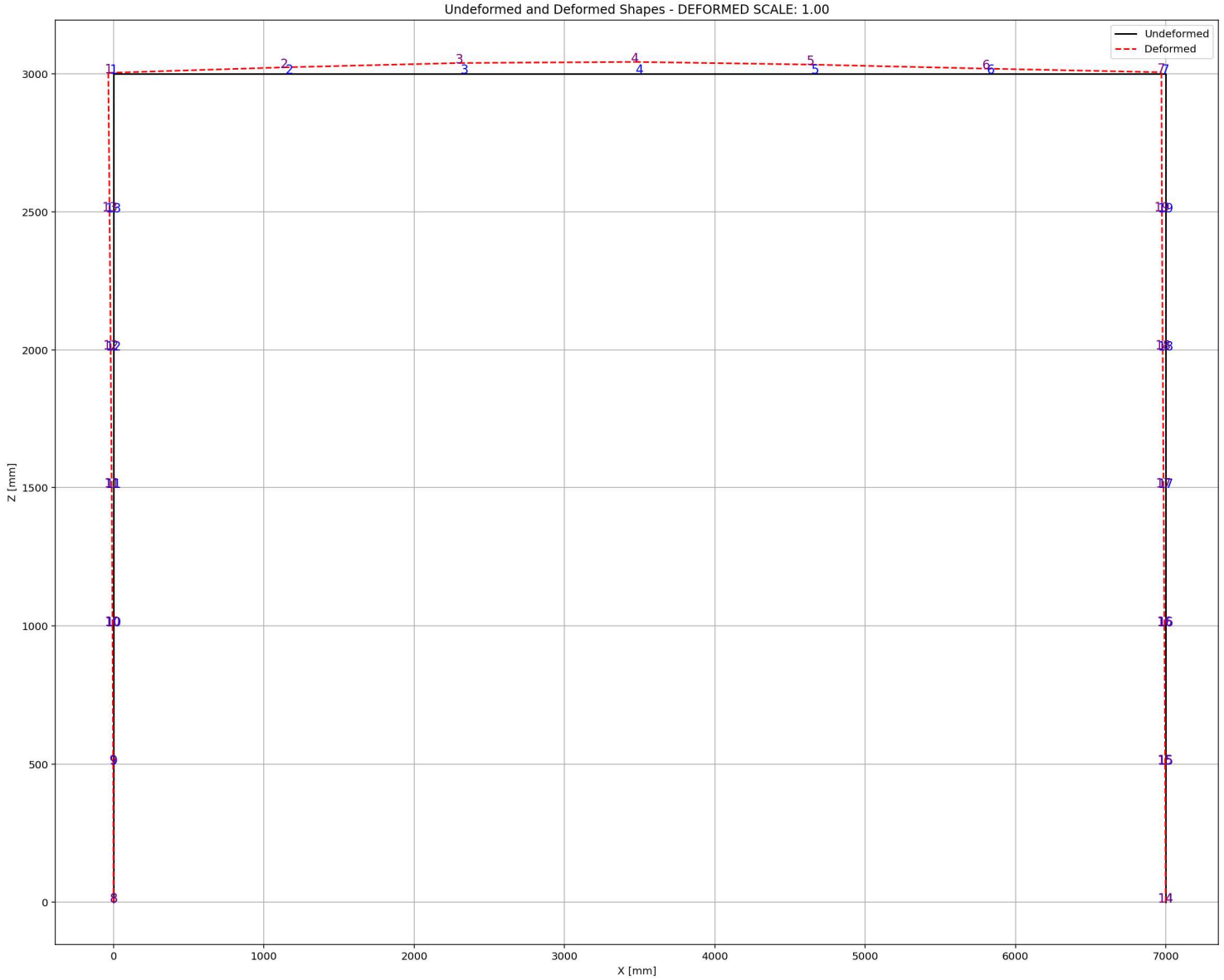

Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
250000
200000
150000
100000
50000
0
−35
−30
−25
−20
−15
−10
−5
0
Displacement [mm]


Period of Structure During Pushover Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
Displacement [mm]
70
60
50
40
30
20
10
0
−35
−30
−25
−20
−15
−10
−5
0


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 2.759
TOP FIBER - Stress: 2.729
Stress (N/mm^2)
Strain (mm/mm)
4
3
2
1
0
−1
−2
−3
−4
−0.0001
0.0000
0.0001
0.0002


Neutral Axis Curve During Analysis
TOP FIBER CRACK DEPTH: 189.312
Neutral Axis (mm)
210
205
200
195
190
−35
−30
−25
−20
−15
−10
−5
0
Displacment (mm)


Crack Depth Curve During Analysis
TOP FIBER CRACK DEPTH: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−35
−30
−25
−20
−15
−10
−5
0
Displacment (mm)


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000

CYCLIC DISPLACEMENT ANALYSIS

Spyder (Python 3.12)

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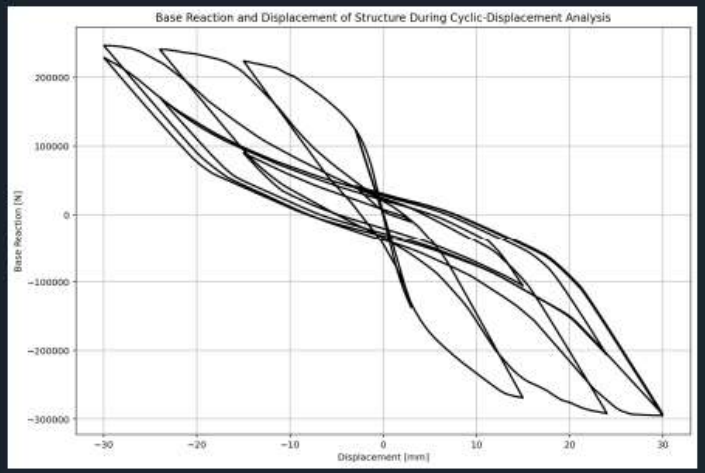
...ENERGY_DISSIPATION_CAPACITY_INDEX\12_CONCRETE_FRAME_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CONCRETE_FRAME_8_ANA\P(t)_CONCRETE_FRAME_8_ANA.py

P(t)_CONCRETE_FRAME_8_ANA.py | CONCRETE_SECTION_FUN_NEW.py

```
500     if ANAL_TYPE == 'CYCLIC_DISPLACEMENT': # STATIC TIME-HISTORY ANALYSIS
501         IDctrlDOF = 1   # 1: Horizental Dispalcement - 2: Vertical Dispalcement
502         DMAX = -30.0    # [mm] Max. Displacement
503         n_points = 10000
504         # 4. CYCLIC DISPALCEMENT PROTOCOL
505         # Key strain points (same logic as your protocol)
506         key_disp = np.array([
507             0.0,
508             0.1*DMAX,   -0.1*DMAX,
509             0.5*DMAX,   -0.5*DMAX,
510             0.8*DMAX,   -0.8*DMAX,
511             DMAX,       -DMAX,
512             0.1*DMAX,   -0.1*DMAX,
513             0.5*DMAX,   -0.5*DMAX,
514             0.8*DMAX,   -0.8*DMAX,
515             DMAX,       -DMAX,
516             0.0
517         ])
518
519         # Generate 10000-point displacement protocol
520         disp_protocol = np.interp(
521             np.linspace(0, len(key_disp) - 1, n_points),
522             np.arange(len(key_disp)),
523             key_disp
524         )
525
526         ops.analysis('Static')
527         STEP = 0.0
528         for target_disp in disp_protocol:
529             current_disp = ops.nodeDisp(center_node, 1) # DISPALCEMENT APPLIED IN NODE IN
530             dU = target_disp - current_disp
531             ops.integrator('DisplacementControl', center_node, IDctrlDOF, dU)
532             OK = ops.analyze(1)
533             S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
```

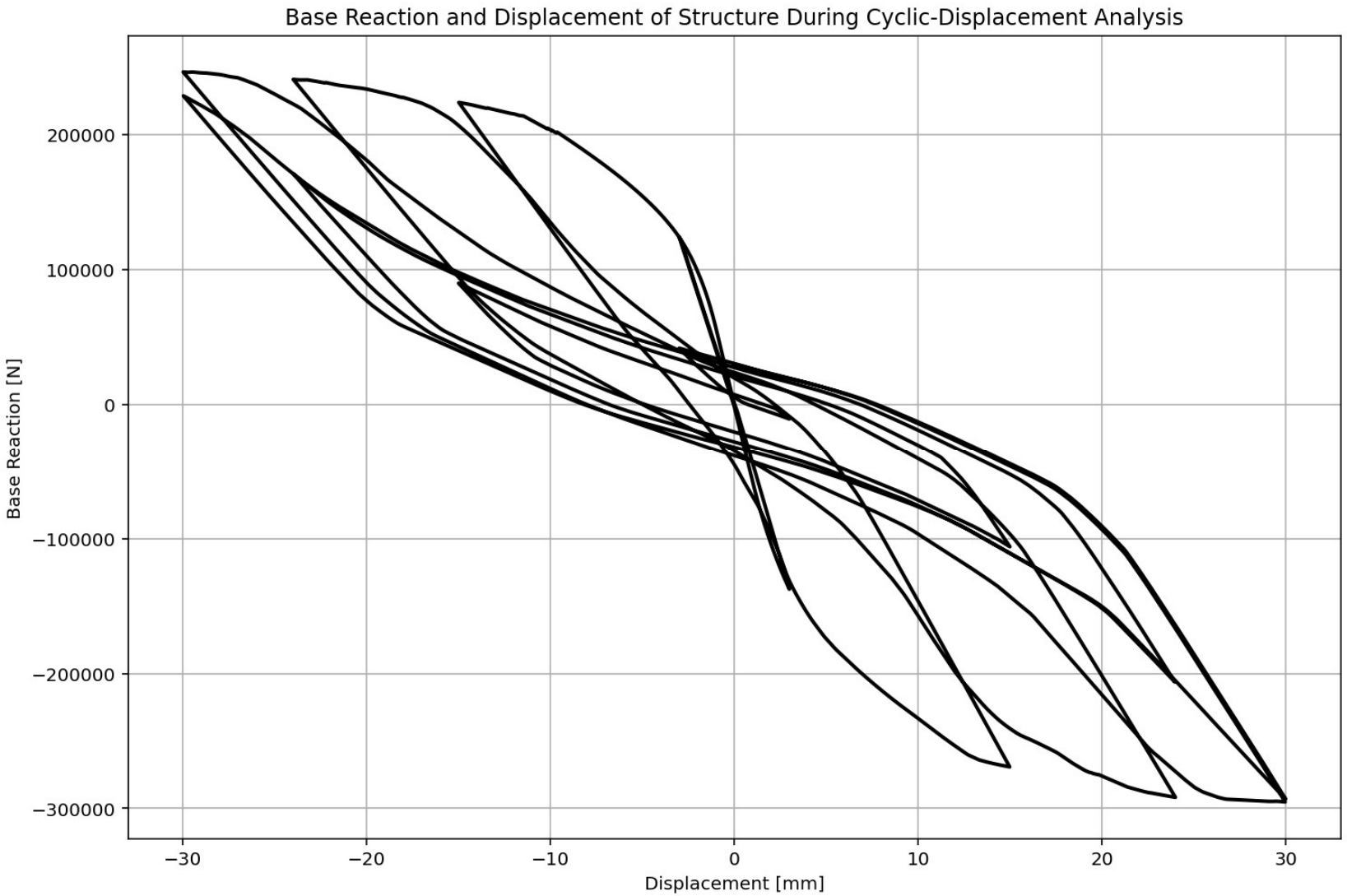
30 %

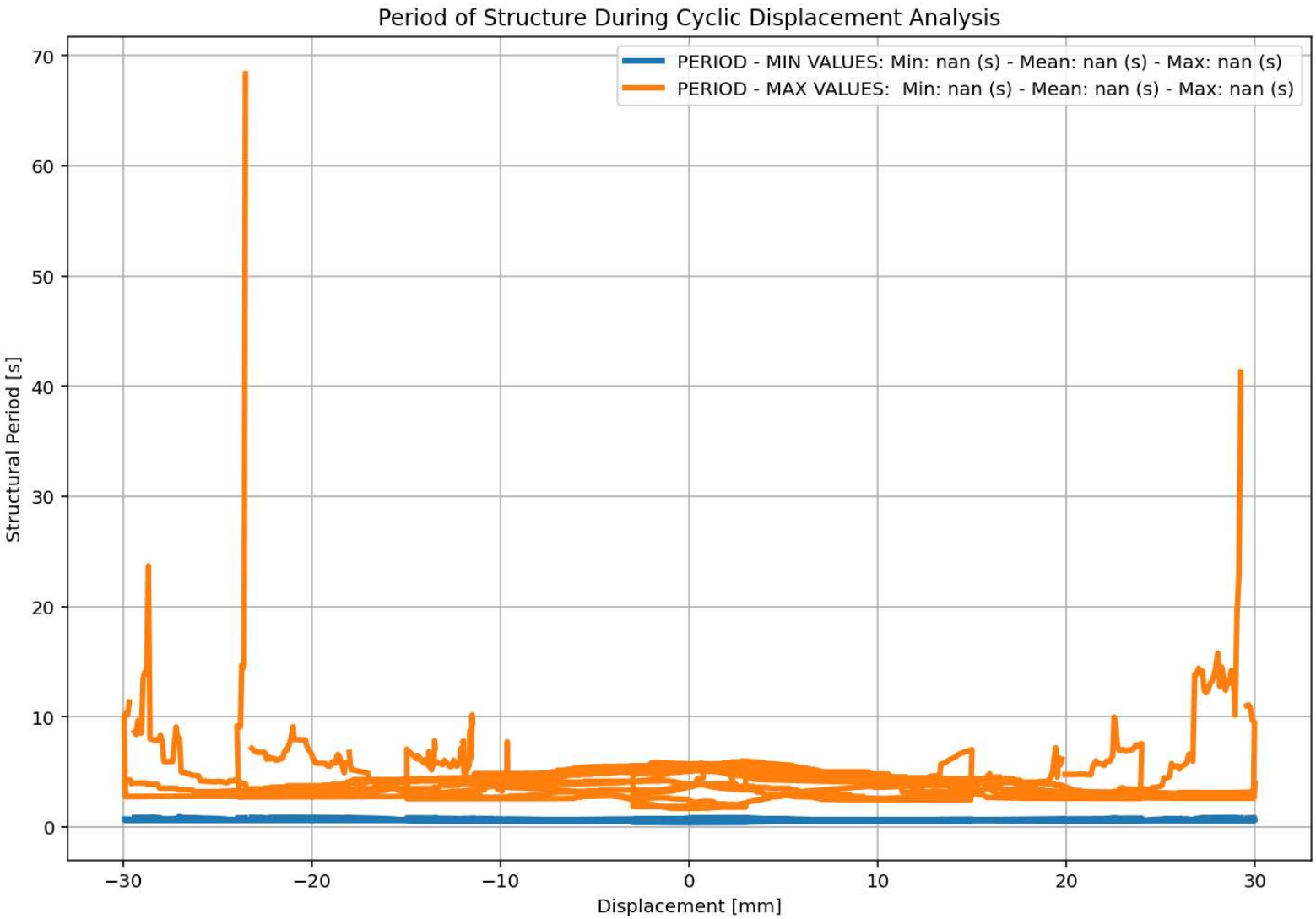


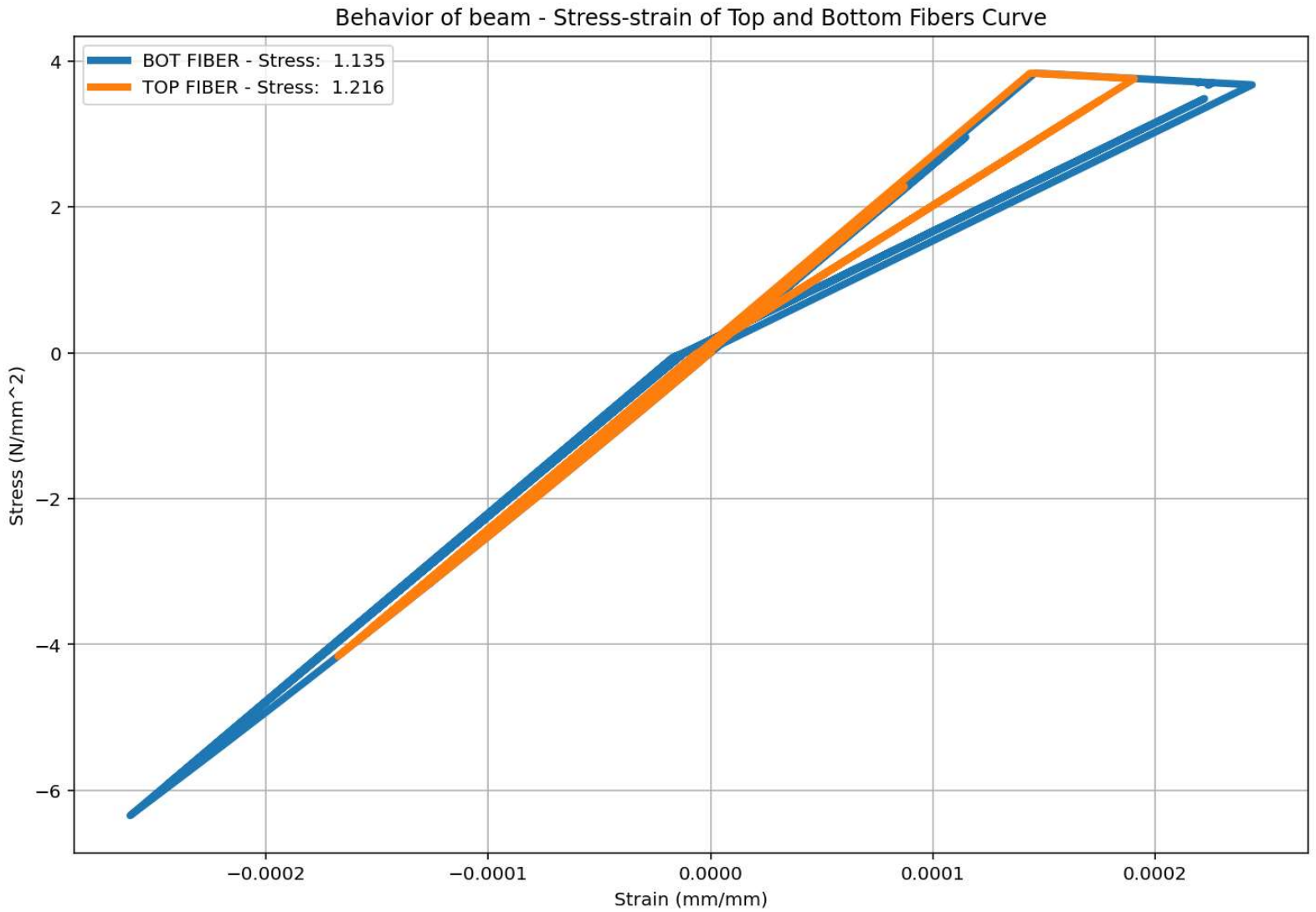


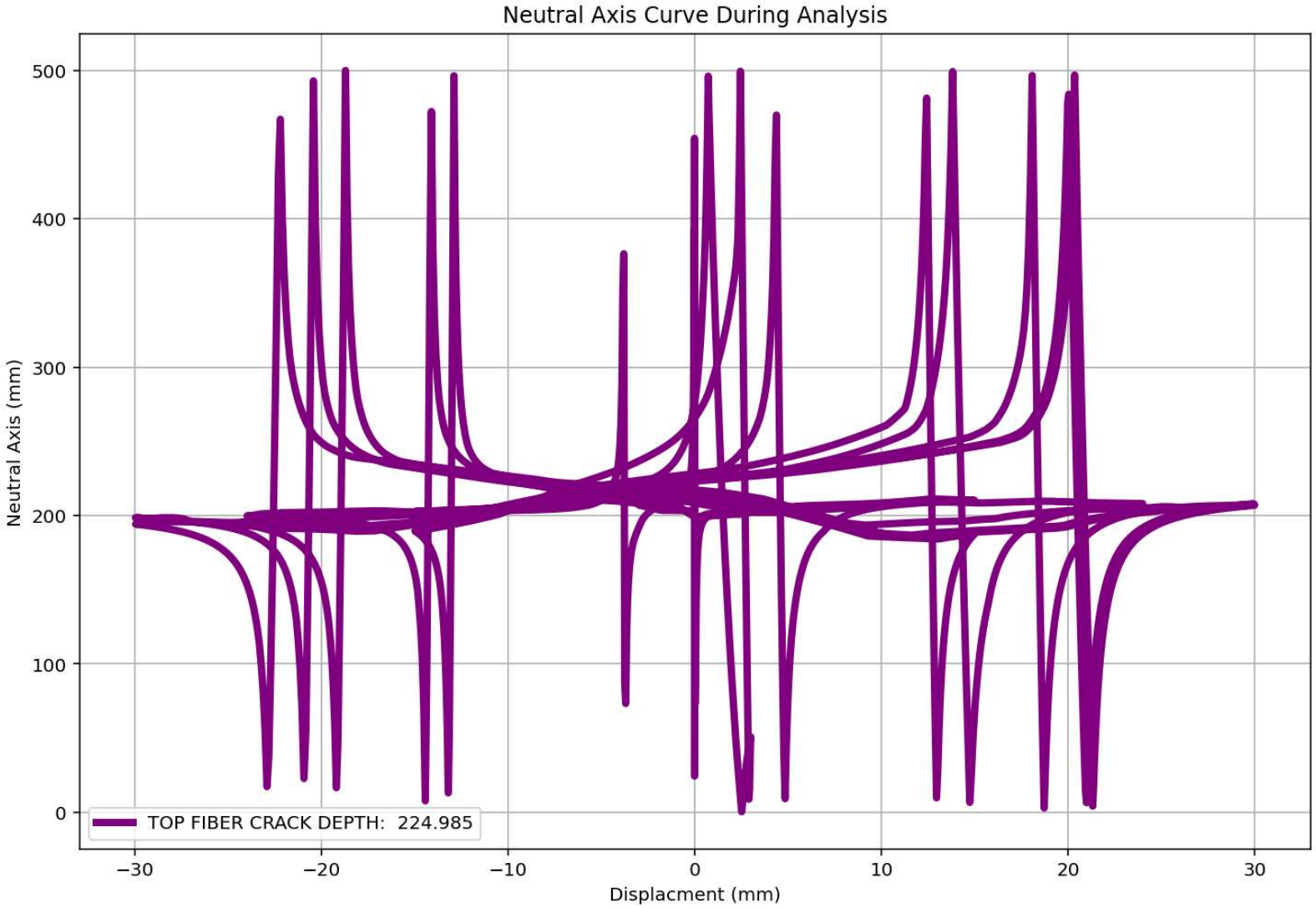
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

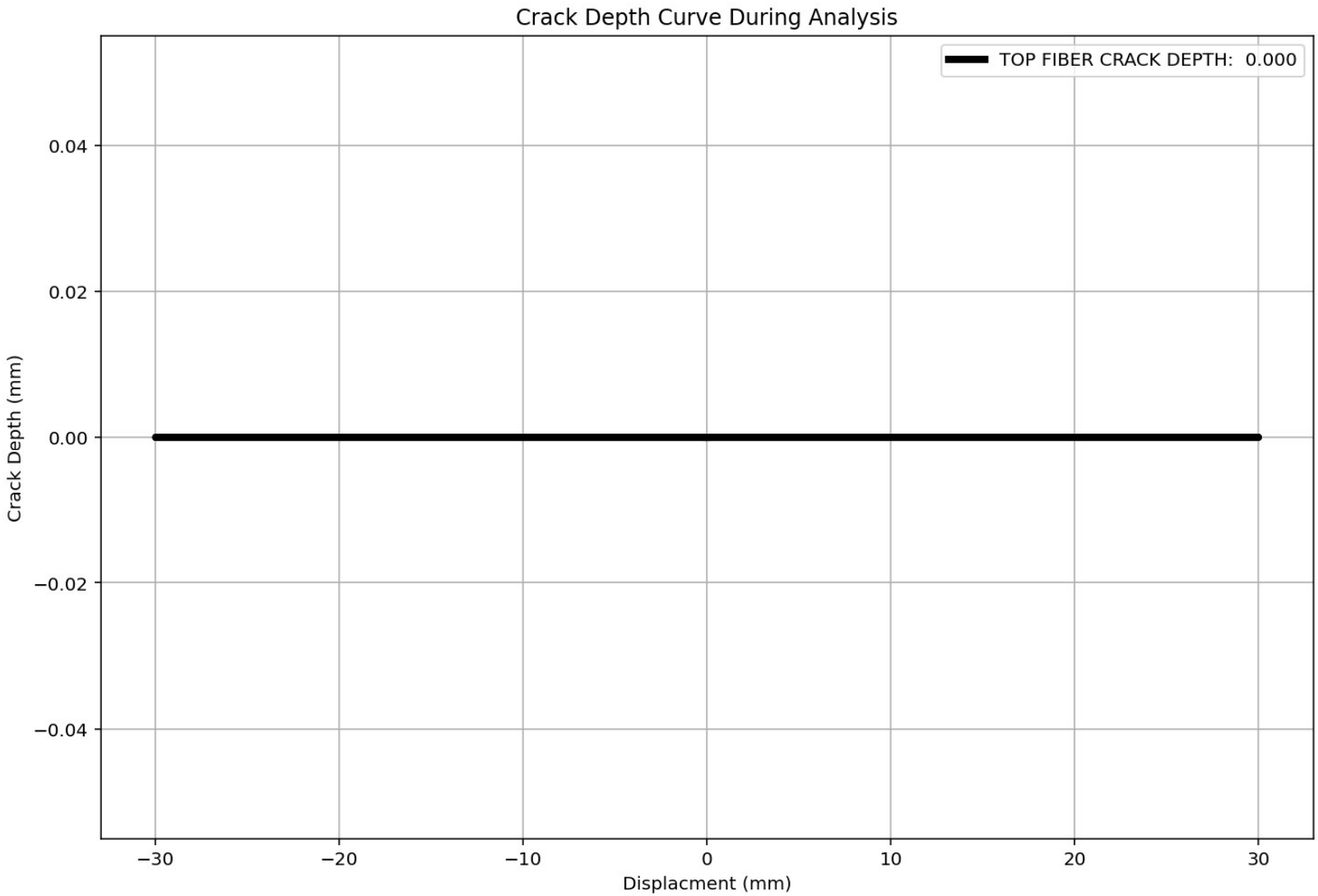
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 502, Col 19 | UTF-8 | CRLF | RW | Mem 40%

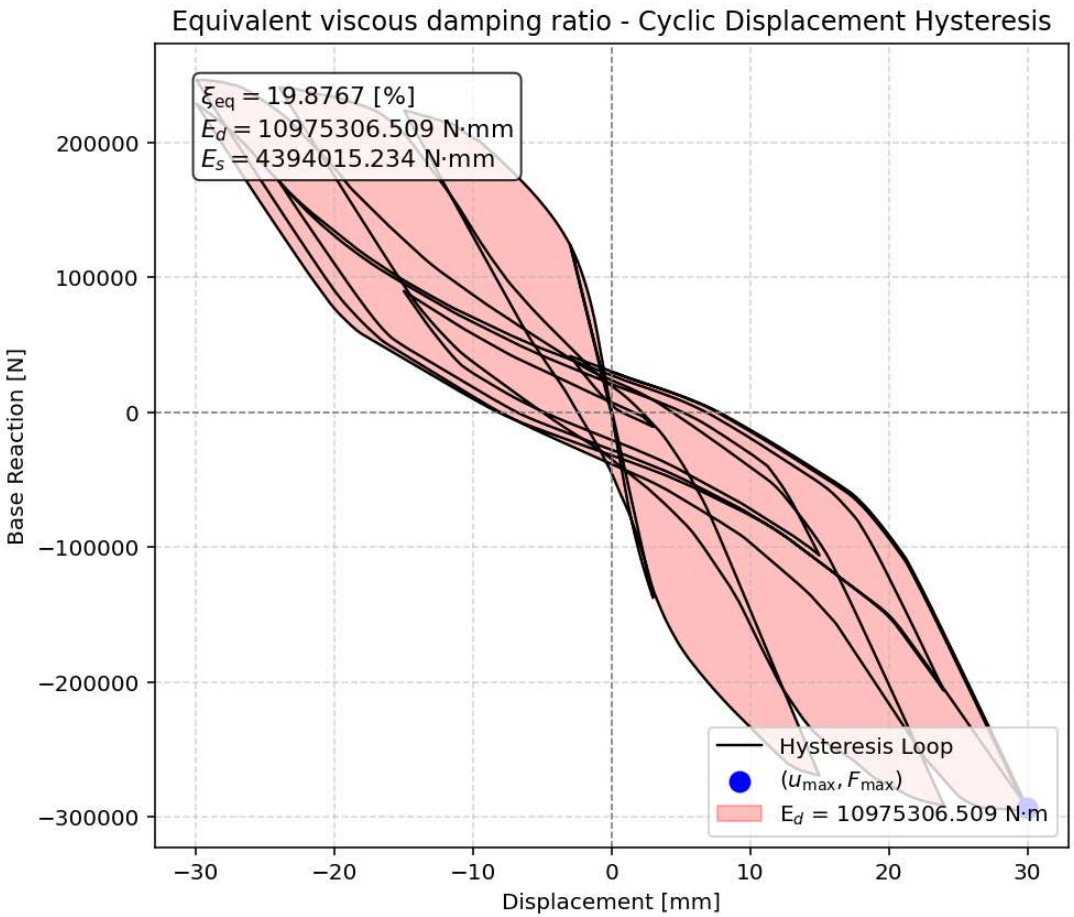

Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
Base Reaction [N]
200000
100000
0
−100000
−200000
−300000
−30
−20
−10
0
10
20
30
Displacement [mm]

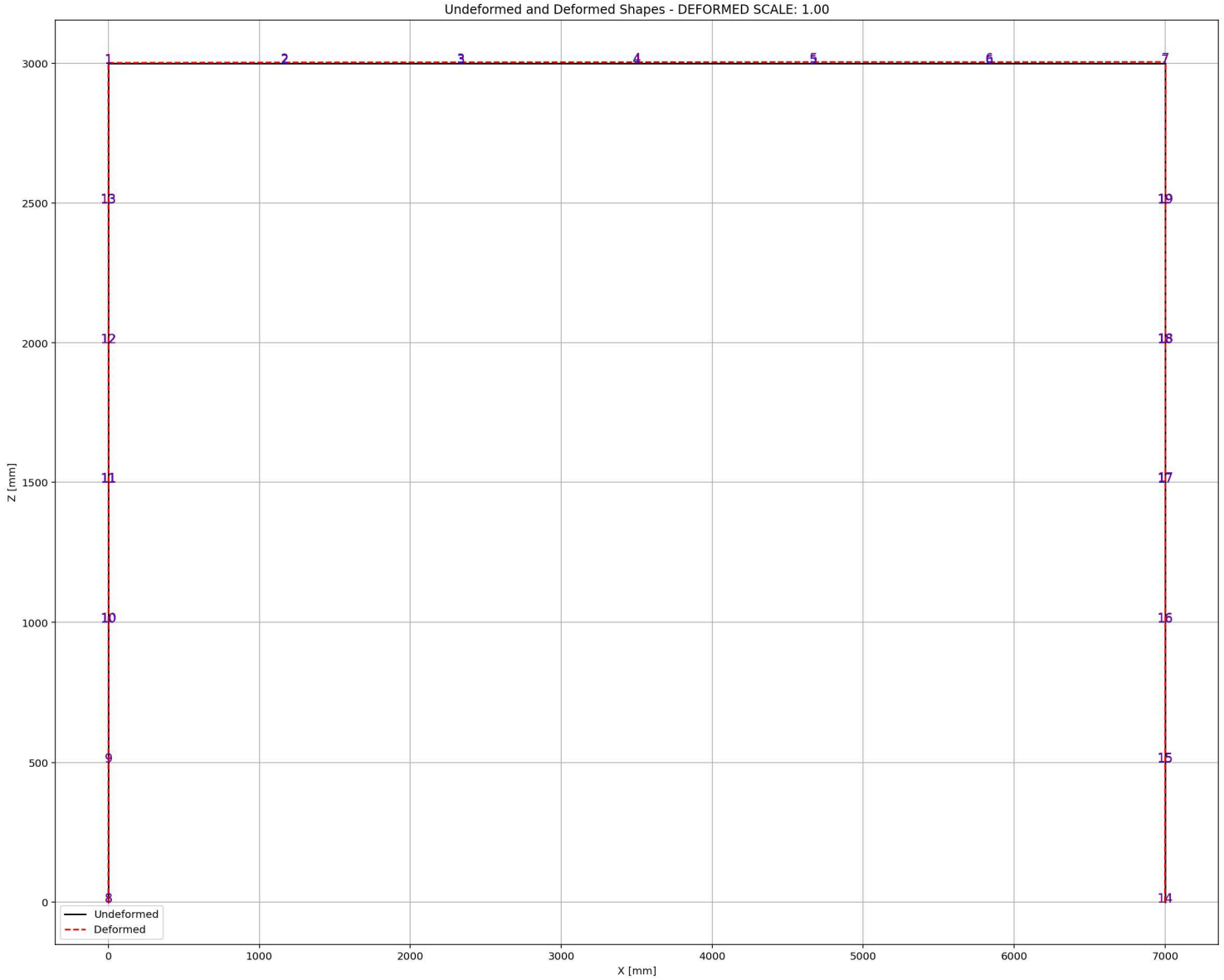

Period of Structure During Cyclic Displacement Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
Displacement [mm]
70
60
50
40
30
20
10
0
−30
−20
−10
0
10
20
30


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 1.135
TOP FIBER - Stress: 1.216
Stress (N/mm^2)
Strain (mm/mm)
4
2
0
−2
−4
−6
−0.0002
−0.0001
0.0000
0.0001
0.0002


Neutral Axis Curve During Analysis
Neutral Axis (mm)
500
400
300
200
100
0
−30
−20
−10
0
10
20
30
Displacment (mm)
TOP FIBER CRACK DEPTH: 224.985


Crack Depth Curve During Analysis
TOP FIBER CRACK DEPTH: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−30
−20
−10
0
10
20
30
Displacment (mm)


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
$\xi_{eq} = 19.8767$ [%]
$E_d = 10975306.509$ N·mm
$E_s = 4394015.234$ N·mm
Base Reaction [N]
Displacement [mm]
200000
100000
0
−100000
−200000
−300000
−30
−20
−10
0
10
20
30
Hysteresis Loop
(u_{max}, F_{max})
E_d = 10975306.509 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
Undeformed
Deformed

STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

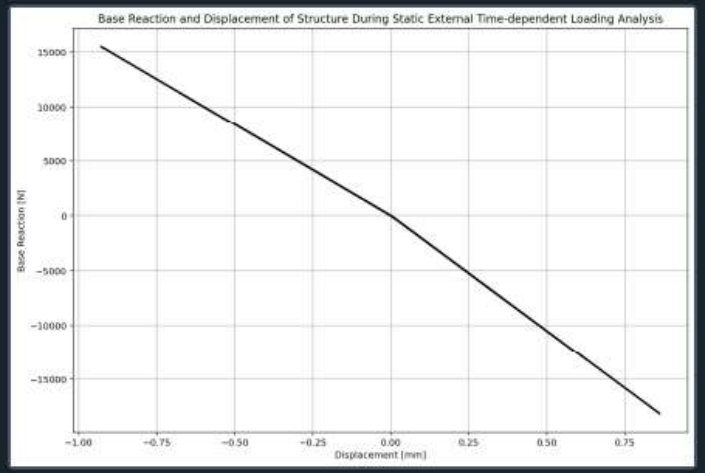
...ENERGY_DISSIPATION_CAPACITY_INDEX\12_CONCRETE_FRAME_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CONCRETE_FRAME_8_ANA\P(t)_CONCRETE_FRAME_8_ANA.py

P(t)_CONCRETE_FRAME_8_ANA.py | CONCRETE_SECTION_FUN_NEW.py

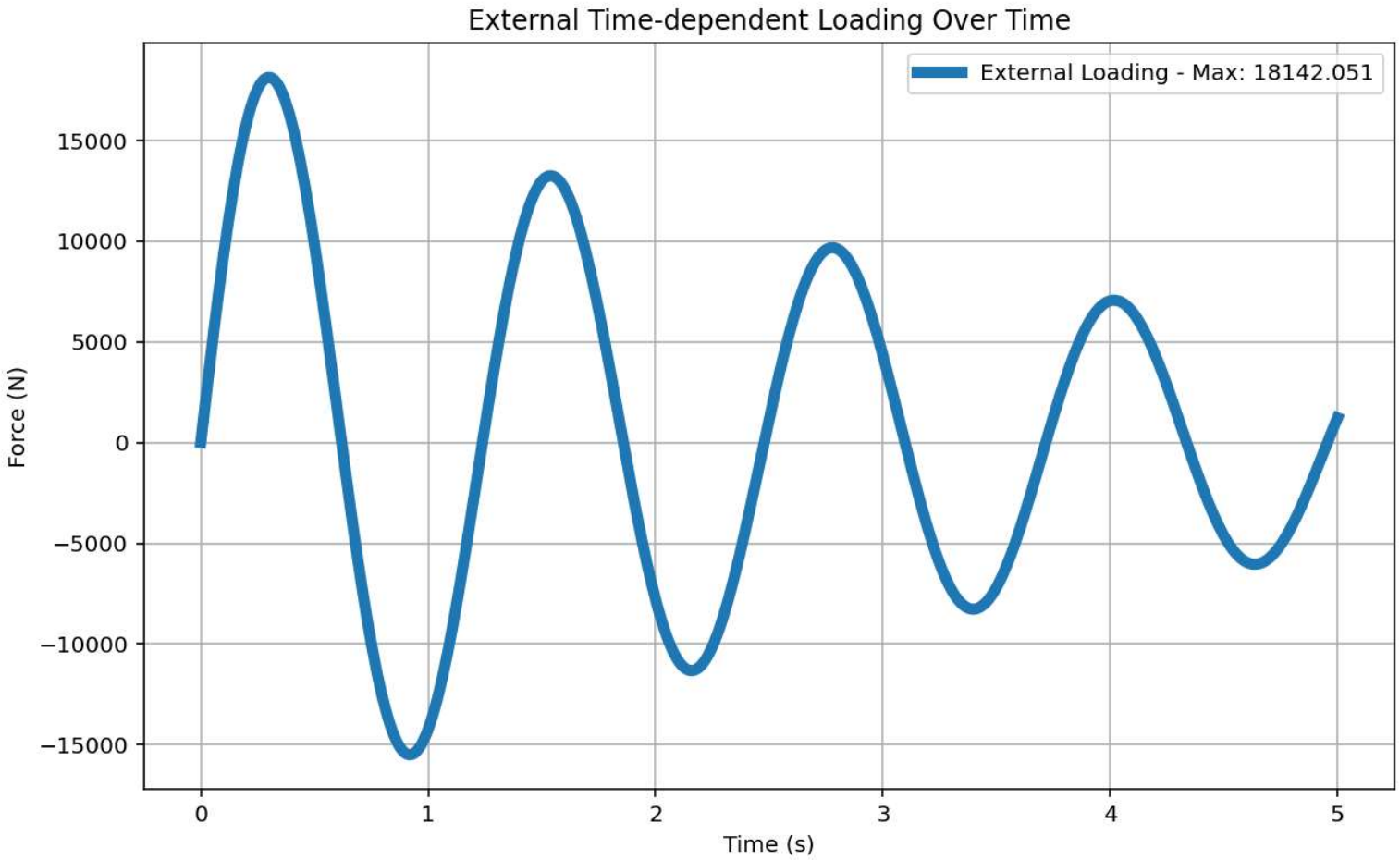
```
1269  # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1270  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1271  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIES
1272  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1273  ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1274
1275
1276  DATA = CONCRETE_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1277
1278  (reactionX_ETDLS, reactionY_ETDLS, disp_mid_ETDLS,
1279   ele_axialforce_ETDLS, ele_shearforce_ETDLS, ele_momentforce_ETDLS,
1280   node_displacementsX_ETDLS, node_displacementsY_ETDLS,
1281    dispX_ETDLS, dispY_ETDLS,
1282   PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS,
1283   STRESSb_ETDLS, STRAINb_ETDLS, STRESSt_ETDLS, STRAINt_ETDLS, CURVATURE_ETDLS) = DATA
1284
1285
1286  XDATA = disp_mid_ETDLS
1287  YDATA = reactionX_ETDLS
1288  XLABEL = 'Displacement [mm]'
1289  YLABEL = 'Base Reaction [N]'
1290  TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent
1291  COLOR = 'black'
1292  SEMILOGY = False
1293  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1294
1295  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1296  plt.figure(0, figsize=(12, 8))
1297  plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
1298  plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
1299  plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
1300  plt.ylabel('Structural Period [s]')
1301  plt.xlabel('Displacement [mm]')
1302  #plt.semilogy()
```

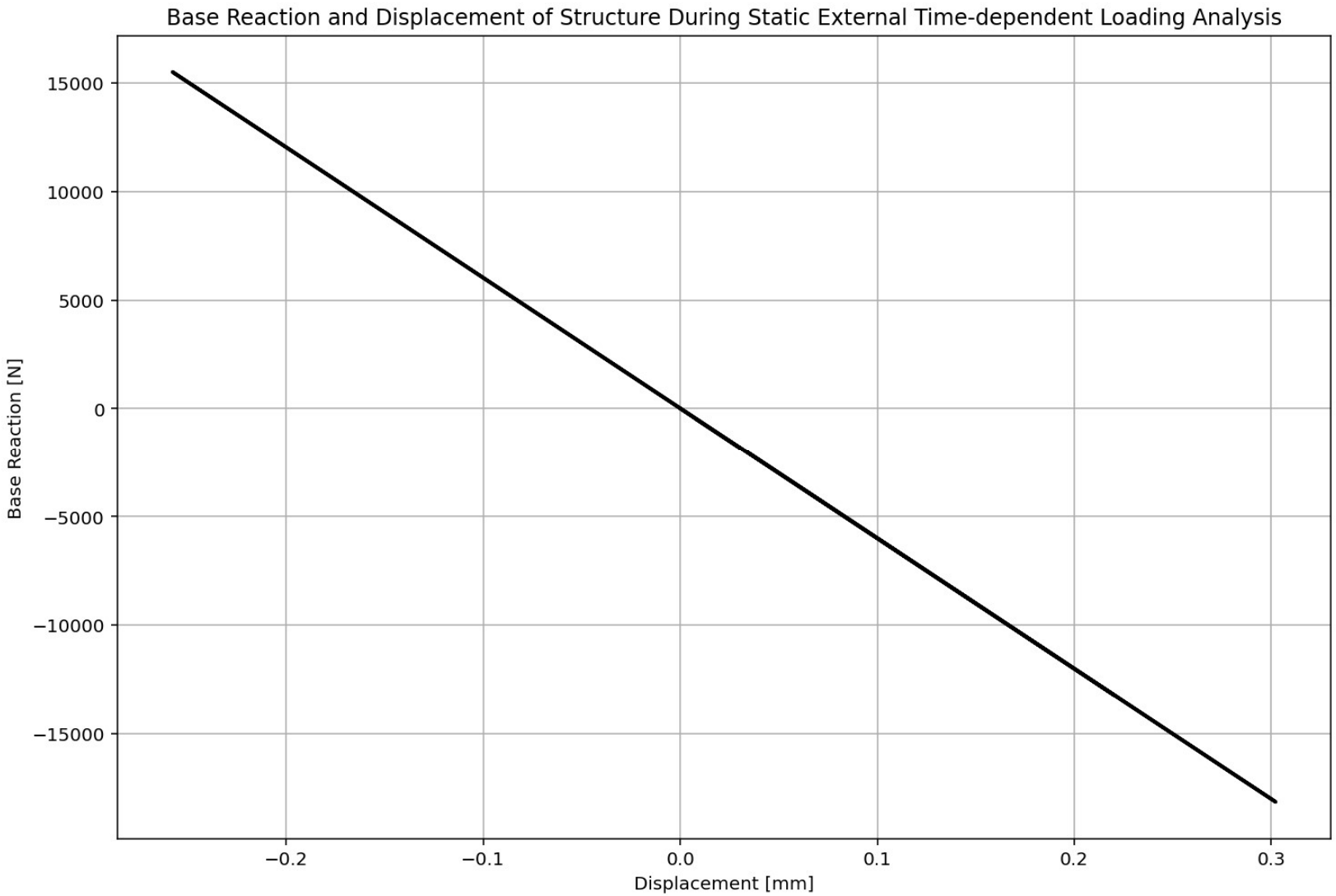
30 %

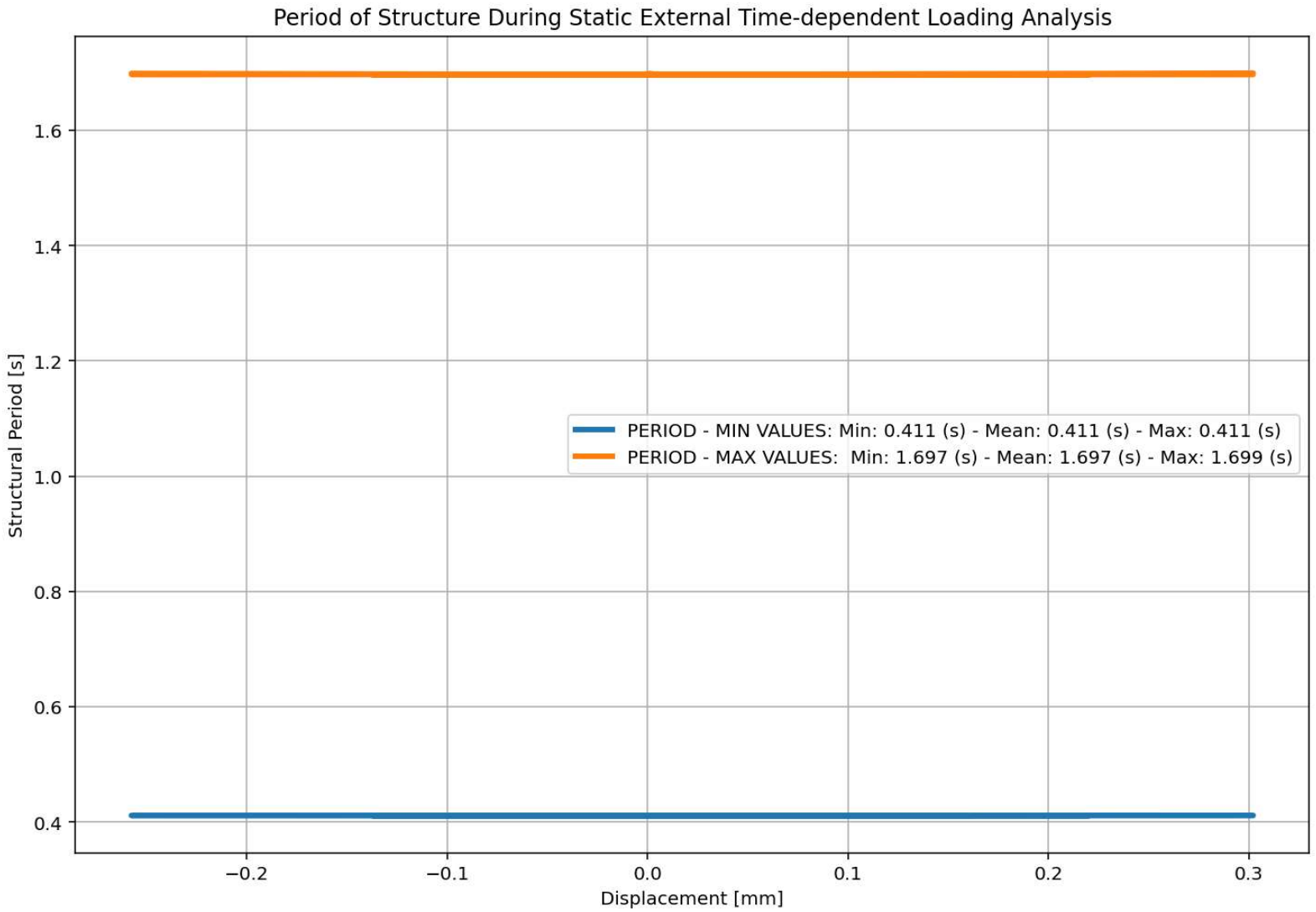


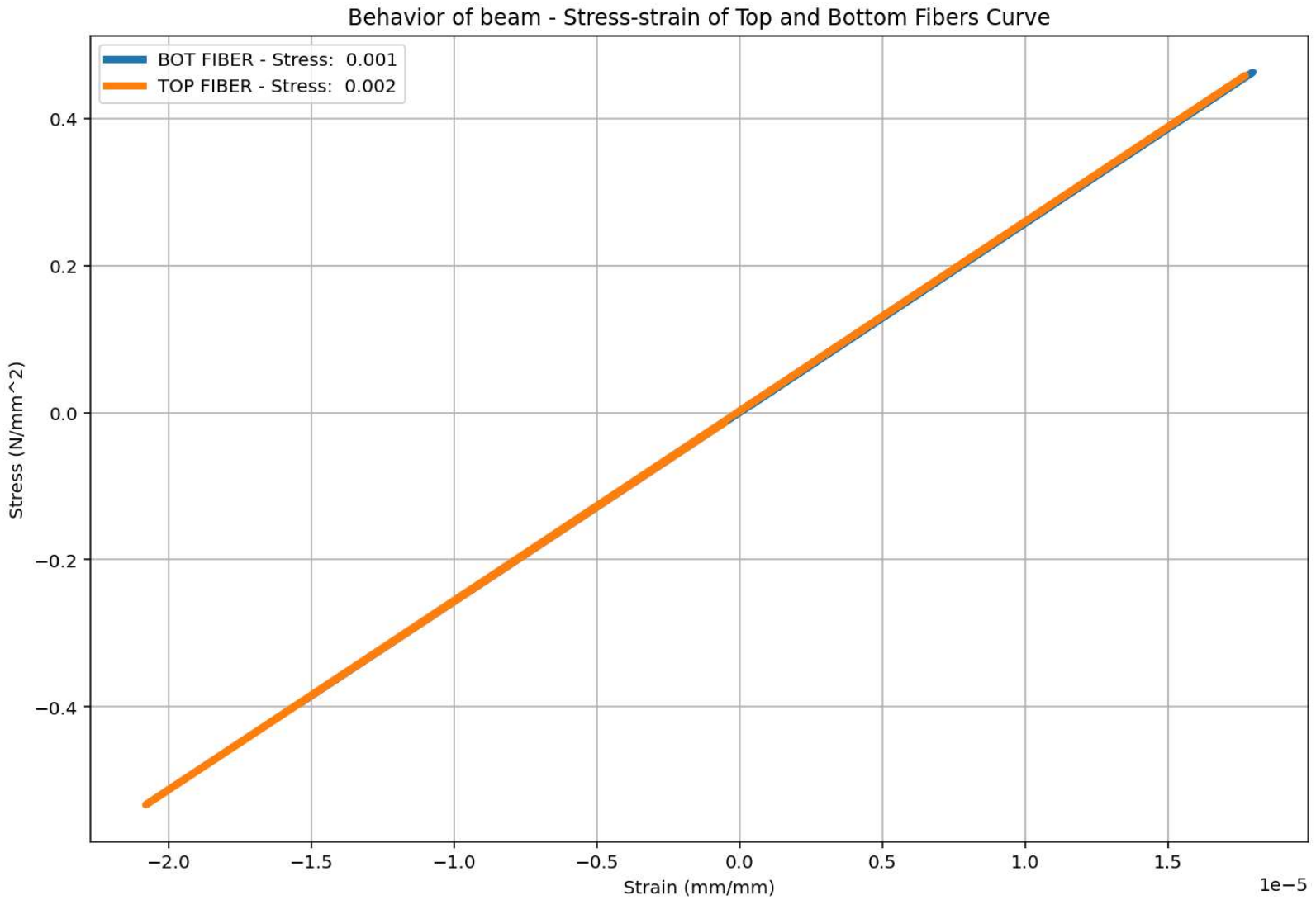
IPython Console | Files | Help | Variable Explorer | Debugger | History | Plots

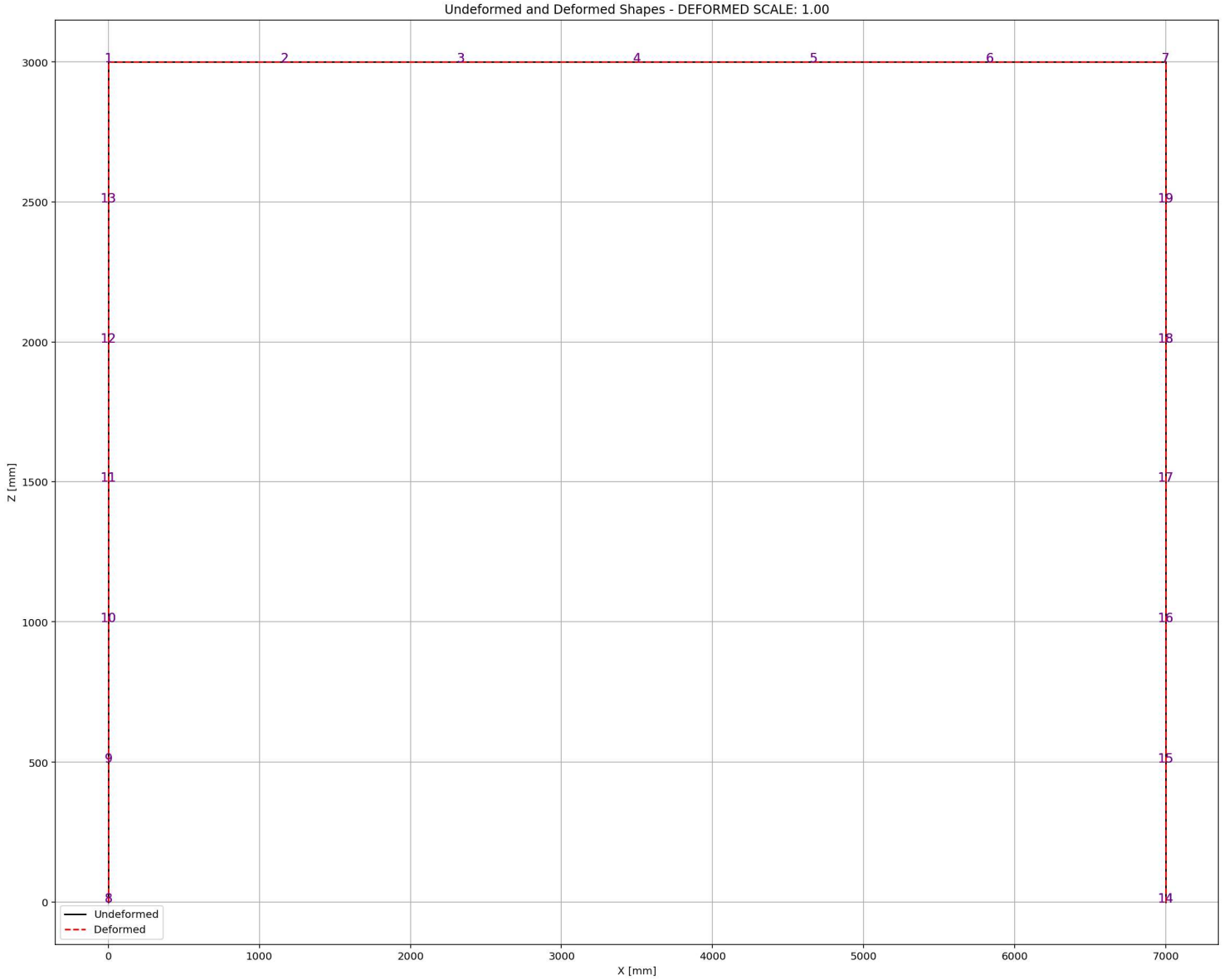
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 112, Col 81 UTF-8 CRLF RW Mem 46%


External Time-dependent Loading Over Time
External Loading - Max: 18142.051
Force (N)
15000
10000
5000
0
−5000
−10000
−15000
0
1
2
3
4
5
Time (s)


Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
15000
10000
5000
0
−5000
−10000
−15000
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [mm]


Period of Structure During Static External Time-dependent Loading Analysis
Structural Period [s]
1.6
1.4
1.2
1.0
0.8
0.6
0.4
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [mm]
PERIOD - MIN VALUES: Min: 0.411 (s) - Mean: 0.411 (s) - Max: 0.411 (s)
PERIOD - MAX VALUES: Min: 1.697 (s) - Mean: 1.697 (s) - Max: 1.699 (s)


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.001
TOP FIBER - Stress: 0.002
0.4
0.2
0.0
−0.2
−0.4
Stress (N/mm^2)
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Strain (mm/mm)
1e−5


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
Undeformed
Deformed

DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

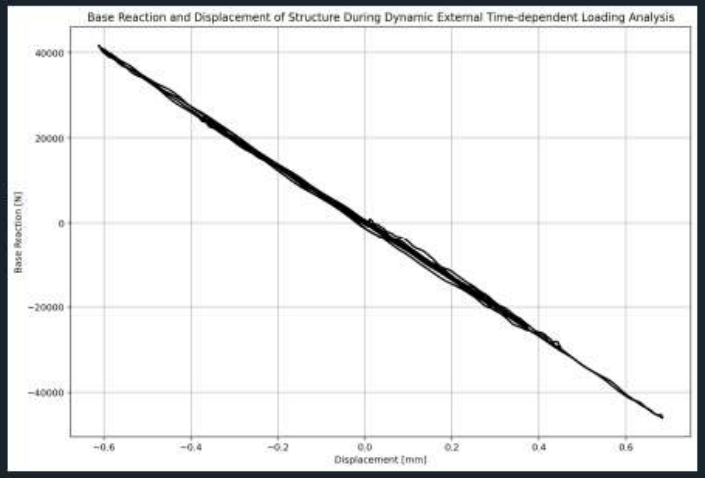
File Edit Search Source Run Debug Consoles Projects Tools View Help

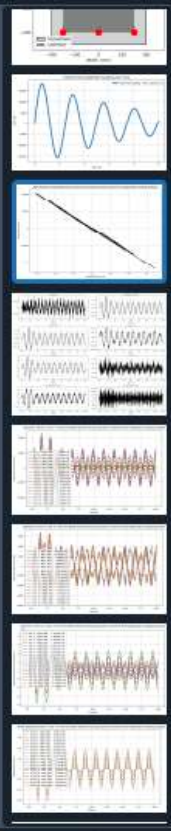
...ENERGY_DISSIPATION_CAPACITY_INDEX\12_CONCRETE_FRAME_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CONCRETE_FRAME_8_ANA\P(t)_CONCRETE_FRAME_8_ANA.py

P(t)_CONCRETE_FRAME_8_ANA.py | COMPOSITE_REC_I_SECTION_FUN.py

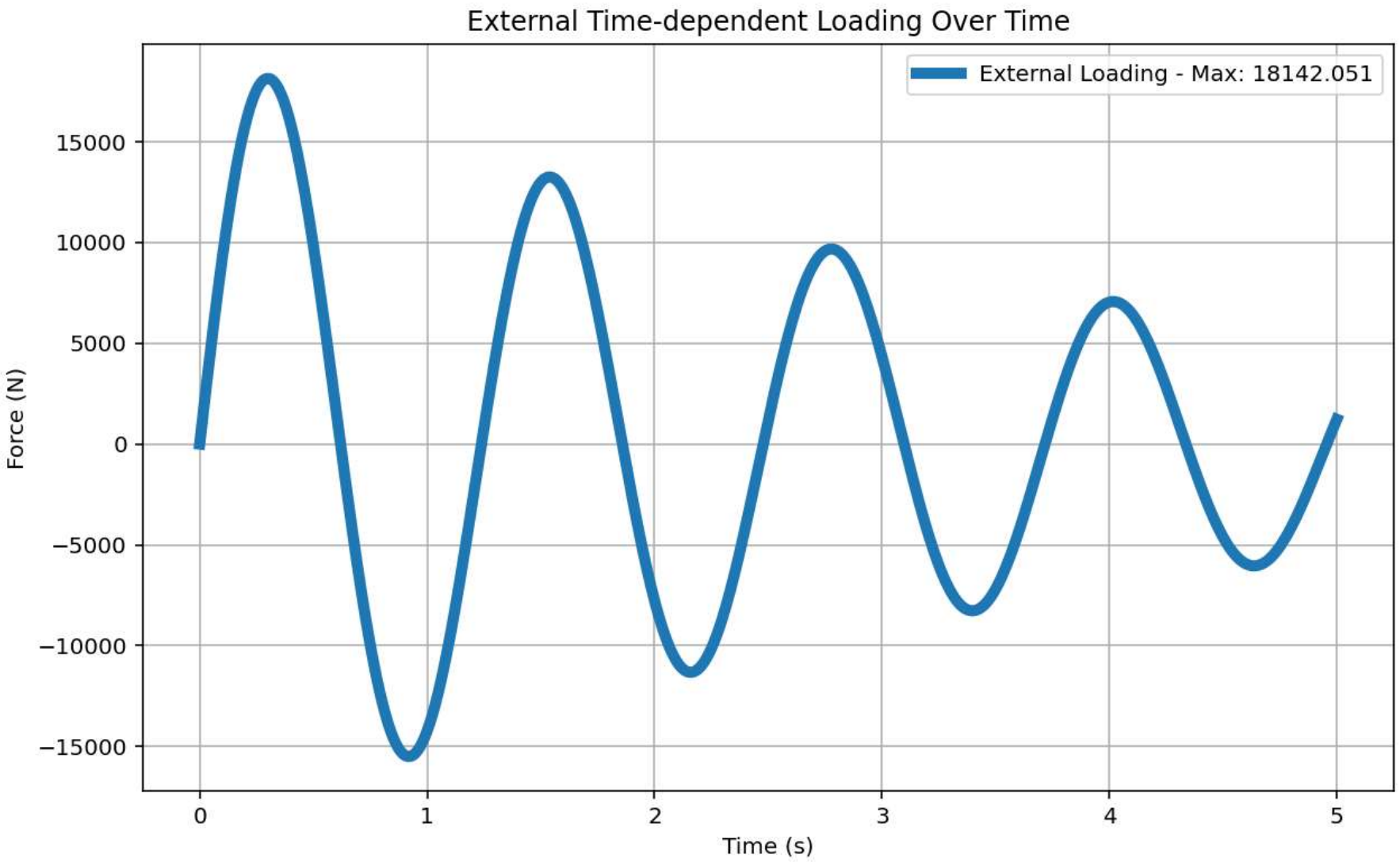
```
724     if ANAL_TYPE == 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING': # DYNAMIC TIME-HISTORY ANAL
725         #%% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
726         duration = 20.0             # [s] Analysis duration
727         dt = 0.005                  # [s] Time step
728         DT = dt                     # [s] Time step
729         DT_time = 5.0               # [s] Total external Load Analysis Durations [*******]
730         force_amplitude = 19600.0   # [N] Amplitude Force
731         omega_DT = 5.0715           # [rad/s] Natural angular frequency
732         DR = 0.03                   # Damping Ratio
733
734         # Check function
735         def CHECK_FUN(DT_time ,duration):
736             if DT_time > duration:
737                 print('\n\nAnalysis Duration Must be greater than External Load Duration\n
738                 exit()
739             return -1
740
741         CHECK_FUN(DT_time ,duration)
742
743         def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time): # P(t) = P0 s
744             import numpy as np
745             import matplotlib.pyplot as plt
746             # External Load Durations
747             num_steps = int(DT_time / DT)
748             load_time = np.linspace(0, DT_time, num_steps)
749             target_frequency = 1.0 * omega_DT  # Target excitation frequency
750             DT_load = force_amplitude * np.sin(target_frequency * load_time)
751             # Plot External Time-dependent Loading
752             plt.figure(figsize=(10, 6))
753             plt.plot(load_time, DT_load, label=f'External Loading - Max: {np.max(DT_load):
754             plt.title('External Time-dependent Loading Over Time')
755             plt.xlabel('Time (s)')
756             plt.ylabel('Force (N)')
757             plt.grid(True)
```

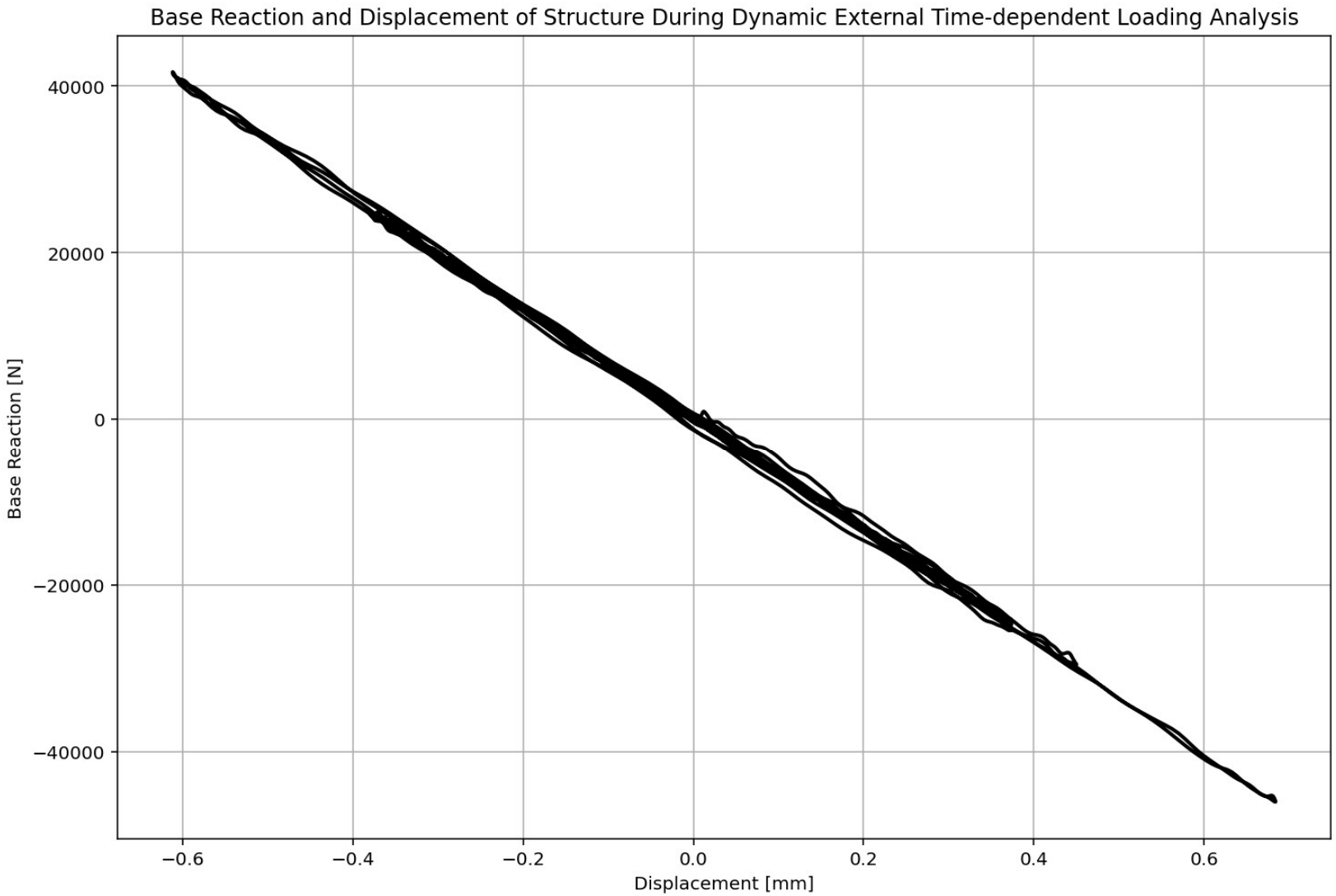


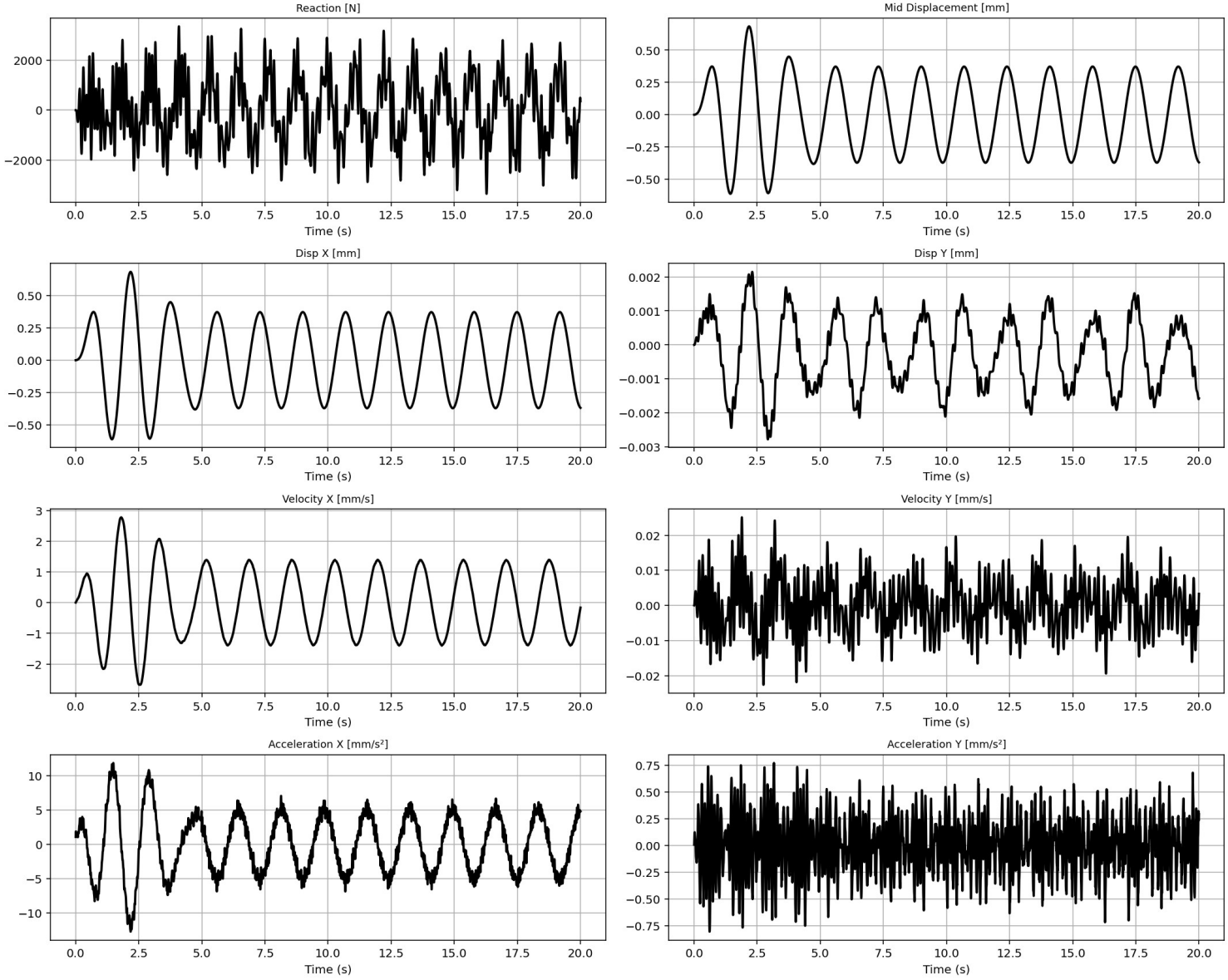


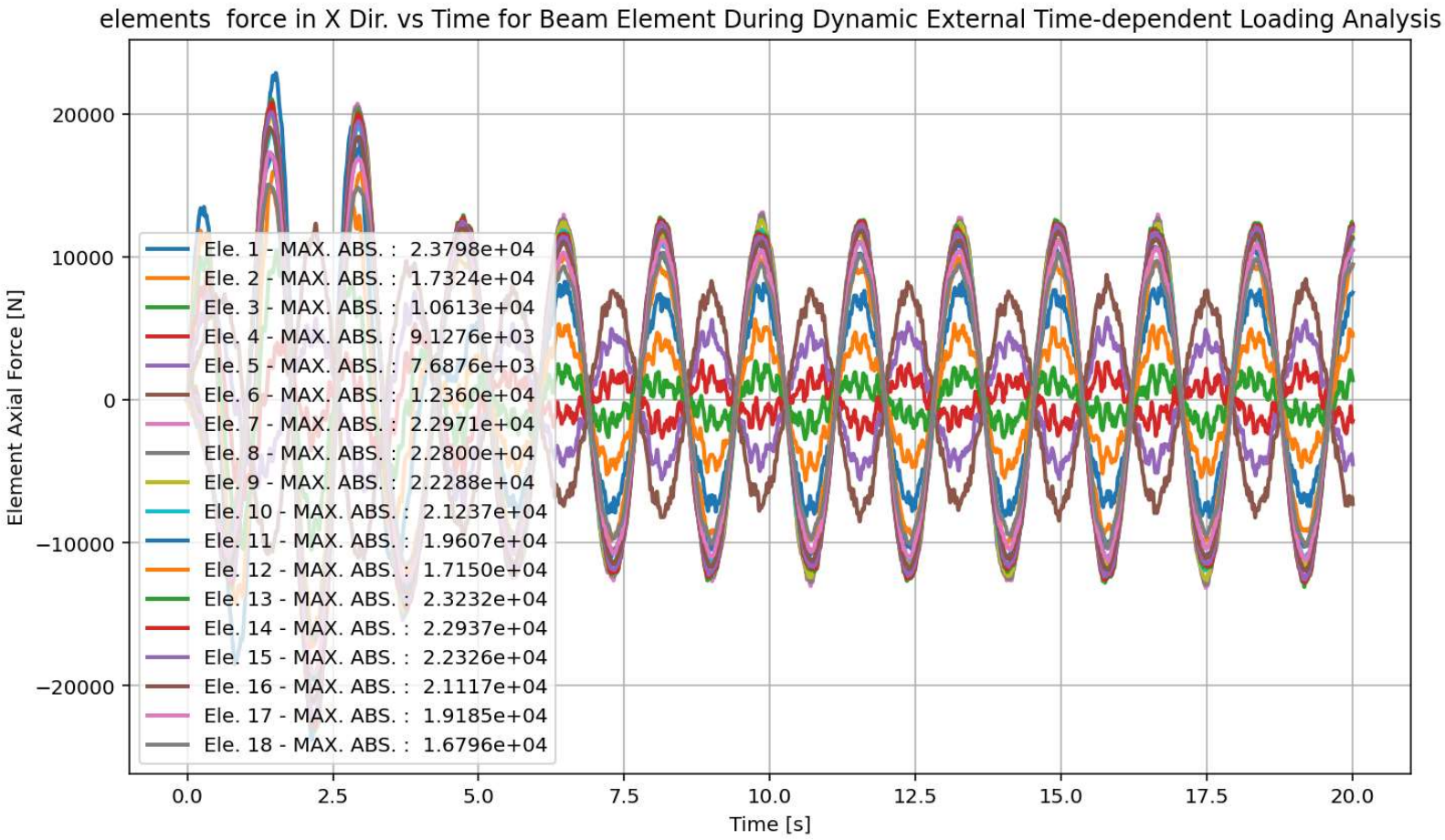
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

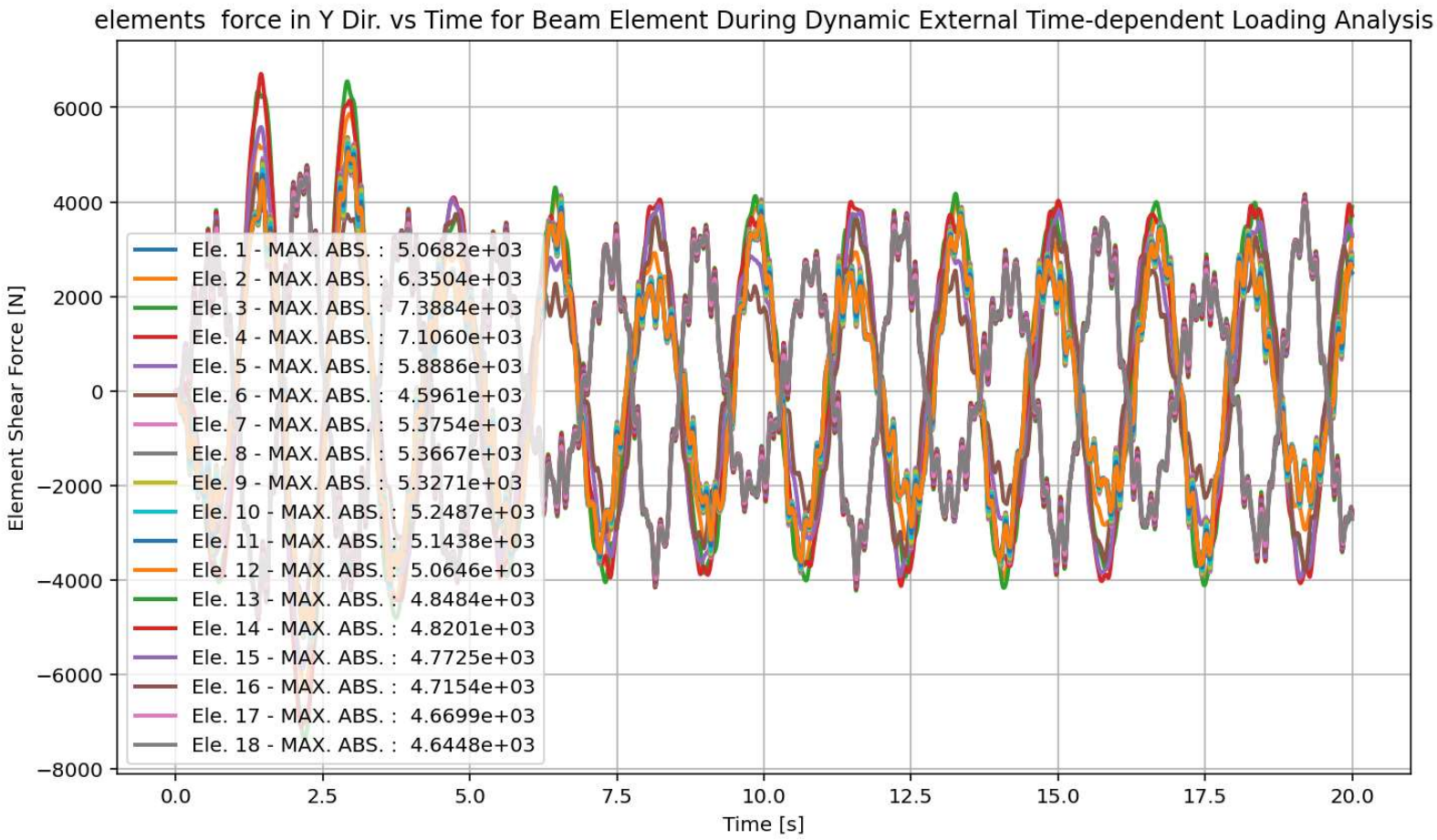
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 1341, Col 62 | UTF-8 | CRLF | RW | Mem 32%

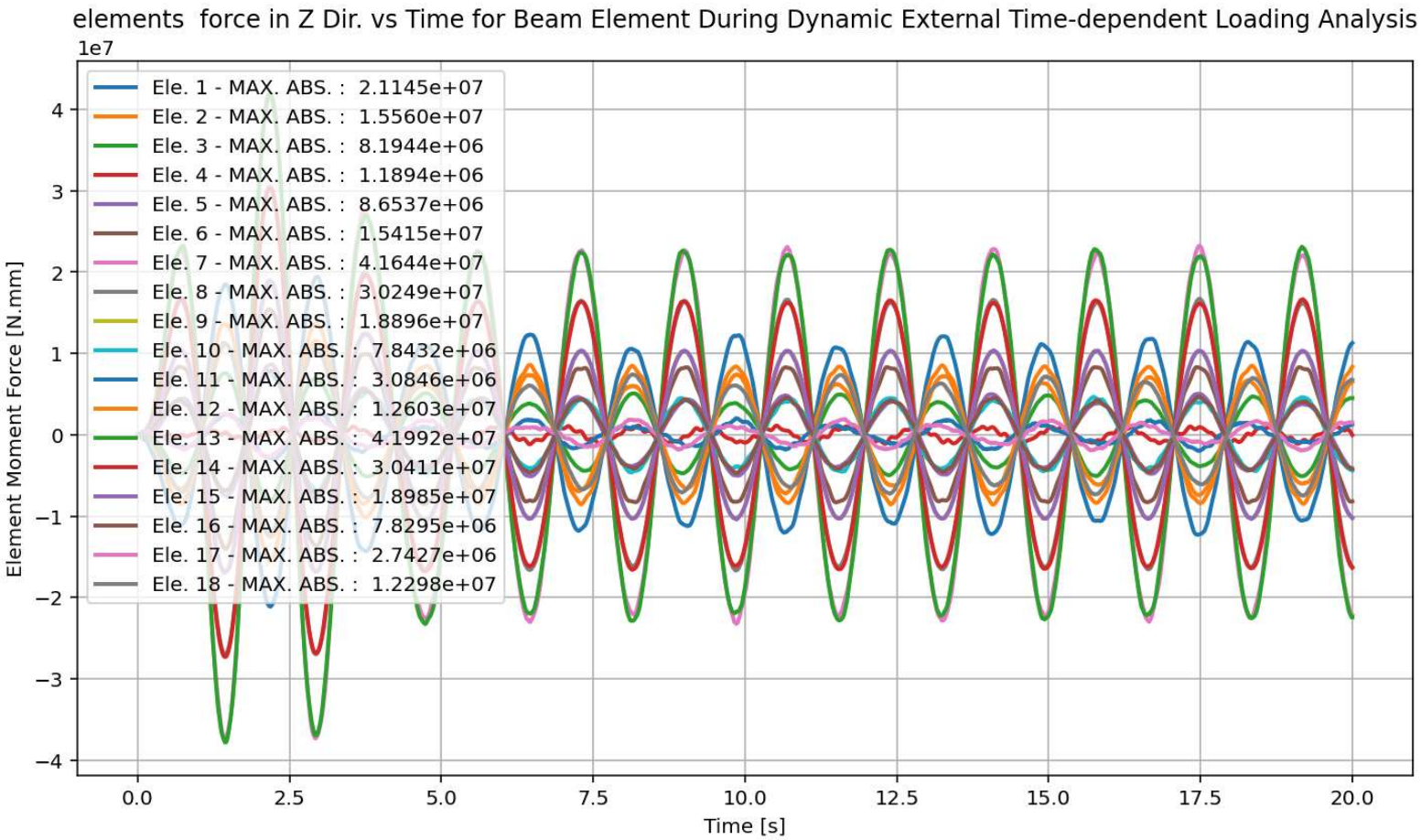

External Time-dependent Loading Over Time
External Loading - Max: 18142.051
Force (N)
15000
10000
5000
0
−5000
−10000
−15000
0
1
2
3
4
5
Time (s)

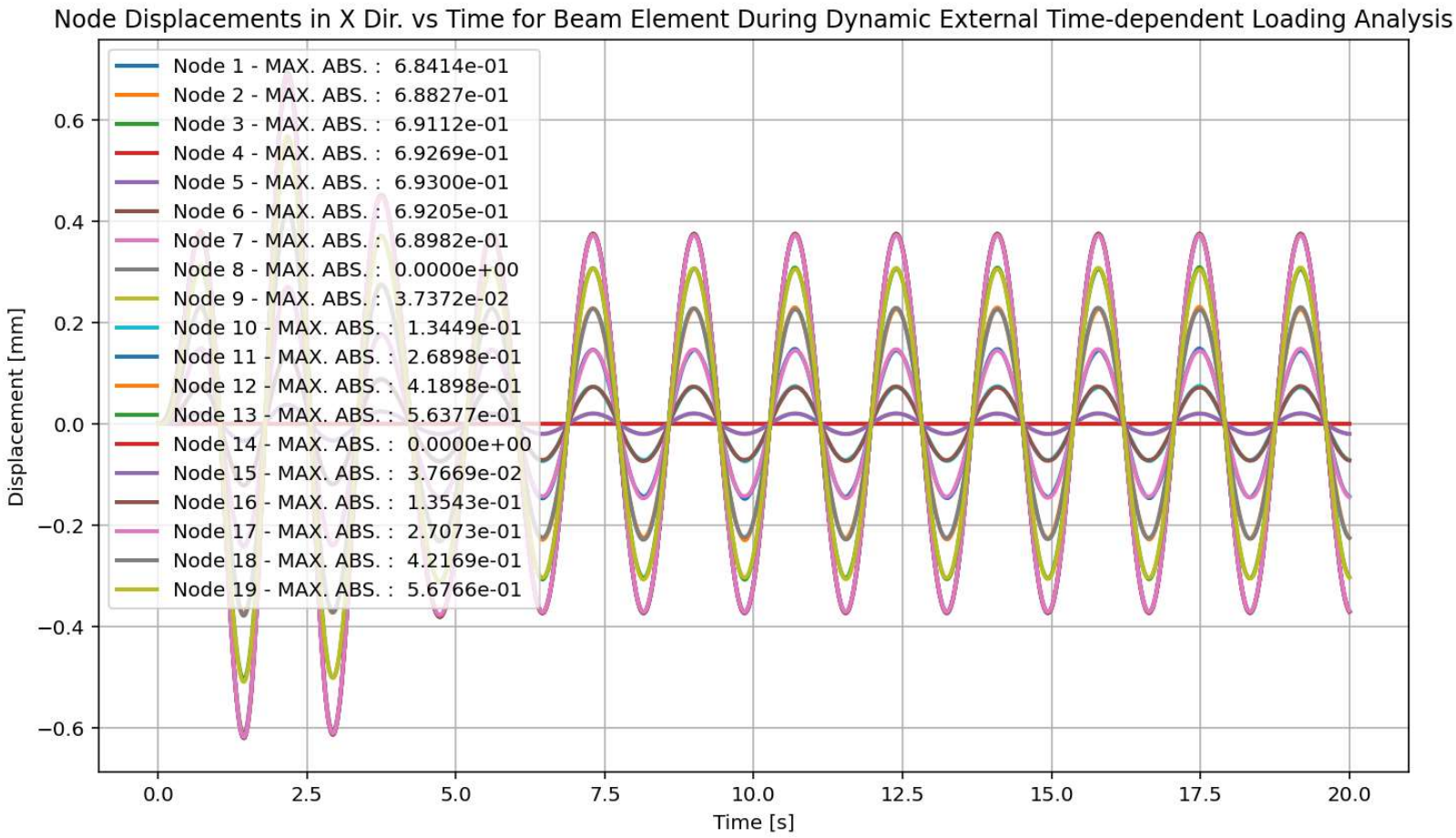

Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Reaction [N]
40000
20000
0
−20000
−40000
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
Displacement [mm]

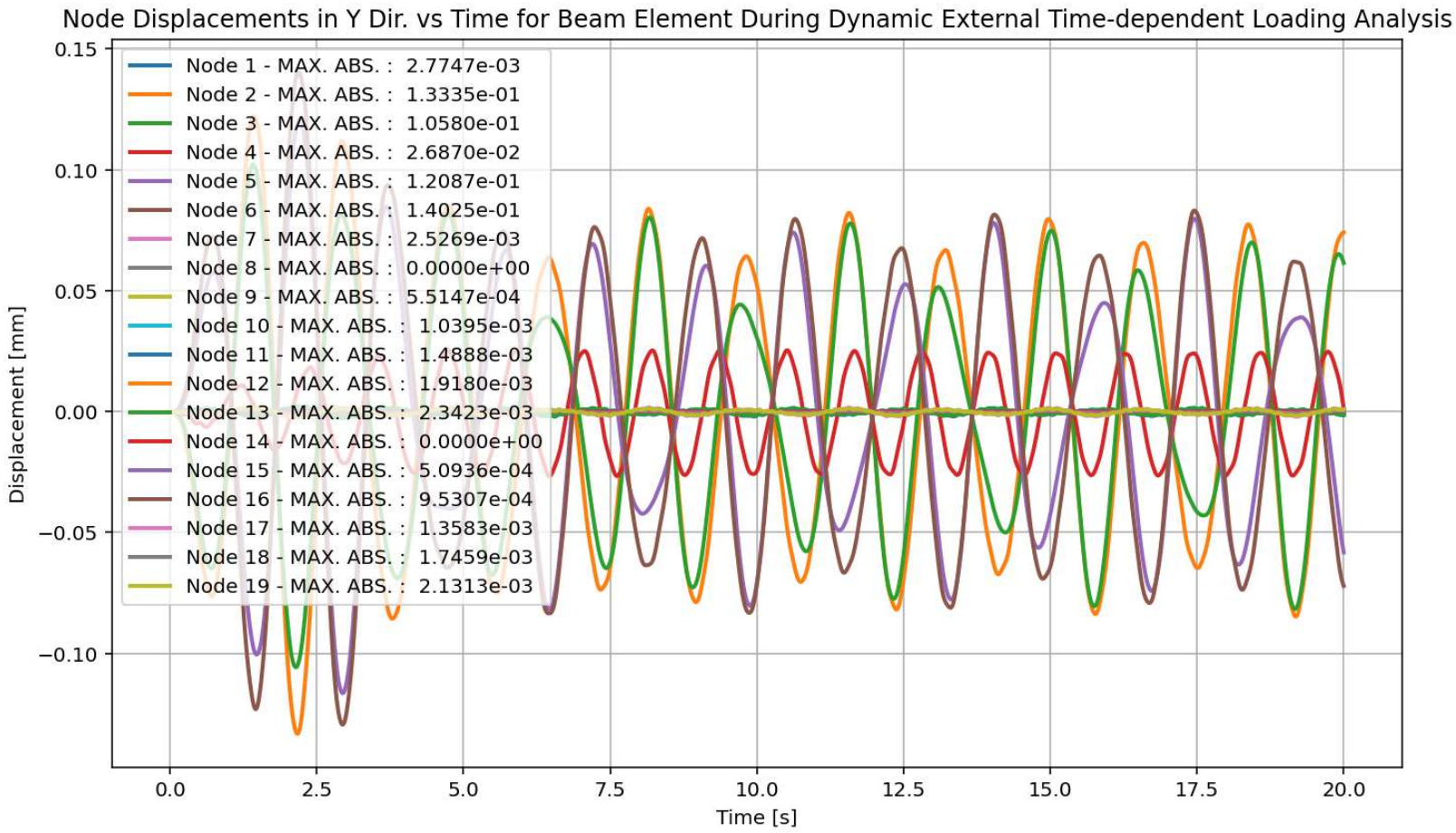

Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
Time (s)
Acceleration Y [mm/s²]
Time (s)

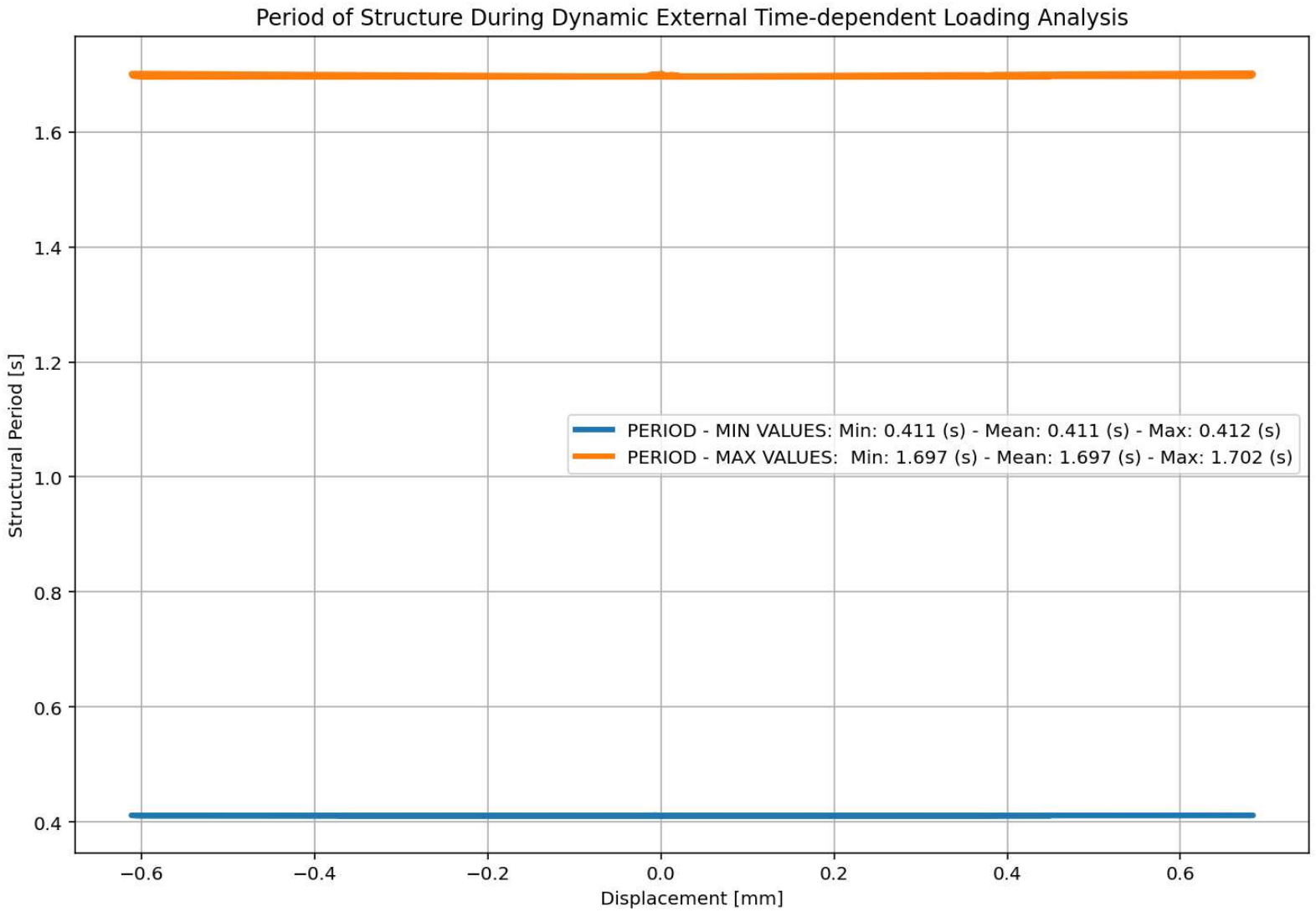

elements force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Element Axial Force [N]
Time [s]
20000
10000
0
−10000
−20000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 2.3798e+04
Ele. 2 - MAX. ABS. : 1.7324e+04
Ele. 3 - MAX. ABS. : 1.0613e+04
Ele. 4 - MAX. ABS. : 9.1276e+03
Ele. 5 - MAX. ABS. : 7.6876e+03
Ele. 6 - MAX. ABS. : 1.2360e+04
Ele. 7 - MAX. ABS. : 2.2971e+04
Ele. 8 - MAX. ABS. : 2.2800e+04
Ele. 9 - MAX. ABS. : 2.2288e+04
Ele. 10 - MAX. ABS. : 2.1237e+04
Ele. 11 - MAX. ABS. : 1.9607e+04
Ele. 12 - MAX. ABS. : 1.7150e+04
Ele. 13 - MAX. ABS. : 2.3232e+04
Ele. 14 - MAX. ABS. : 2.2937e+04
Ele. 15 - MAX. ABS. : 2.2326e+04
Ele. 16 - MAX. ABS. : 2.1117e+04
Ele. 17 - MAX. ABS. : 1.9185e+04
Ele. 18 - MAX. ABS. : 1.6796e+04

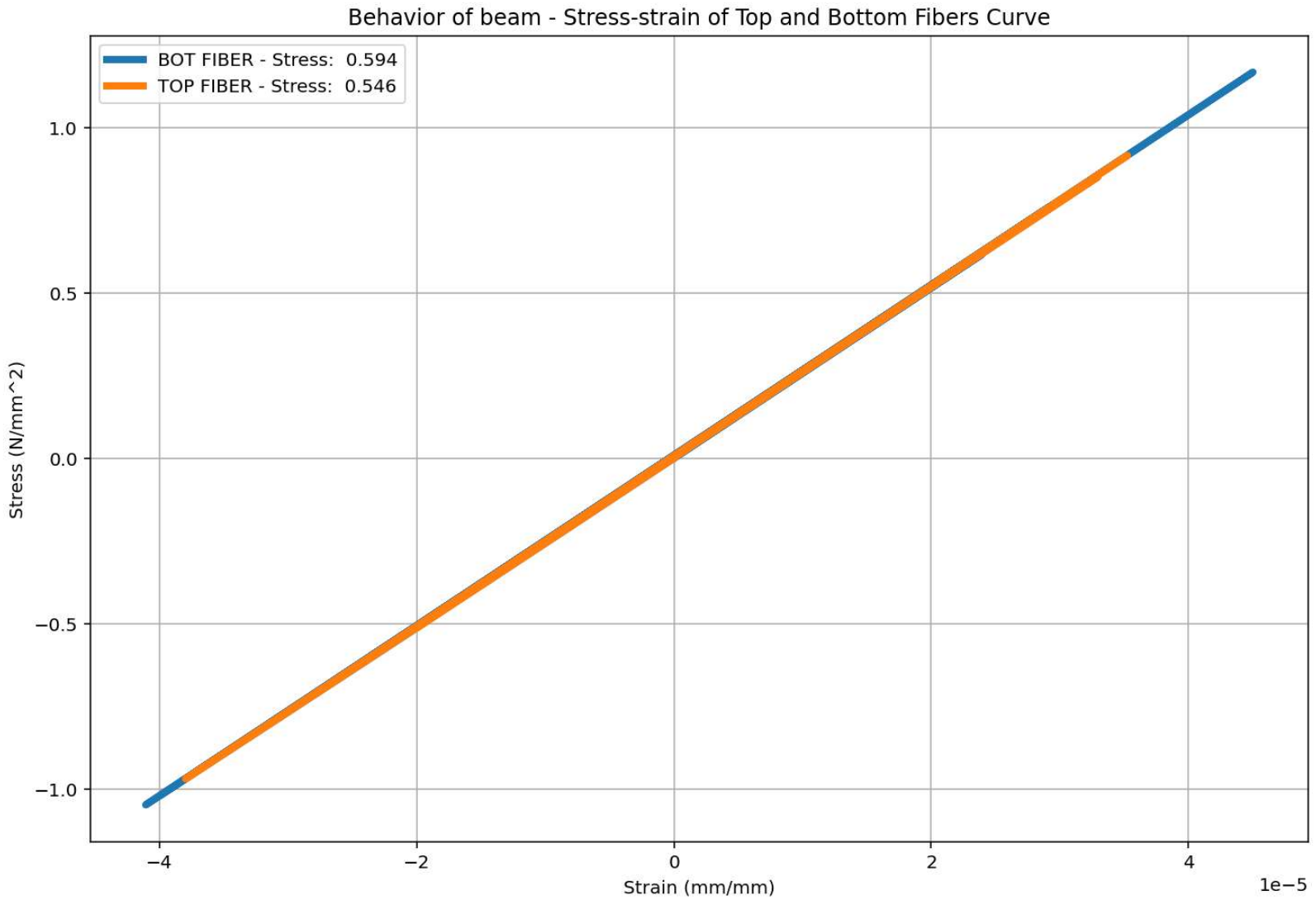

elements force in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Element Shear Force [N]
Time [s]
6000
4000
2000
0
−2000
−4000
−6000
−8000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 5.0682e+03
Ele. 2 - MAX. ABS. : 6.3504e+03
Ele. 3 - MAX. ABS. : 7.3884e+03
Ele. 4 - MAX. ABS. : 7.1060e+03
Ele. 5 - MAX. ABS. : 5.8886e+03
Ele. 6 - MAX. ABS. : 4.5961e+03
Ele. 7 - MAX. ABS. : 5.3754e+03
Ele. 8 - MAX. ABS. : 5.3667e+03
Ele. 9 - MAX. ABS. : 5.3271e+03
Ele. 10 - MAX. ABS. : 5.2487e+03
Ele. 11 - MAX. ABS. : 5.1438e+03
Ele. 12 - MAX. ABS. : 5.0646e+03
Ele. 13 - MAX. ABS. : 4.8484e+03
Ele. 14 - MAX. ABS. : 4.8201e+03
Ele. 15 - MAX. ABS. : 4.7725e+03
Ele. 16 - MAX. ABS. : 4.7154e+03
Ele. 17 - MAX. ABS. : 4.6699e+03
Ele. 18 - MAX. ABS. : 4.6448e+03

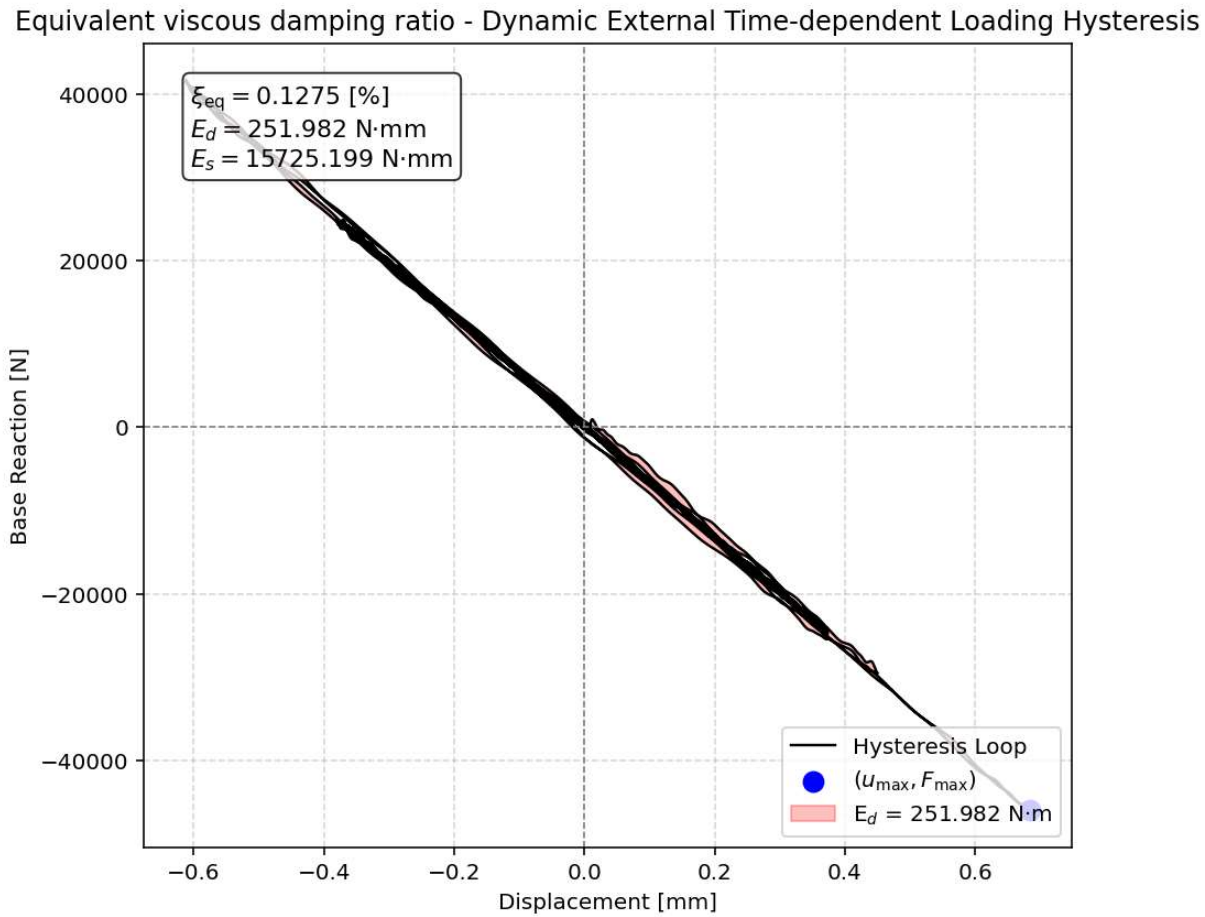

elements force in Z Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
1e7
Ele. 1 - MAX. ABS. : 2.1145e+07
Ele. 2 - MAX. ABS. : 1.5560e+07
Ele. 3 - MAX. ABS. : 8.1944e+06
Ele. 4 - MAX. ABS. : 1.1894e+06
Ele. 5 - MAX. ABS. : 8.6537e+06
Ele. 6 - MAX. ABS. : 1.5415e+07
Ele. 7 - MAX. ABS. : 4.1644e+07
Ele. 8 - MAX. ABS. : 3.0249e+07
Ele. 9 - MAX. ABS. : 1.8896e+07
Ele. 10 - MAX. ABS. : 7.8432e+06
Ele. 11 - MAX. ABS. : 3.0846e+06
Ele. 12 - MAX. ABS. : 1.2603e+07
Ele. 13 - MAX. ABS. : 4.1992e+07
Ele. 14 - MAX. ABS. : 3.0411e+07
Ele. 15 - MAX. ABS. : 1.8985e+07
Ele. 16 - MAX. ABS. : 7.8295e+06
Ele. 17 - MAX. ABS. : 2.7427e+06
Ele. 18 - MAX. ABS. : 1.2298e+07
Element Moment Force [N.mm]
Time [s]
4
3
2
1
0
−1
−2
−3
−4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

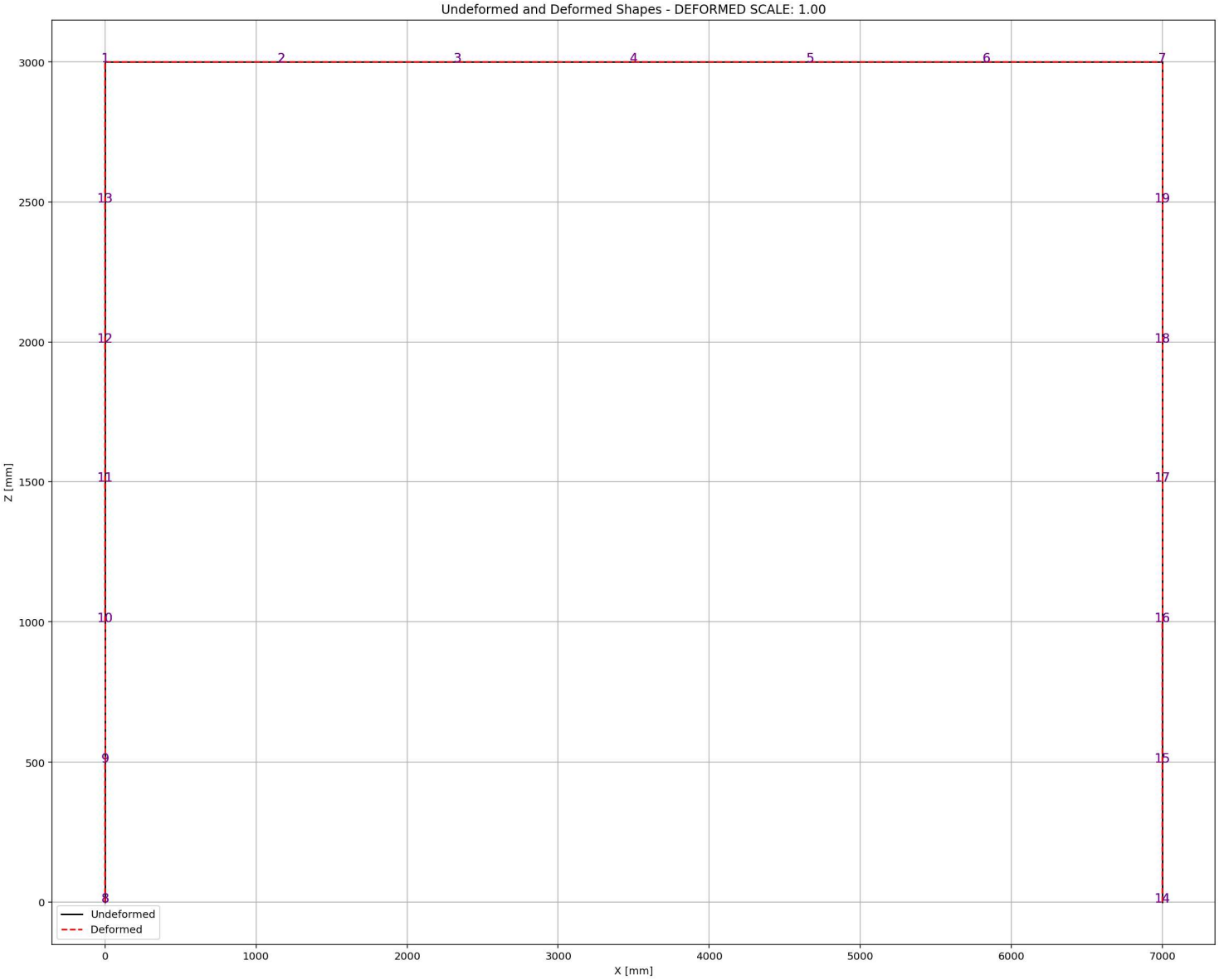

Node Displacements in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 6.8414e-01
Node 2 - MAX. ABS. : 6.8827e-01
Node 3 - MAX. ABS. : 6.9112e-01
Node 4 - MAX. ABS. : 6.9269e-01
Node 5 - MAX. ABS. : 6.9300e-01
Node 6 - MAX. ABS. : 6.9205e-01
Node 7 - MAX. ABS. : 6.8982e-01
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 3.7372e-02
Node 10 - MAX. ABS. : 1.3449e-01
Node 11 - MAX. ABS. : 2.6898e-01
Node 12 - MAX. ABS. : 4.1898e-01
Node 13 - MAX. ABS. : 5.6377e-01
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 3.7669e-02
Node 16 - MAX. ABS. : 1.3543e-01
Node 17 - MAX. ABS. : 2.7073e-01
Node 18 - MAX. ABS. : 4.2169e-01
Node 19 - MAX. ABS. : 5.6766e-01
Displacement [mm]
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 2.7747e-03
Node 2 - MAX. ABS. : 1.3335e-01
Node 3 - MAX. ABS. : 1.0580e-01
Node 4 - MAX. ABS. : 2.6870e-02
Node 5 - MAX. ABS. : 1.2087e-01
Node 6 - MAX. ABS. : 1.4025e-01
Node 7 - MAX. ABS. : 2.5269e-03
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 5.5147e-04
Node 10 - MAX. ABS. : 1.0395e-03
Node 11 - MAX. ABS. : 1.4888e-03
Node 12 - MAX. ABS. : 1.9180e-03
Node 13 - MAX. ABS. : 2.3423e-03
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 5.0936e-04
Node 16 - MAX. ABS. : 9.5307e-04
Node 17 - MAX. ABS. : 1.3583e-03
Node 18 - MAX. ABS. : 1.7459e-03
Node 19 - MAX. ABS. : 2.1313e-03
Displacement [mm]
0.15
0.10
0.05
0.00
−0.05
−0.10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Period of Structure During Dynamic External Time-dependent Loading Analysis
Structural Period [s]
Displacement [mm]
PERIOD - MIN VALUES: Min: 0.411 (s) - Mean: 0.411 (s) - Max: 0.412 (s)
PERIOD - MAX VALUES: Min: 1.697 (s) - Mean: 1.697 (s) - Max: 1.702 (s)
1.6
1.4
1.2
1.0
0.8
0.6
0.4
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.594
TOP FIBER - Stress: 0.546
Stress (N/mm^2)
1.0
0.5
0.0
−0.5
−1.0
−4
−2
0
2
4
Strain (mm/mm)
1e−5


Equivalent viscous damping ratio - Dynamic External Time-dependent Loading Hysteresis
$\xi_{eq} = 0.1275$ [%]
$E_d = 251.982$ N·mm
$E_s = 15725.199$ N·mm
Base Reaction [N]
Displacement [mm]
40000
20000
0
−20000
−40000
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
Hysteresis Loop
(u_{max}, F_{max})
E_d = 251.982 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000

FREE-VIBRATION ANALYSIS

Spyder (Python 3.12)

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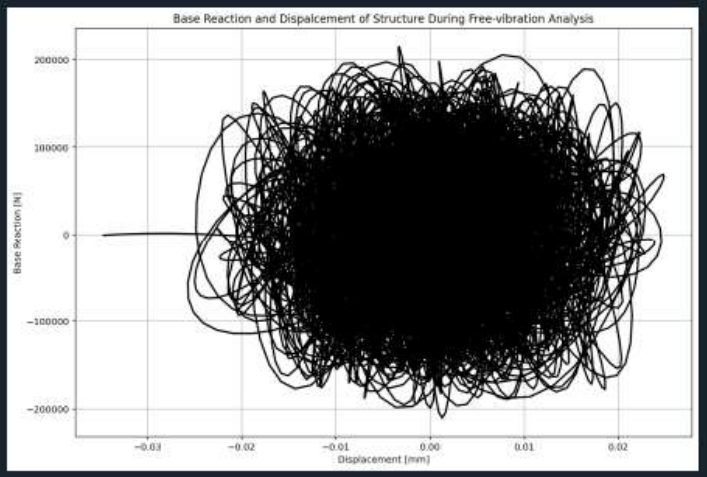
...ENERGY_DISSIPATION_CAPACITY_INDEX\12_CONCRETE_FRAME_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CONCRETE_FRAME_8_ANA\P(t)_CONCRETE_FRAME_8_ANA.py

P(t)_CONCRETE_FRAME_8_ANA.py | COMPOSITE_REC_I_SECTION_FUN.py

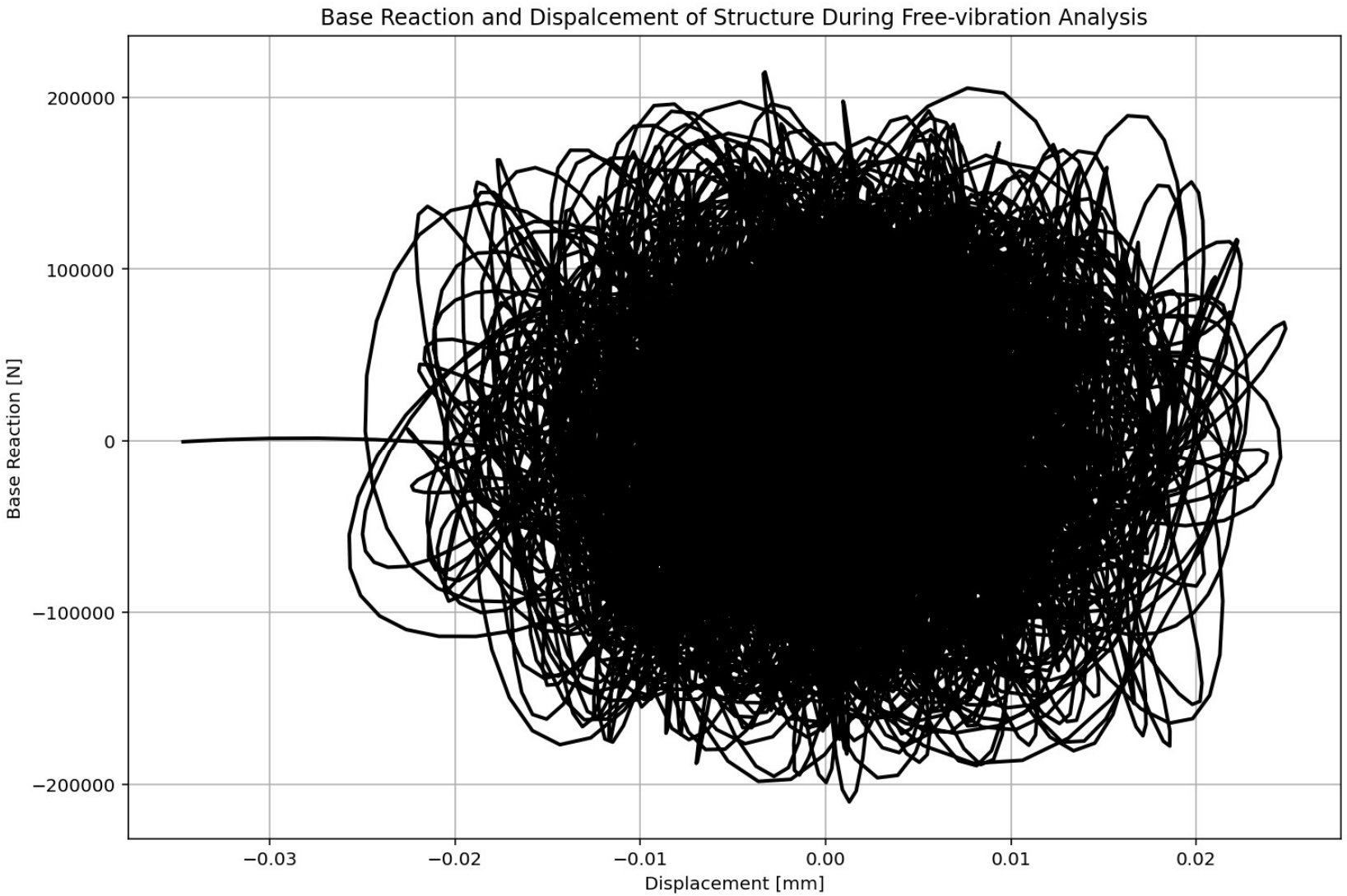
```
879         return  DATA
880
881     if ANAL_TYPE == 'FREE-VIBRATION': # DYNAMIC TIME-HISTORY ANALYSIS
882         #%% DEFINE PARAMETERS FOR FREE-VIBRATION ANALYSIS
883         u0 = -0.035                         # [mm] Initial displacement
884         v0 = 0.15                           # [mm/s] Initial velocity
885         a0 = 0.065                          # [mm/s^2] Initial acceleration
886         IU = True                           # Free Vibration with Initial Displacement
887         IV = True                           # Free Vibration with Initial Velocity
888         IA = True                           # Free Vibration with Initial Acceleration
889         duration = 20.0                     # [s] Analysis duration
890         dt = 0.001                          # [s] Time step
891         DR = 0.03                           # Damping Ratio
892
893         if IU == True:
894             # Define initial displacment
895             ops.setNodeDisp(center_node, 1, u0, '-commit') # INFO LINK: https://openseespy
896         if IV == True:
897             # Define initial velocity
898             ops.setNodeVel(center_node, 1, v0, '-commit')  # INFO LINK: https://openseespy
899         if IA == True:
900             # Define initial  acceleration
901             ops.setNodeAccel(center_node, 1, a0, '-commit') # INFO LINK: https://openseesp
902
903         ops.constraints('Plain')
904         ops.numberer('Plain')
905         ops.system('BandGeneral')
906         ops.test('NormDispIncr', MAX_TOLERANCE, MAX_ITERATIONS) # INFO LINK: https://opens
907         #ops.integrator('CentralDifference')  # JUST FOR LINEAR ANALYSIS - INFO LINK: http
908         alpha=0.5; beta=0.25;
909         ops.integrator('Newmark', alpha, beta) # INFO LINK: https://openseespydoc.readthed
910         #alpha=2/3;gamma=1.5-alpha; gamma=1.5-alpha;beta=(2-alpha)**2/4;
911         #ops.integrator('HHT', alpha, gamma, beta) # INFO LINK: https://openseespydoc.read
912         ops.algorithm('Newton')  # INFO LINK: https://openseespydoc.readthedocs.io/en/late
```

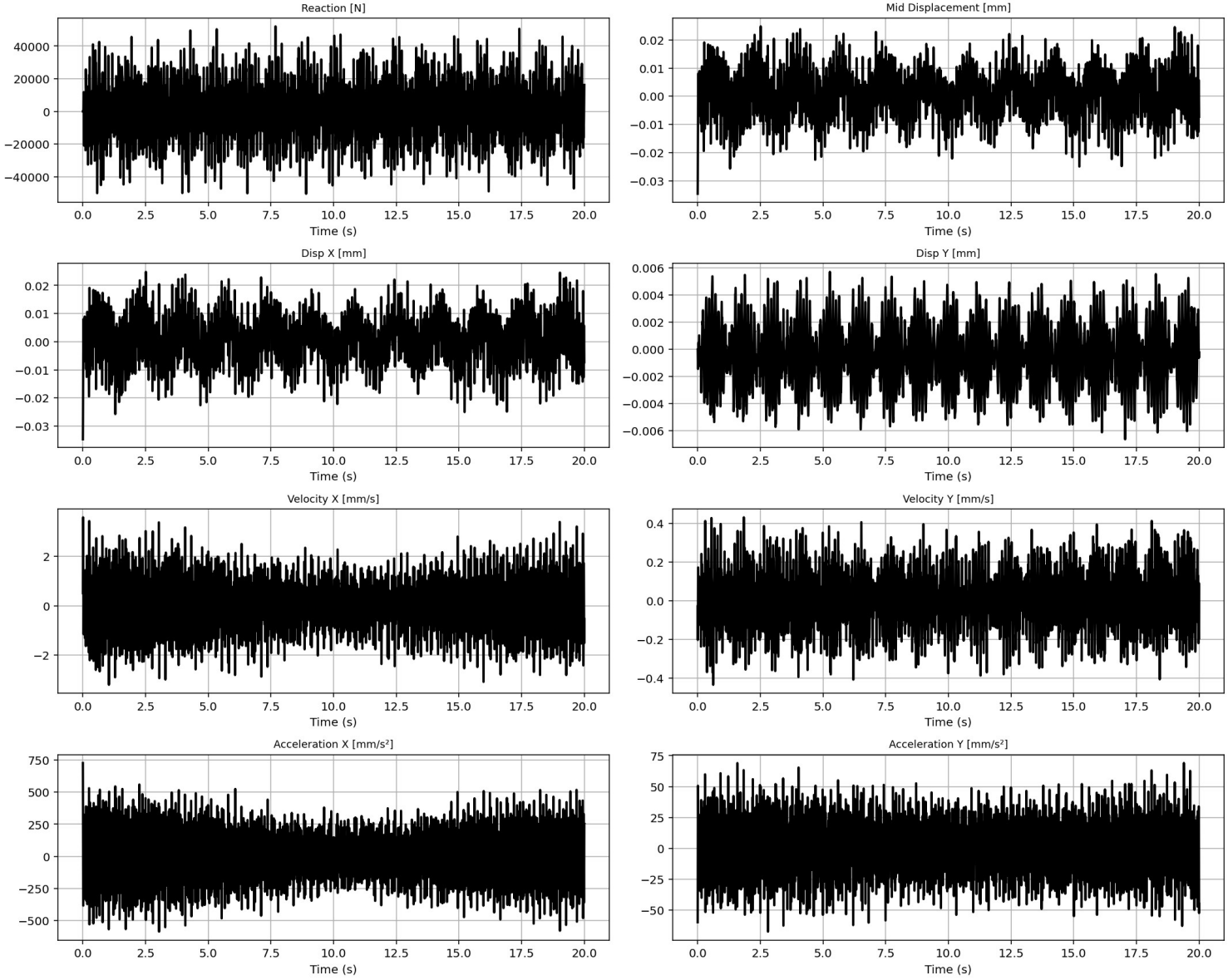
30 %

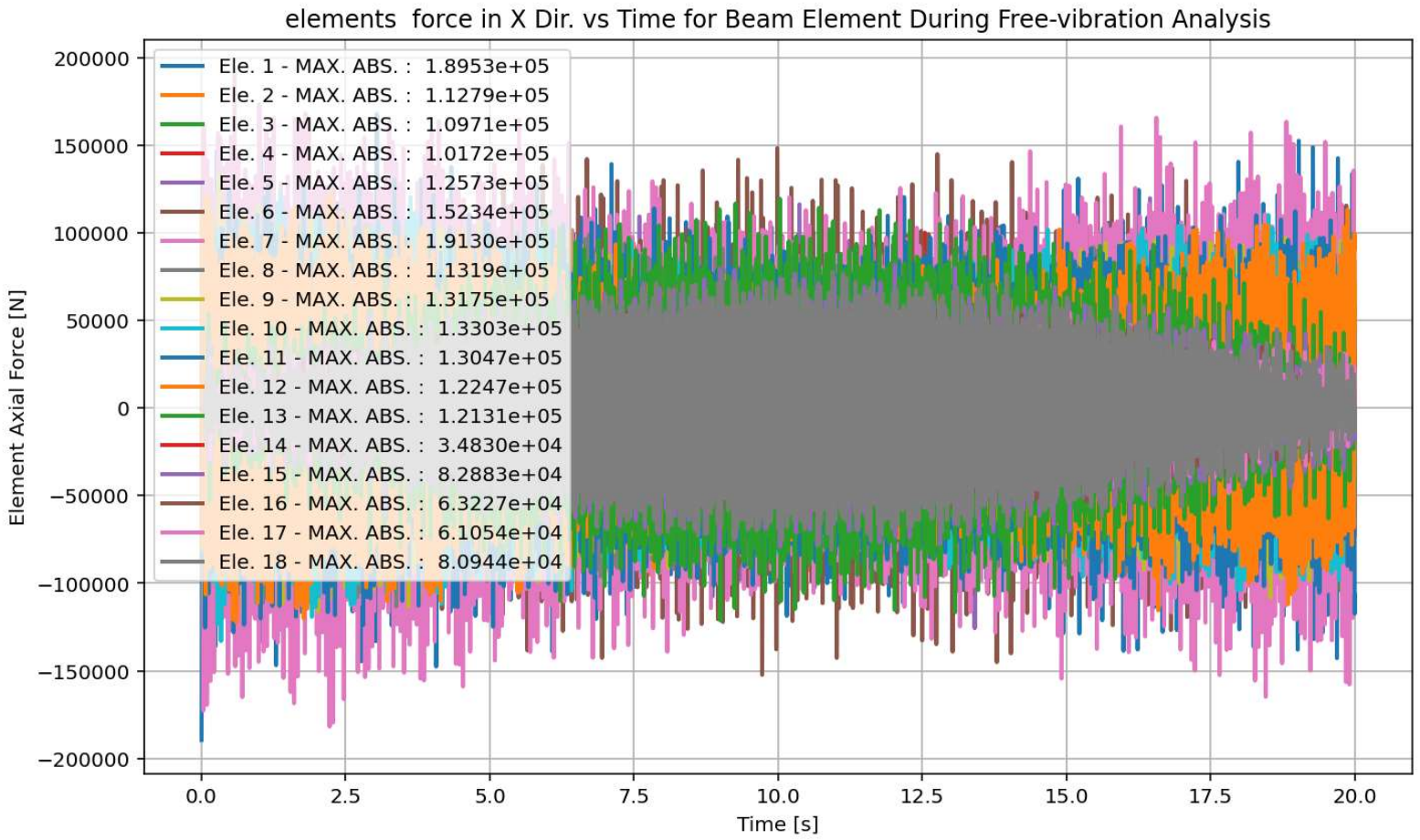


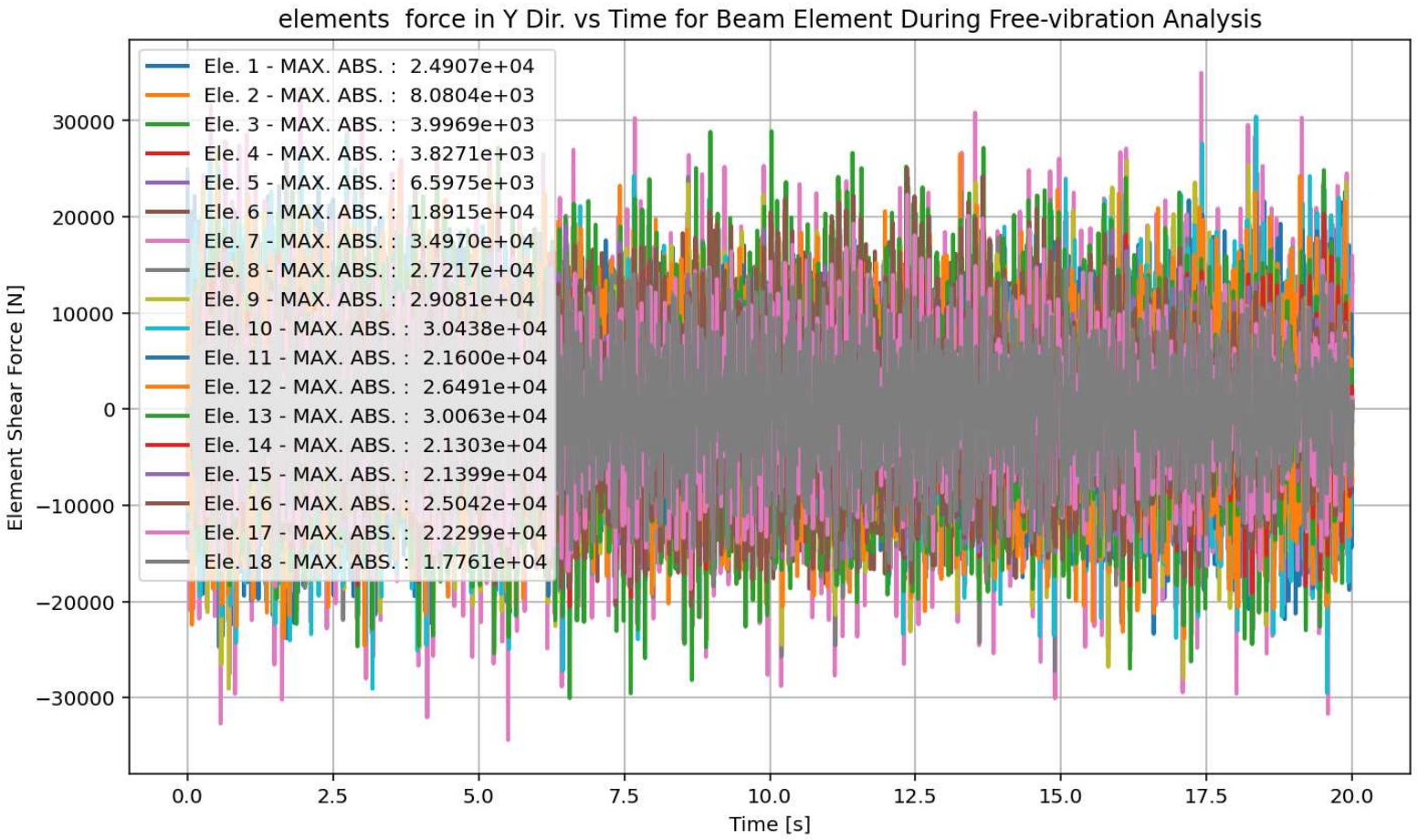
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

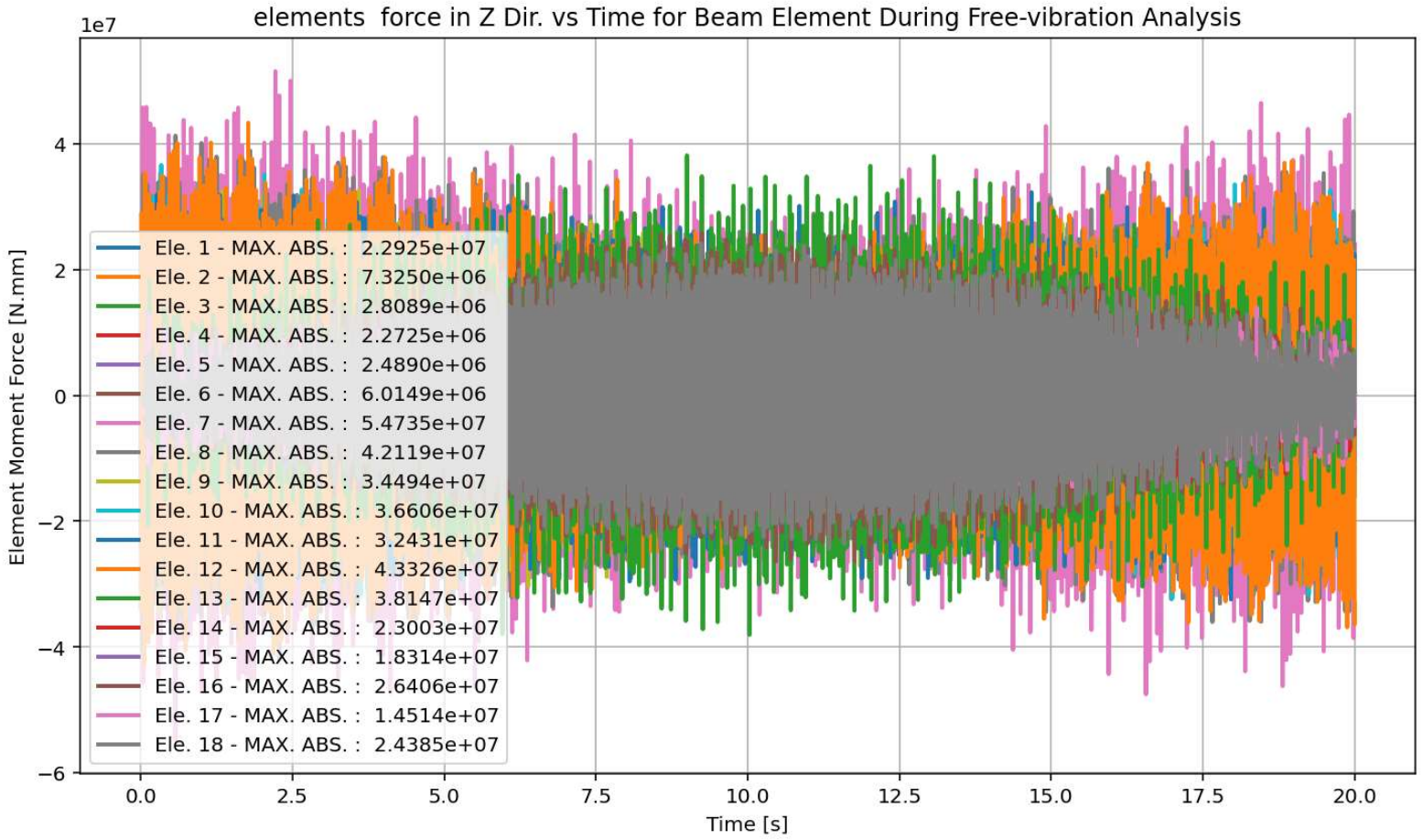
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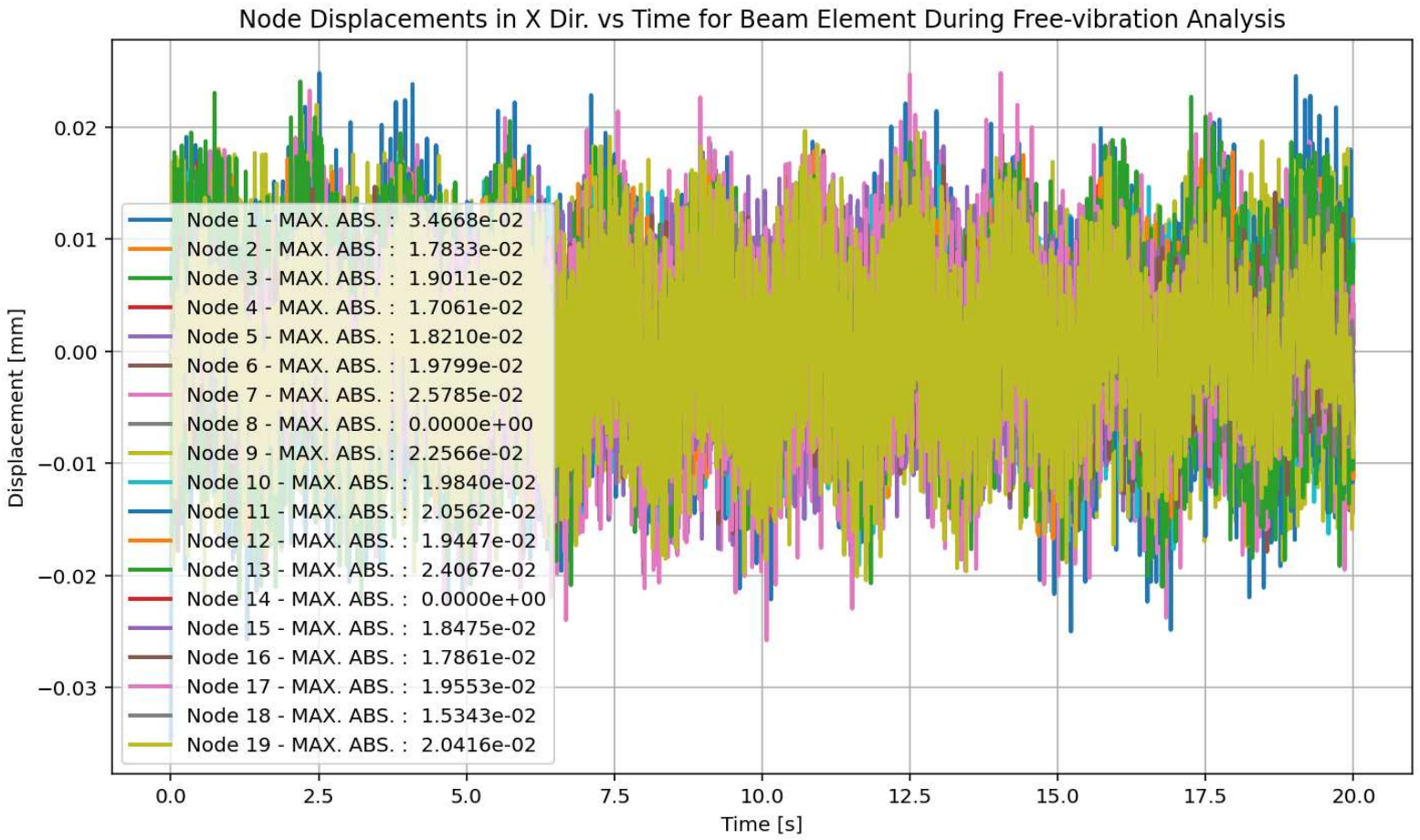

Base Reaction and Dispalcement of Structure During Free-vibration Analysis
200000
100000
0
−100000
−200000
Base Reaction [N]
−0.03
−0.02
−0.01
0.00
0.01
0.02
Displacement [mm]

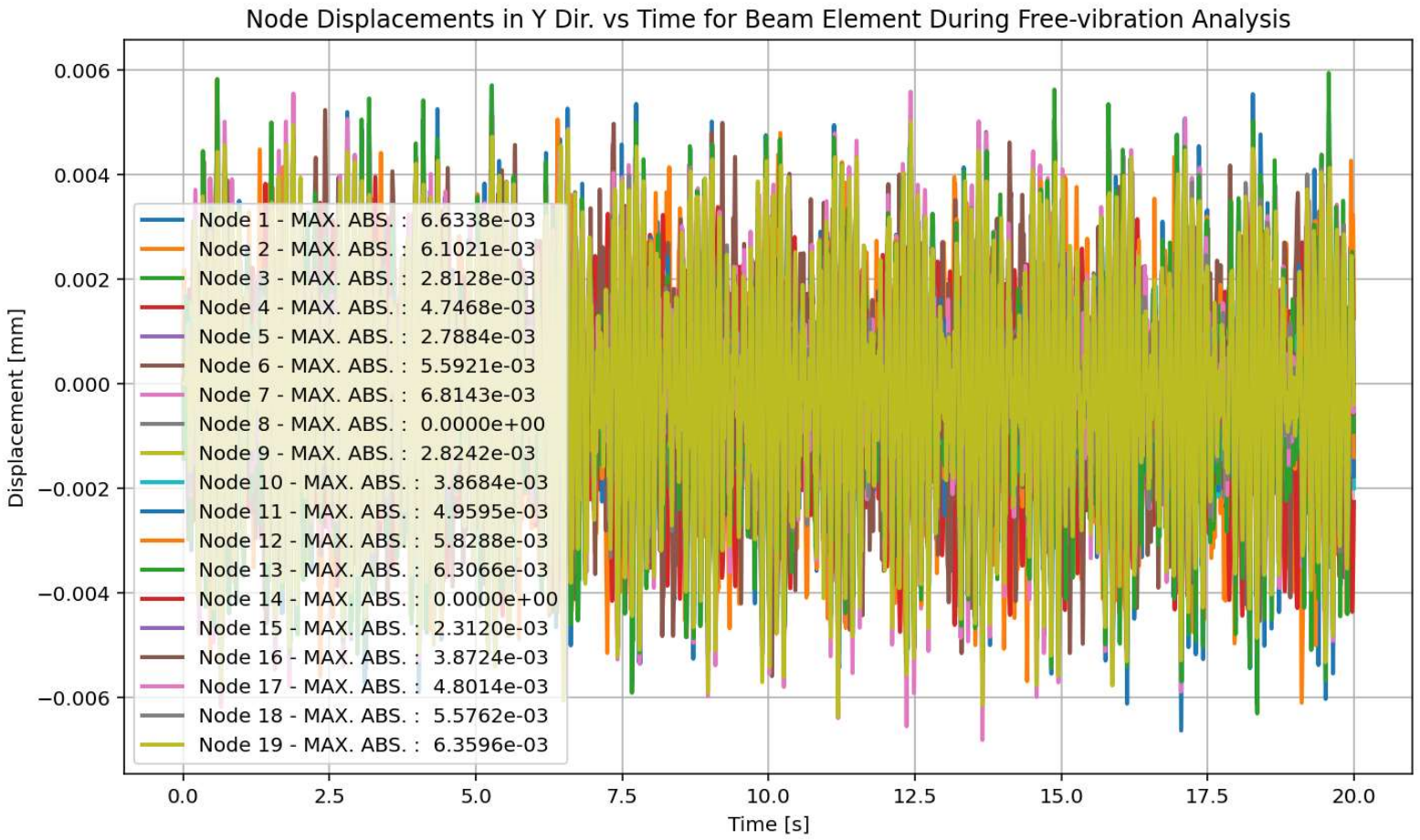

Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
Time (s)
Acceleration Y [mm/s²]
Time (s)

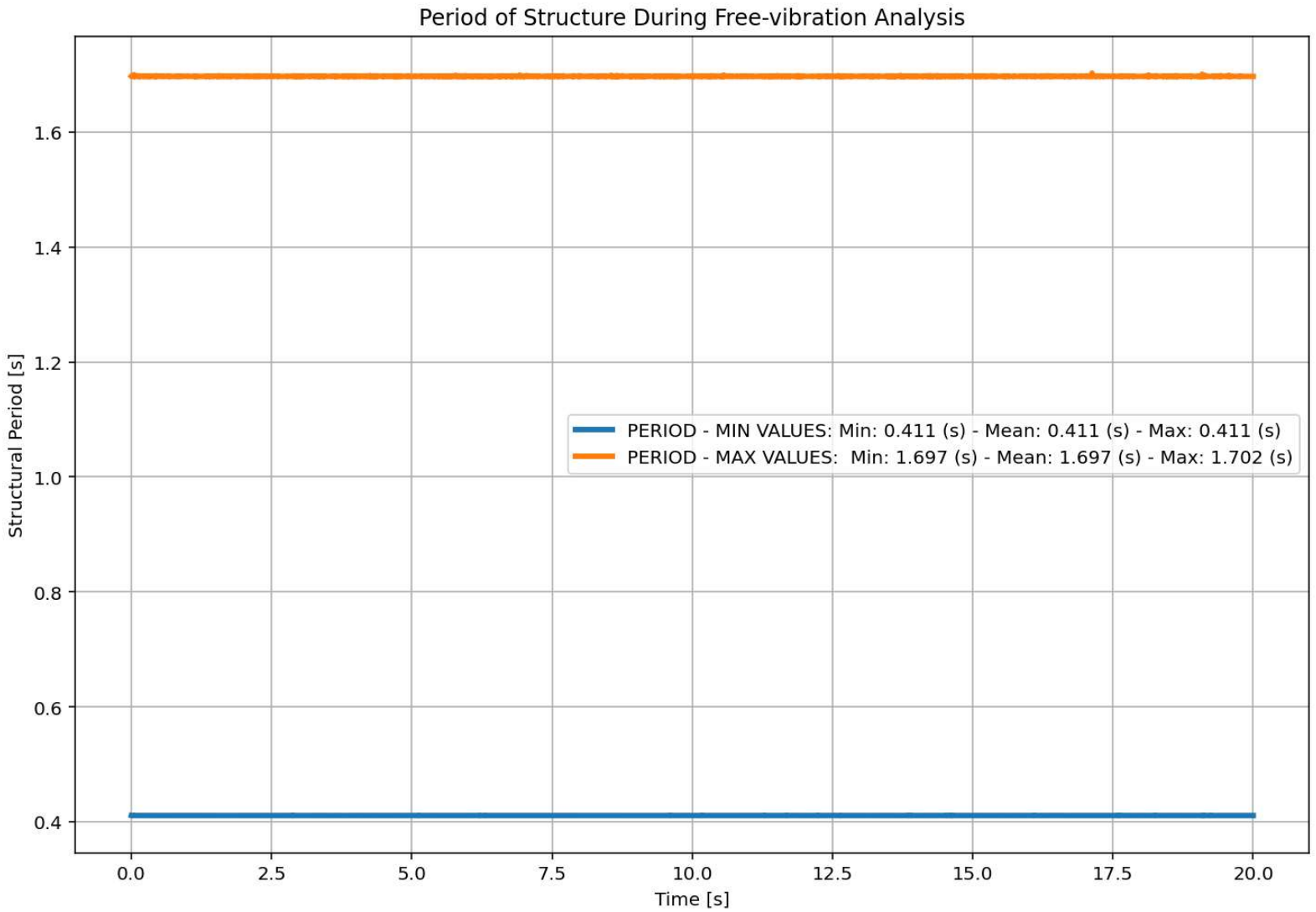

elements force in X Dir. vs Time for Beam Element During Free-vibration Analysis
Ele. 1 - MAX. ABS. : 1.8953e+05
Ele. 2 - MAX. ABS. : 1.1279e+05
Ele. 3 - MAX. ABS. : 1.0971e+05
Ele. 4 - MAX. ABS. : 1.0172e+05
Ele. 5 - MAX. ABS. : 1.2573e+05
Ele. 6 - MAX. ABS. : 1.5234e+05
Ele. 7 - MAX. ABS. : 1.9130e+05
Ele. 8 - MAX. ABS. : 1.1319e+05
Ele. 9 - MAX. ABS. : 1.3175e+05
Ele. 10 - MAX. ABS. : 1.3303e+05
Ele. 11 - MAX. ABS. : 1.3047e+05
Ele. 12 - MAX. ABS. : 1.2247e+05
Ele. 13 - MAX. ABS. : 1.2131e+05
Ele. 14 - MAX. ABS. : 3.4830e+04
Ele. 15 - MAX. ABS. : 8.2883e+04
Ele. 16 - MAX. ABS. : 6.3227e+04
Ele. 17 - MAX. ABS. : 6.1054e+04
Ele. 18 - MAX. ABS. : 8.0944e+04
Element Axial Force [N]
200000
150000
100000
50000
0
−50000
−100000
−150000
−200000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

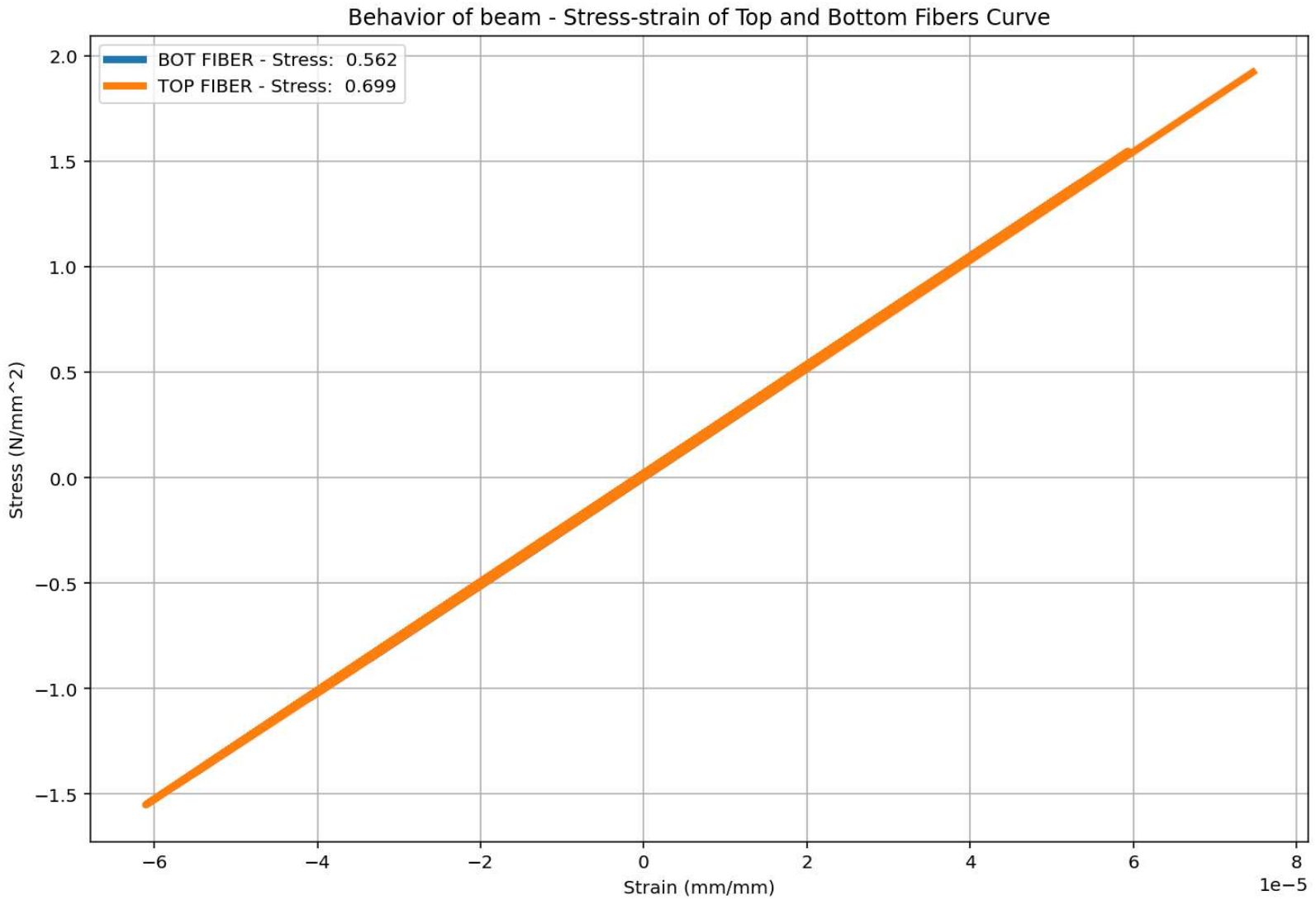

elements force in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Ele. 1 - MAX. ABS. : 2.4907e+04
Ele. 2 - MAX. ABS. : 8.0804e+03
Ele. 3 - MAX. ABS. : 3.9969e+03
Ele. 4 - MAX. ABS. : 3.8271e+03
Ele. 5 - MAX. ABS. : 6.5975e+03
Ele. 6 - MAX. ABS. : 1.8915e+04
Ele. 7 - MAX. ABS. : 3.4970e+04
Ele. 8 - MAX. ABS. : 2.7217e+04
Ele. 9 - MAX. ABS. : 2.9081e+04
Ele. 10 - MAX. ABS. : 3.0438e+04
Ele. 11 - MAX. ABS. : 2.1600e+04
Ele. 12 - MAX. ABS. : 2.6491e+04
Ele. 13 - MAX. ABS. : 3.0063e+04
Ele. 14 - MAX. ABS. : 2.1303e+04
Ele. 15 - MAX. ABS. : 2.1399e+04
Ele. 16 - MAX. ABS. : 2.5042e+04
Ele. 17 - MAX. ABS. : 2.2299e+04
Ele. 18 - MAX. ABS. : 1.7761e+04
Element Shear Force [N]
30000
20000
10000
0
−10000
−20000
−30000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

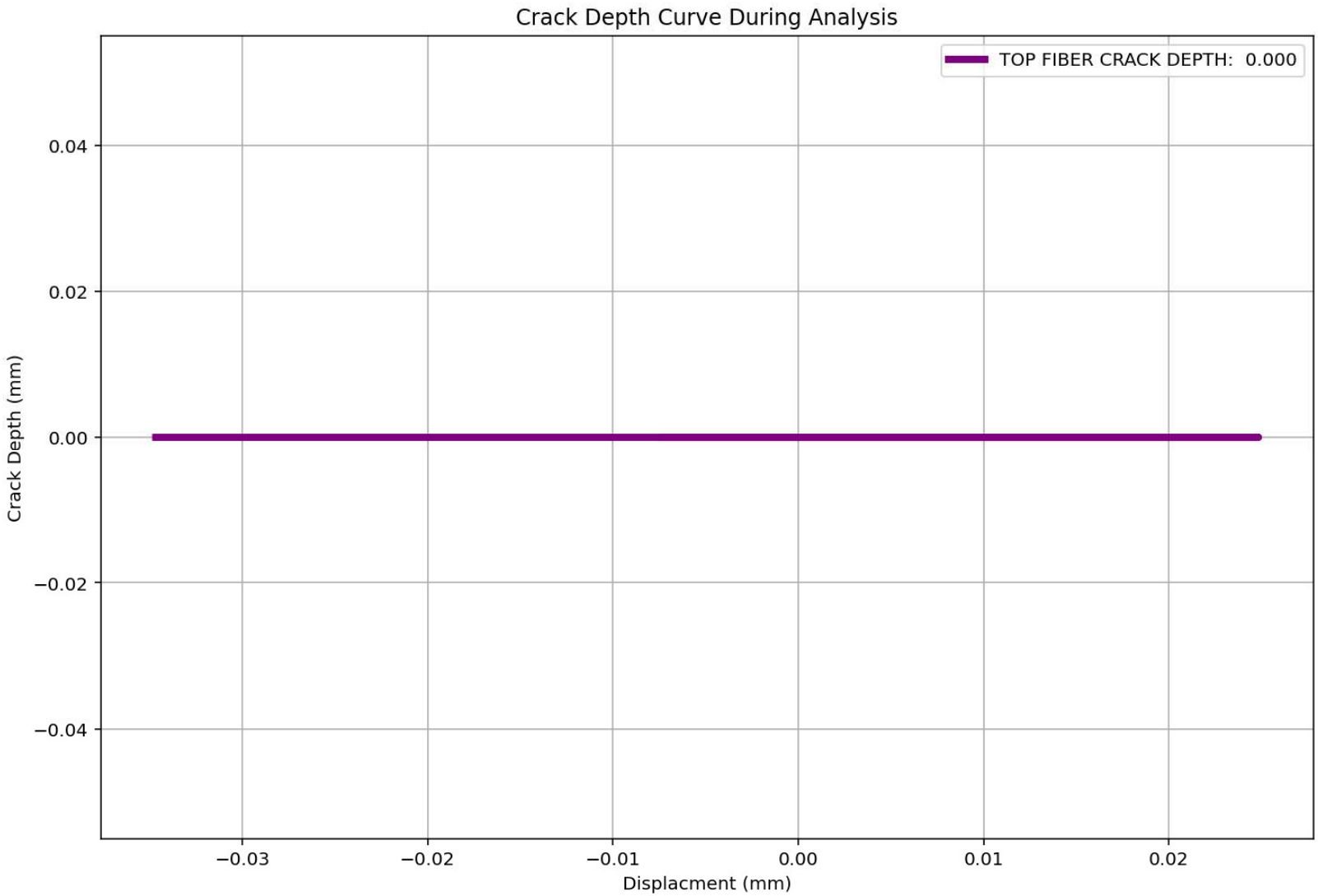

elements force in Z Dir. vs Time for Beam Element During Free-vibration Analysis
1e7
Element Moment Force [N.mm]
Time [s]
Ele. 1 - MAX. ABS. : 2.2925e+07
Ele. 2 - MAX. ABS. : 7.3250e+06
Ele. 3 - MAX. ABS. : 2.8089e+06
Ele. 4 - MAX. ABS. : 2.2725e+06
Ele. 5 - MAX. ABS. : 2.4890e+06
Ele. 6 - MAX. ABS. : 6.0149e+06
Ele. 7 - MAX. ABS. : 5.4735e+07
Ele. 8 - MAX. ABS. : 4.2119e+07
Ele. 9 - MAX. ABS. : 3.4494e+07
Ele. 10 - MAX. ABS. : 3.6606e+07
Ele. 11 - MAX. ABS. : 3.2431e+07
Ele. 12 - MAX. ABS. : 4.3326e+07
Ele. 13 - MAX. ABS. : 3.8147e+07
Ele. 14 - MAX. ABS. : 2.3003e+07
Ele. 15 - MAX. ABS. : 1.8314e+07
Ele. 16 - MAX. ABS. : 2.6406e+07
Ele. 17 - MAX. ABS. : 1.4514e+07
Ele. 18 - MAX. ABS. : 2.4385e+07

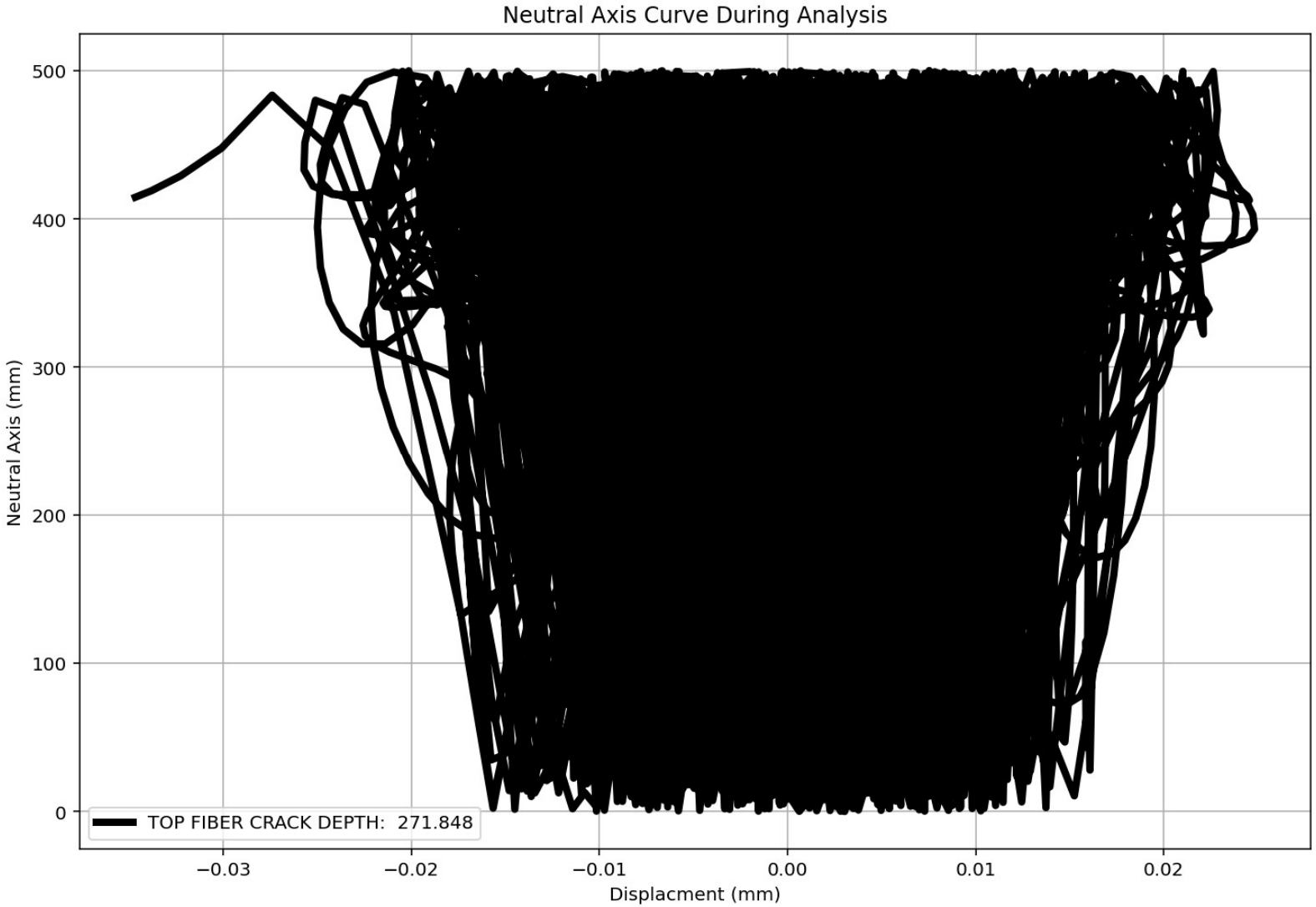

Node Displacements in X Dir. vs Time for Beam Element During Free-vibration Analysis
Displacement [mm]
Time [s]
0.02
0.01
0.00
−0.01
−0.02
−0.03
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 3.4668e-02
Node 2 - MAX. ABS. : 1.7833e-02
Node 3 - MAX. ABS. : 1.9011e-02
Node 4 - MAX. ABS. : 1.7061e-02
Node 5 - MAX. ABS. : 1.8210e-02
Node 6 - MAX. ABS. : 1.9799e-02
Node 7 - MAX. ABS. : 2.5785e-02
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 2.2566e-02
Node 10 - MAX. ABS. : 1.9840e-02
Node 11 - MAX. ABS. : 2.0562e-02
Node 12 - MAX. ABS. : 1.9447e-02
Node 13 - MAX. ABS. : 2.4067e-02
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 1.8475e-02
Node 16 - MAX. ABS. : 1.7861e-02
Node 17 - MAX. ABS. : 1.9553e-02
Node 18 - MAX. ABS. : 1.5343e-02
Node 19 - MAX. ABS. : 2.0416e-02

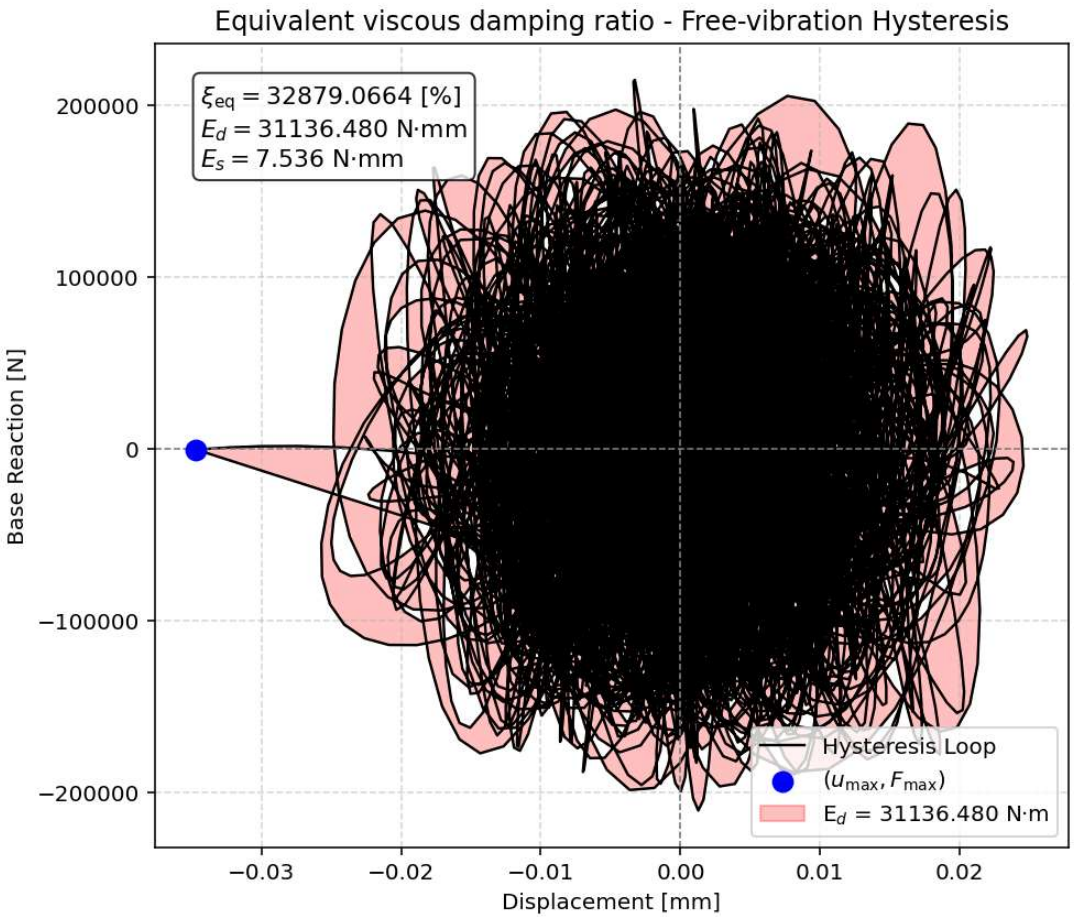

Node Displacements in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Displacement [mm]
Time [s]
0.006
0.004
0.002
0.000
−0.002
−0.004
−0.006
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 6.6338e-03
Node 2 - MAX. ABS. : 6.1021e-03
Node 3 - MAX. ABS. : 2.8128e-03
Node 4 - MAX. ABS. : 4.7468e-03
Node 5 - MAX. ABS. : 2.7884e-03
Node 6 - MAX. ABS. : 5.5921e-03
Node 7 - MAX. ABS. : 6.8143e-03
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 2.8242e-03
Node 10 - MAX. ABS. : 3.8684e-03
Node 11 - MAX. ABS. : 4.9595e-03
Node 12 - MAX. ABS. : 5.8288e-03
Node 13 - MAX. ABS. : 6.3066e-03
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 2.3120e-03
Node 16 - MAX. ABS. : 3.8724e-03
Node 17 - MAX. ABS. : 4.8014e-03
Node 18 - MAX. ABS. : 5.5762e-03
Node 19 - MAX. ABS. : 6.3596e-03

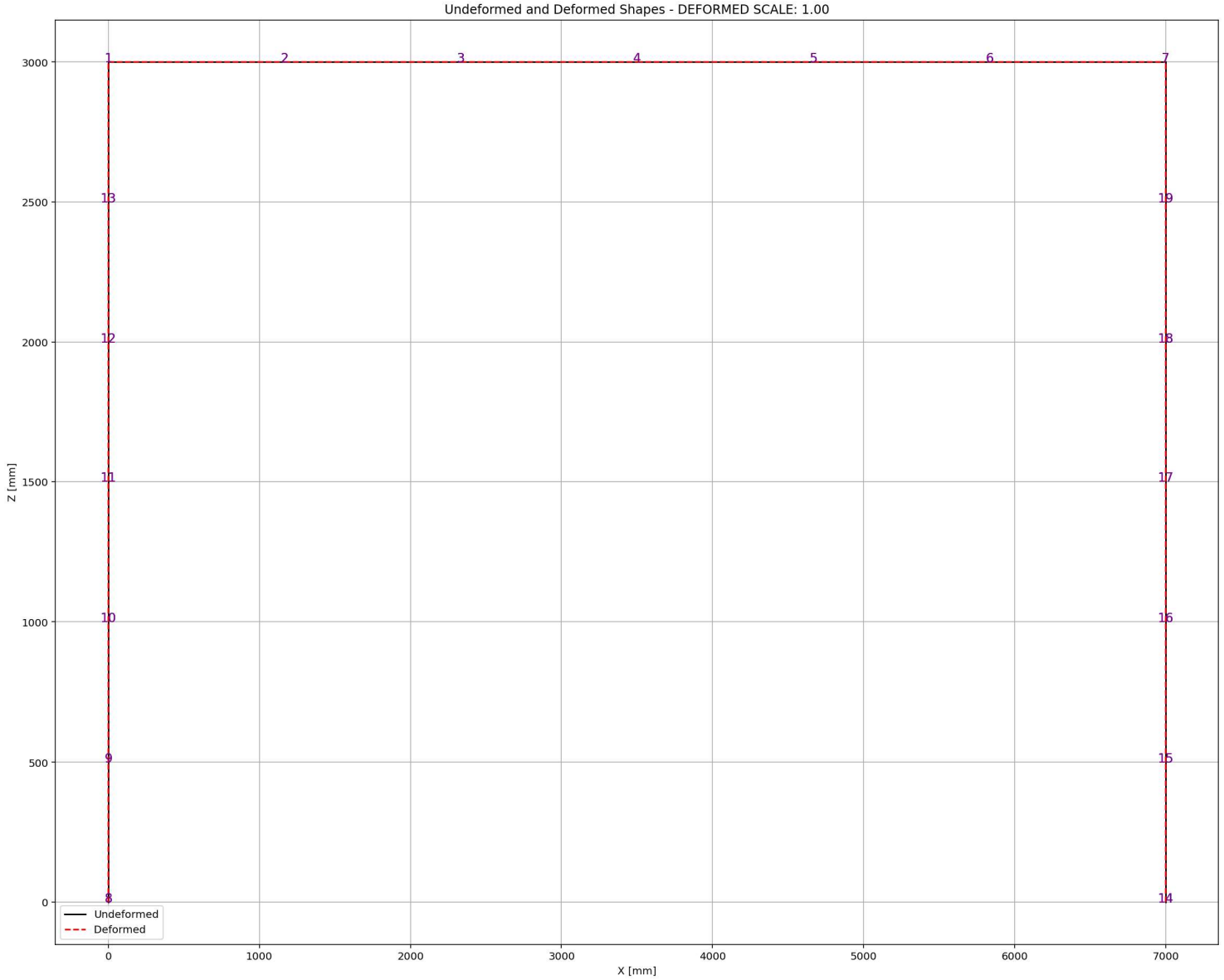

Period of Structure During Free-vibration Analysis
Structural Period [s]
Time [s]
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
PERIOD - MIN VALUES: Min: 0.411 (s) - Mean: 0.411 (s) - Max: 0.411 (s)
PERIOD - MAX VALUES: Min: 1.697 (s) - Mean: 1.697 (s) - Max: 1.702 (s)


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.562
TOP FIBER - Stress: 0.699
Stress (N/mm^2)
Strain (mm/mm)
1e−5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−6
−4
−2
0
2
4
6
8


Crack Depth Curve During Analysis
TOP FIBER CRACK DEPTH: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−0.03
−0.02
−0.01
0.00
0.01
0.02
Displacment (mm)


Neutral Axis Curve During Analysis
Neutral Axis (mm)
500
400
300
200
100
0
−0.03
−0.02
−0.01
0.00
0.01
0.02
Displacment (mm)
TOP FIBER CRACK DEPTH: 271.848


Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 32879.0664$ [%]
$E_d = 31136.480$ N·mm
$E_s = 7.536$ N·mm
200000
100000
0
−100000
−200000
Base Reaction [N]
−0.03
−0.02
−0.01
0.00
0.01
0.02
Displacement [mm]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 31136.480 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000

SEISMIC ANALYSIS

Spyder (Python 3.12)

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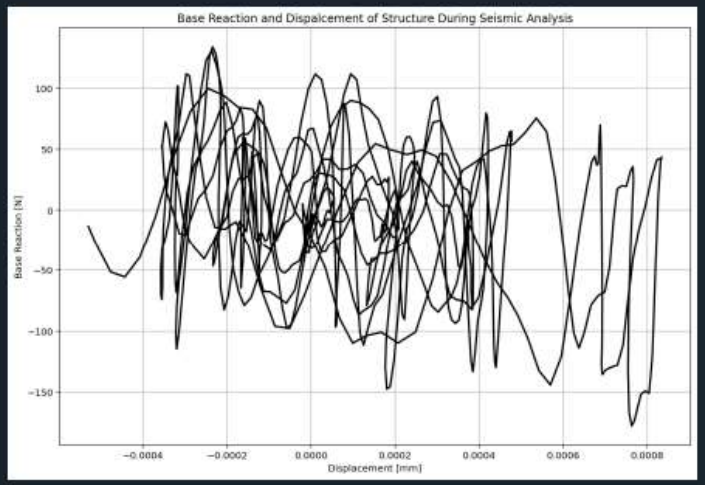
...ENERGY_DISSIPATION_CAPACITY_INDEX\12_CONCRETE_FRAME_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CONCRETE_FRAME_8_ANA\P(t)_CONCRETE_FRAME_8_ANA.py

P(t)_CONCRETE_FRAME_8_ANA.py | COMPOSITE_REC_I_SECTION_FUN.py

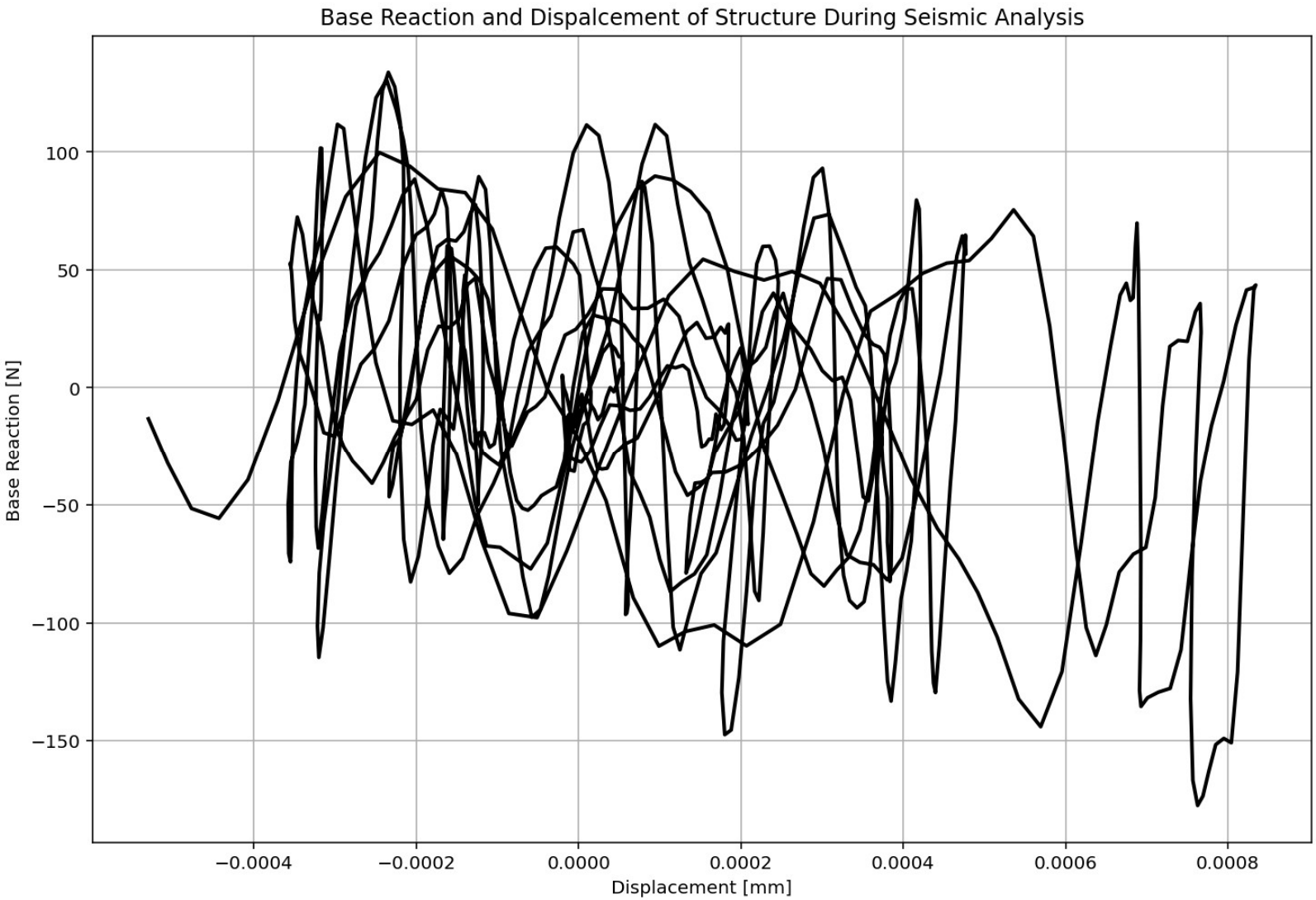
```
989
990     if ANAL_TYPE == 'SEISMIC': # DYNAMIC TIME-HISTORY ANALYSIS
991         #%% DEFINE PARAMETERS FOR SEISMIC ANALYSIS
992         duration = 8.0                  # [s] Analysis duration
993         dt = 0.01                       # [s] Time step
994         GMfact = 9810                   # [mm/s²]standard acceleration of gravity or st
995         SSF_X = 0.0001                  # Seismic Acceleration Scale Factor in X Direct
996         SSF_Y = 0.0001                  # Seismic Acceleration Scale Factor in Y Direct
997         iv0_X = 0.00005                 # [mm/s] Initial velocity applied to the node
998         iv0_Y = 0.00005                 # [mm/s] Initial velocity applied to the node
999         st_iv0 = 0.0                    # [s] Initial velocity applied starting time
1000        SEI = 'X'                       # Seismic Direction
1001        DR = 0.03                       # Damping ratio
1002
1003        # Define time series for input motion (Acceleration time history)
1004        if SEI == 'X':
1005            SEISMIC_TAG_01 = 100
1006            ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Acceler
1007            # Define load patterns
1008            # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fac
1009            ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01,
1010        if SEI == 'Y':
1011            SEISMIC_TAG_02 = 200
1012            ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Acceler
1013            ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02,
1014        if SEI == 'XY':
1015            SEISMIC_TAG_01 = 100
1016            ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Acceler
1017            # Define load patterns
1018            # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fac
1019            ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01,
1020            SEISMIC_TAG_02 = 200
1021            ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Acceler
1022            ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02,
```

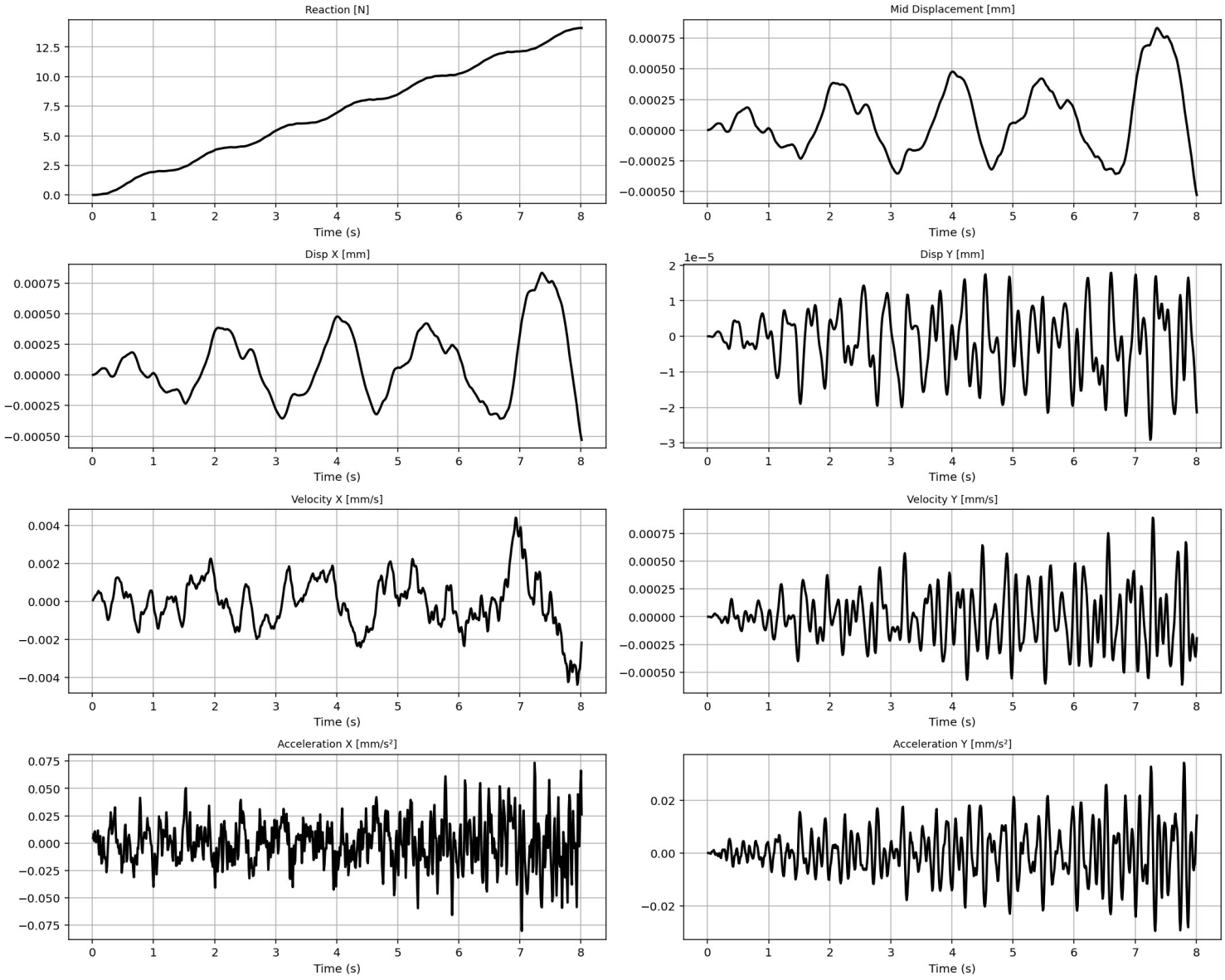
30 %

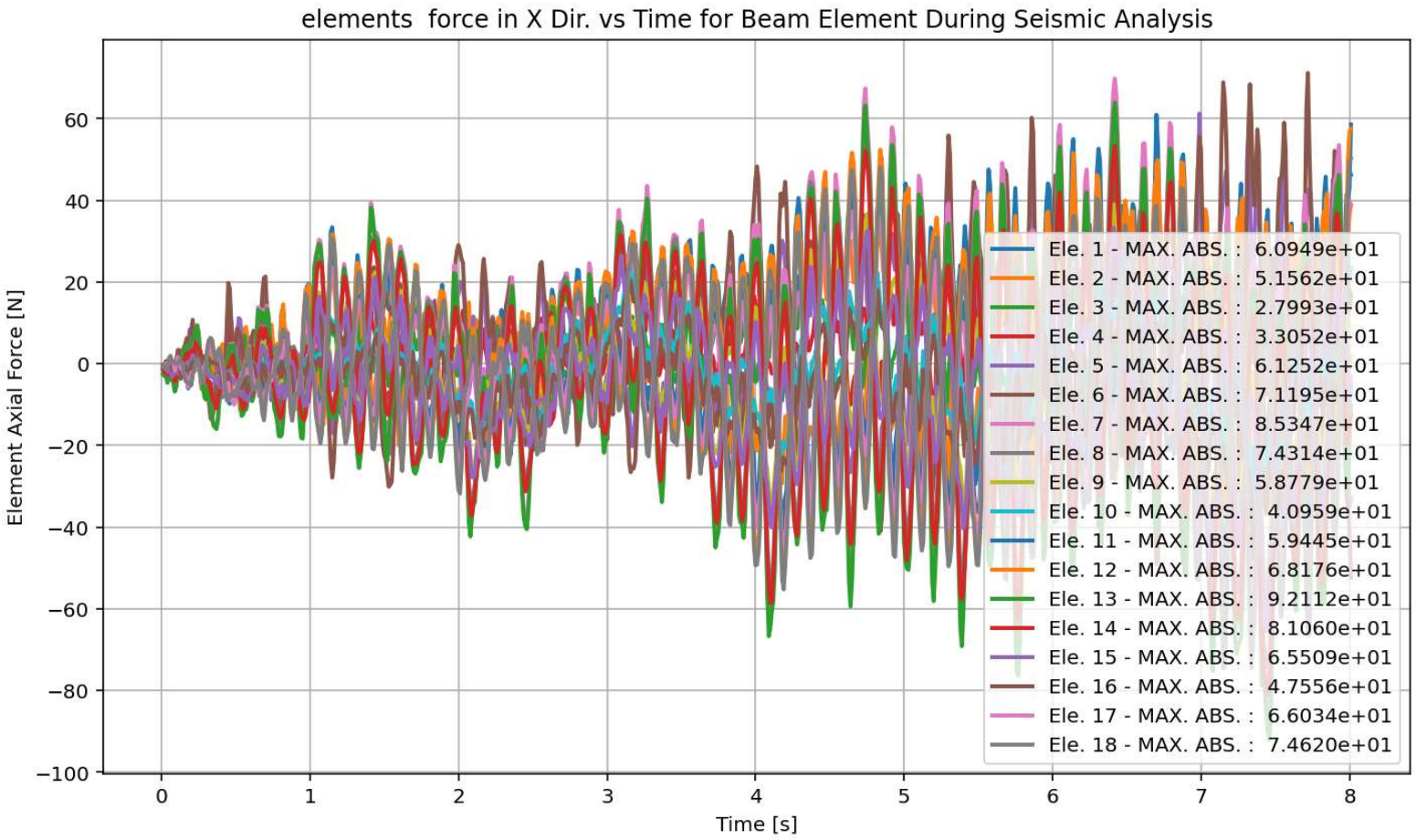


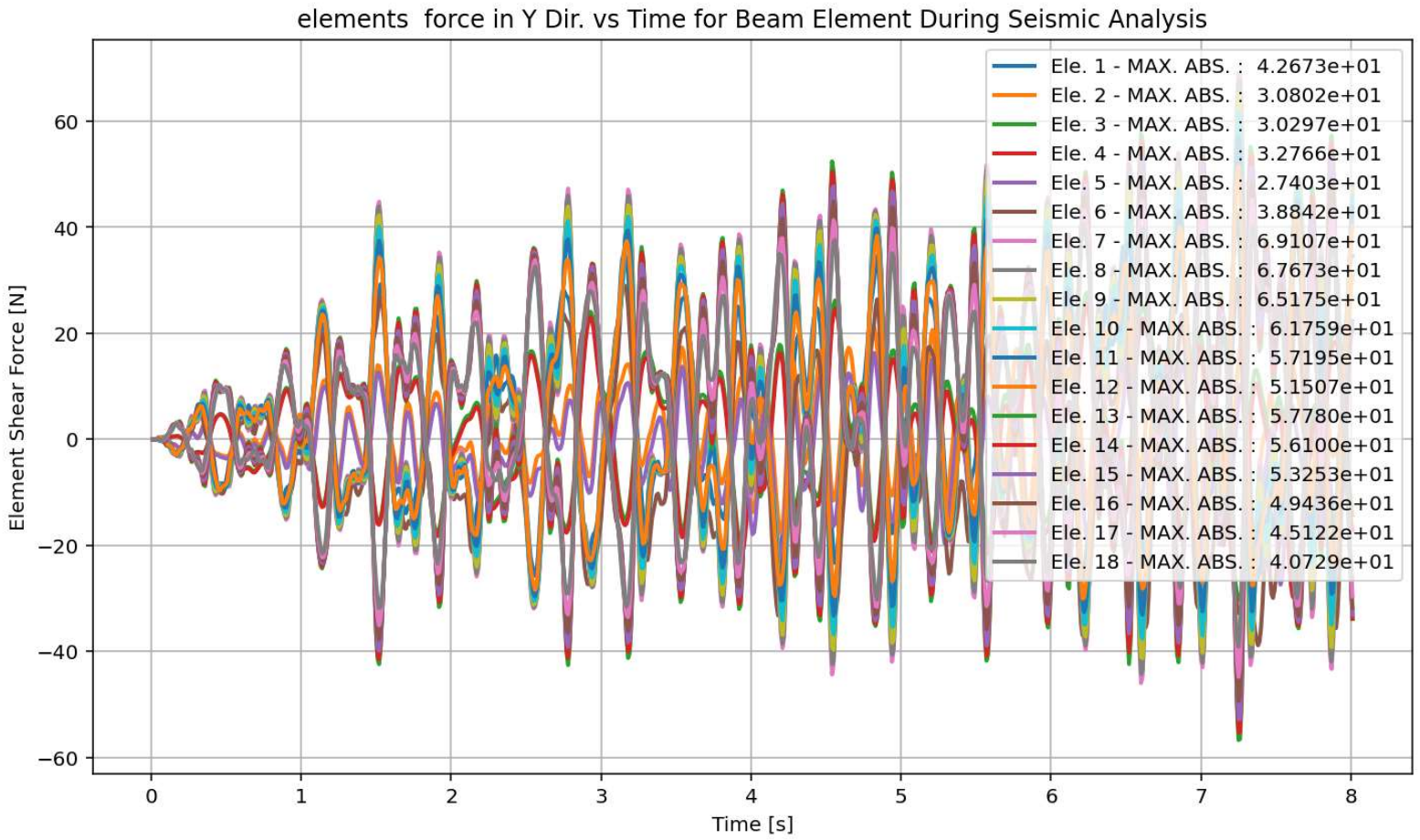
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

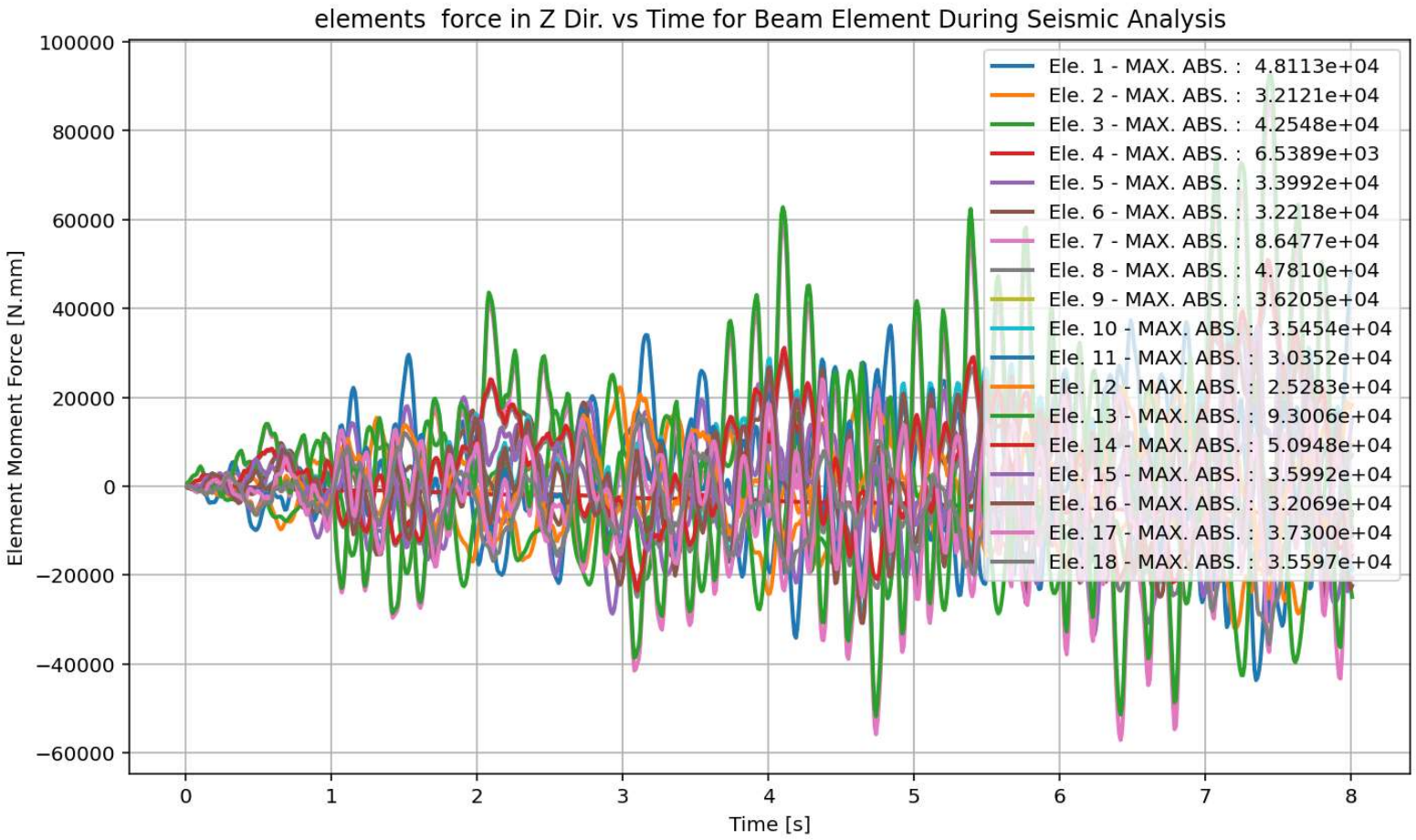
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 989, Col 5 | UTF-8 | CRLF | RW | Mem 38%

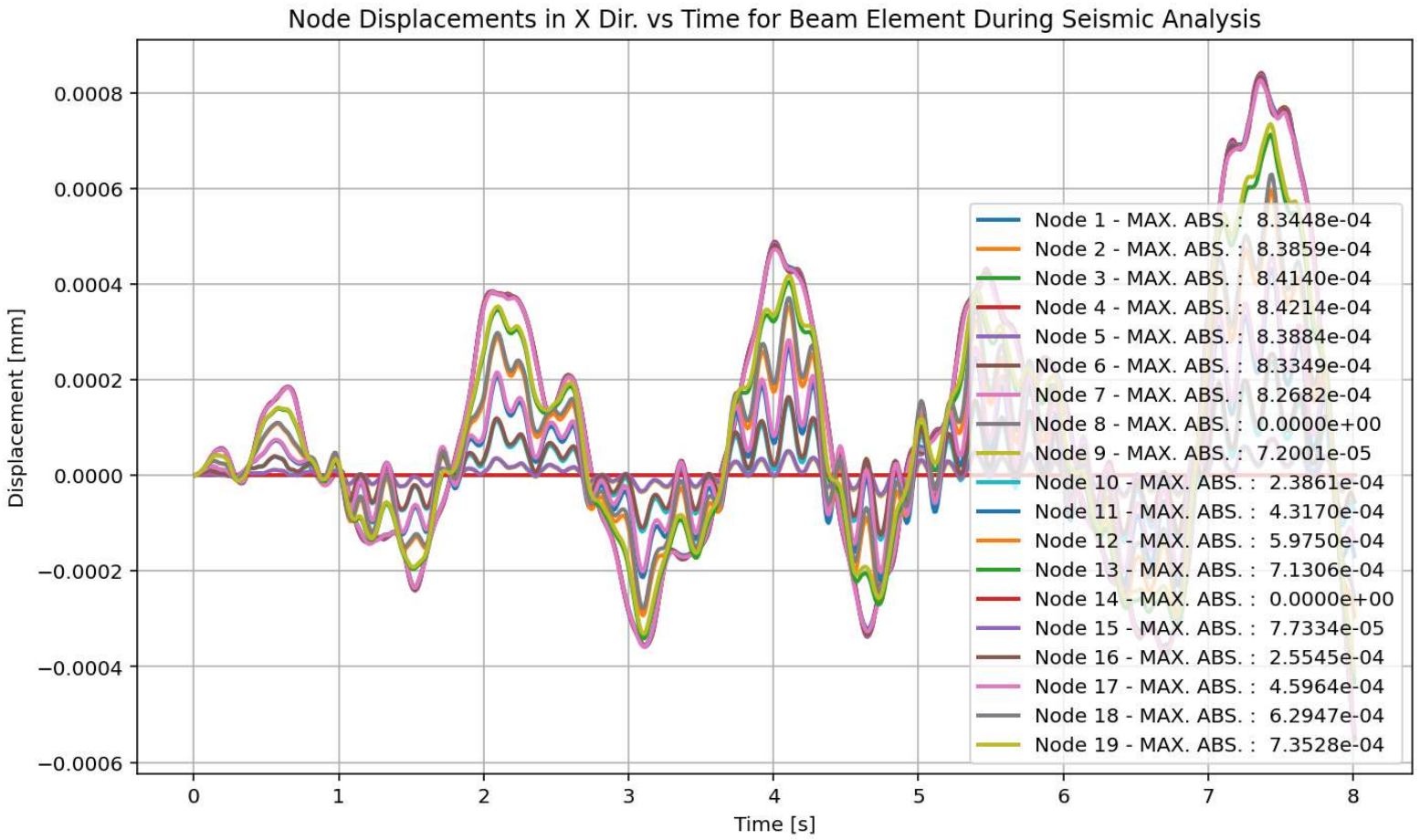

Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
100
50
0
−50
−100
−150
−0.0004
−0.0002
0.0000
0.0002
0.0004
0.0006
0.0008
Displacement [mm]

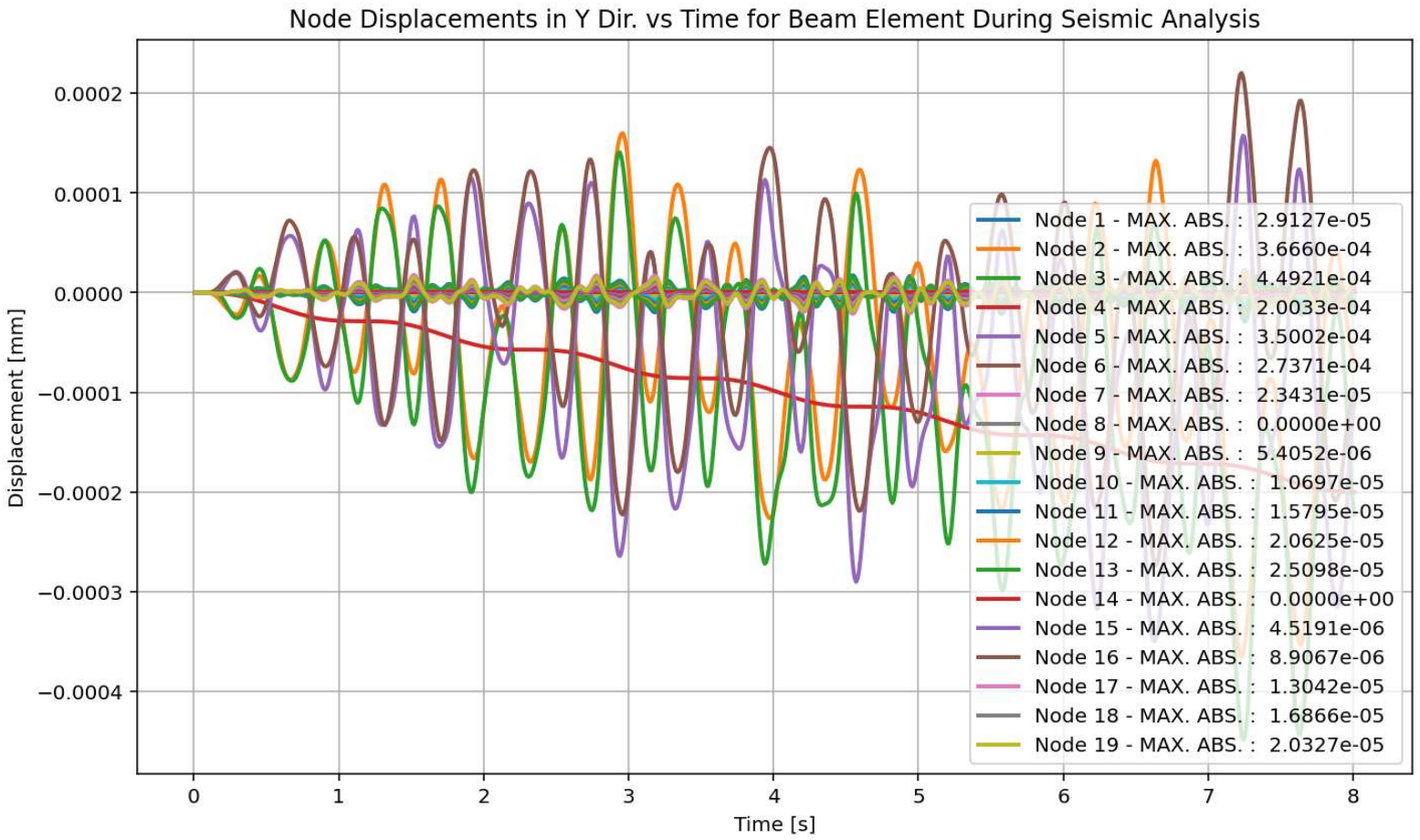

Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
1e−5
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
Time (s)
Acceleration Y [mm/s²]
Time (s)

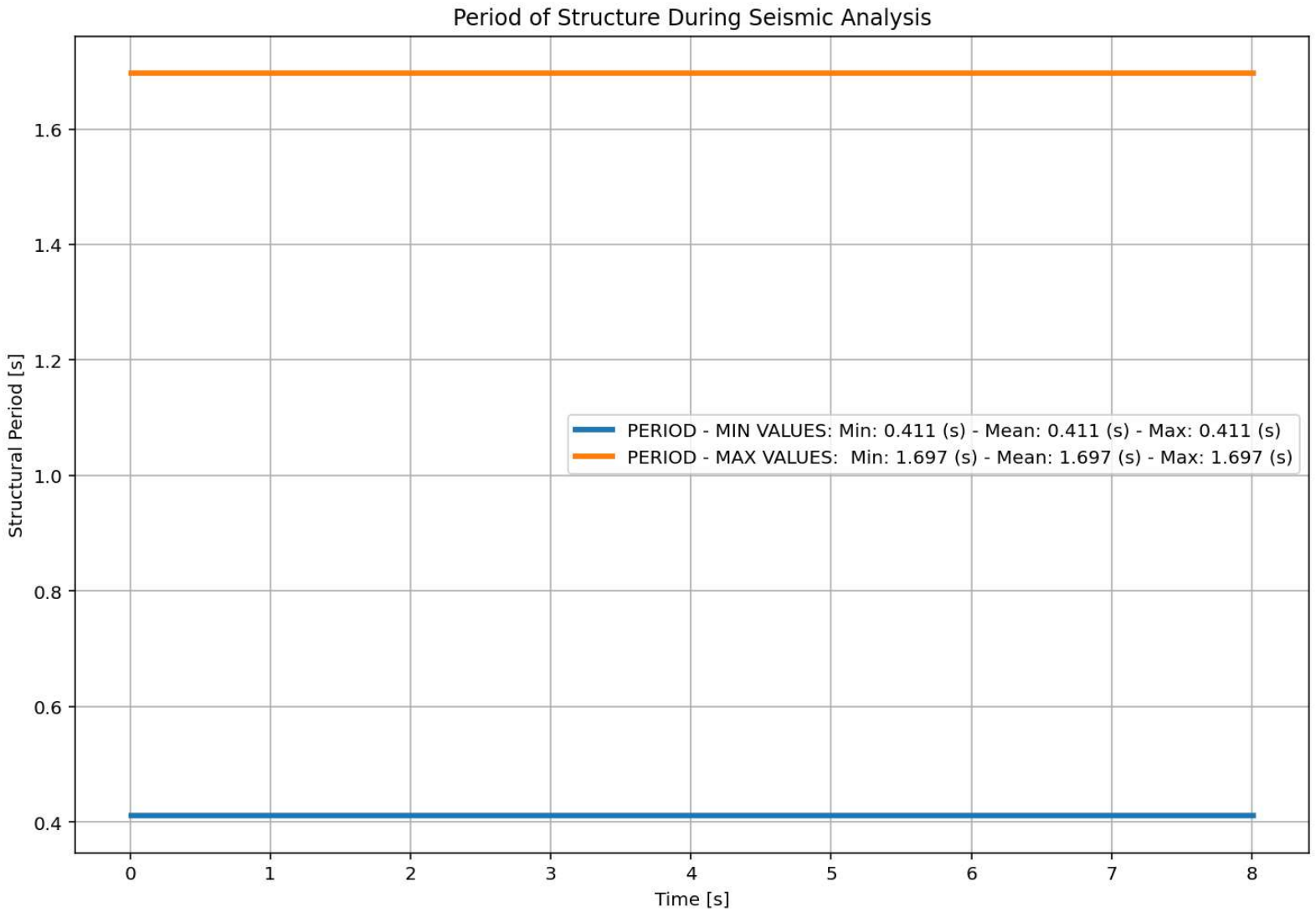

elements force in X Dir. vs Time for Beam Element During Seismic Analysis
Element Axial Force [N]
Time [s]
60
40
20
0
−20
−40
−60
−80
−100
0
1
2
3
4
5
6
7
8
Ele. 1 - MAX. ABS. : 6.0949e+01
Ele. 2 - MAX. ABS. : 5.1562e+01
Ele. 3 - MAX. ABS. : 2.7993e+01
Ele. 4 - MAX. ABS. : 3.3052e+01
Ele. 5 - MAX. ABS. : 6.1252e+01
Ele. 6 - MAX. ABS. : 7.1195e+01
Ele. 7 - MAX. ABS. : 8.5347e+01
Ele. 8 - MAX. ABS. : 7.4314e+01
Ele. 9 - MAX. ABS. : 5.8779e+01
Ele. 10 - MAX. ABS. : 4.0959e+01
Ele. 11 - MAX. ABS. : 5.9445e+01
Ele. 12 - MAX. ABS. : 6.8176e+01
Ele. 13 - MAX. ABS. : 9.2112e+01
Ele. 14 - MAX. ABS. : 8.1060e+01
Ele. 15 - MAX. ABS. : 6.5509e+01
Ele. 16 - MAX. ABS. : 4.7556e+01
Ele. 17 - MAX. ABS. : 6.6034e+01
Ele. 18 - MAX. ABS. : 7.4620e+01

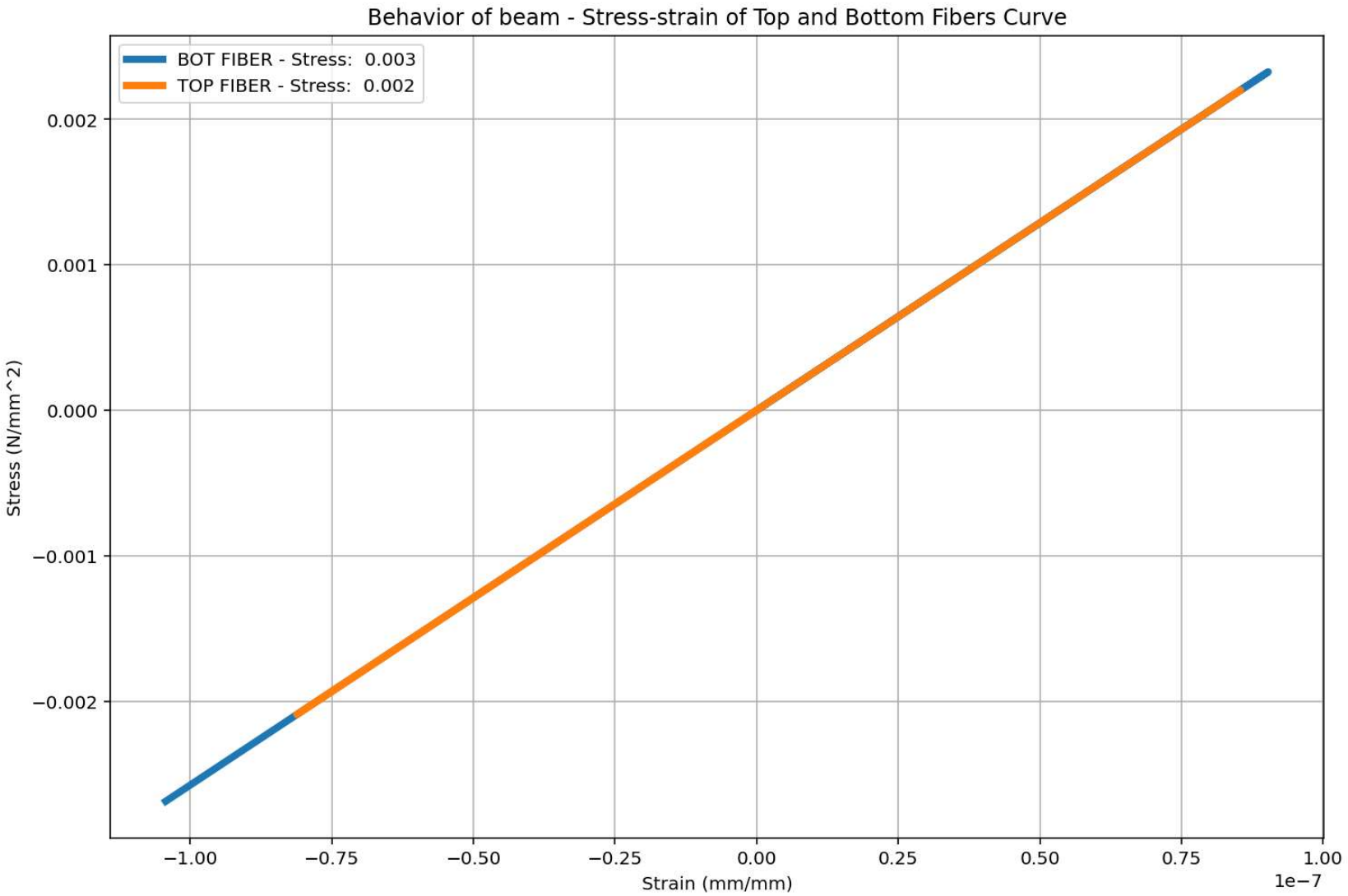

elements force in Y Dir. vs Time for Beam Element During Seismic Analysis
Element Shear Force [N]
Time [s]
60
40
20
0
−20
−40
−60
0
1
2
3
4
5
6
7
8
Ele. 1 - MAX. ABS. : 4.2673e+01
Ele. 2 - MAX. ABS. : 3.0802e+01
Ele. 3 - MAX. ABS. : 3.0297e+01
Ele. 4 - MAX. ABS. : 3.2766e+01
Ele. 5 - MAX. ABS. : 2.7403e+01
Ele. 6 - MAX. ABS. : 3.8842e+01
Ele. 7 - MAX. ABS. : 6.9107e+01
Ele. 8 - MAX. ABS. : 6.7673e+01
Ele. 9 - MAX. ABS. : 6.5175e+01
Ele. 10 - MAX. ABS. : 6.1759e+01
Ele. 11 - MAX. ABS. : 5.7195e+01
Ele. 12 - MAX. ABS. : 5.1507e+01
Ele. 13 - MAX. ABS. : 5.7780e+01
Ele. 14 - MAX. ABS. : 5.6100e+01
Ele. 15 - MAX. ABS. : 5.3253e+01
Ele. 16 - MAX. ABS. : 4.9436e+01
Ele. 17 - MAX. ABS. : 4.5122e+01
Ele. 18 - MAX. ABS. : 4.0729e+01

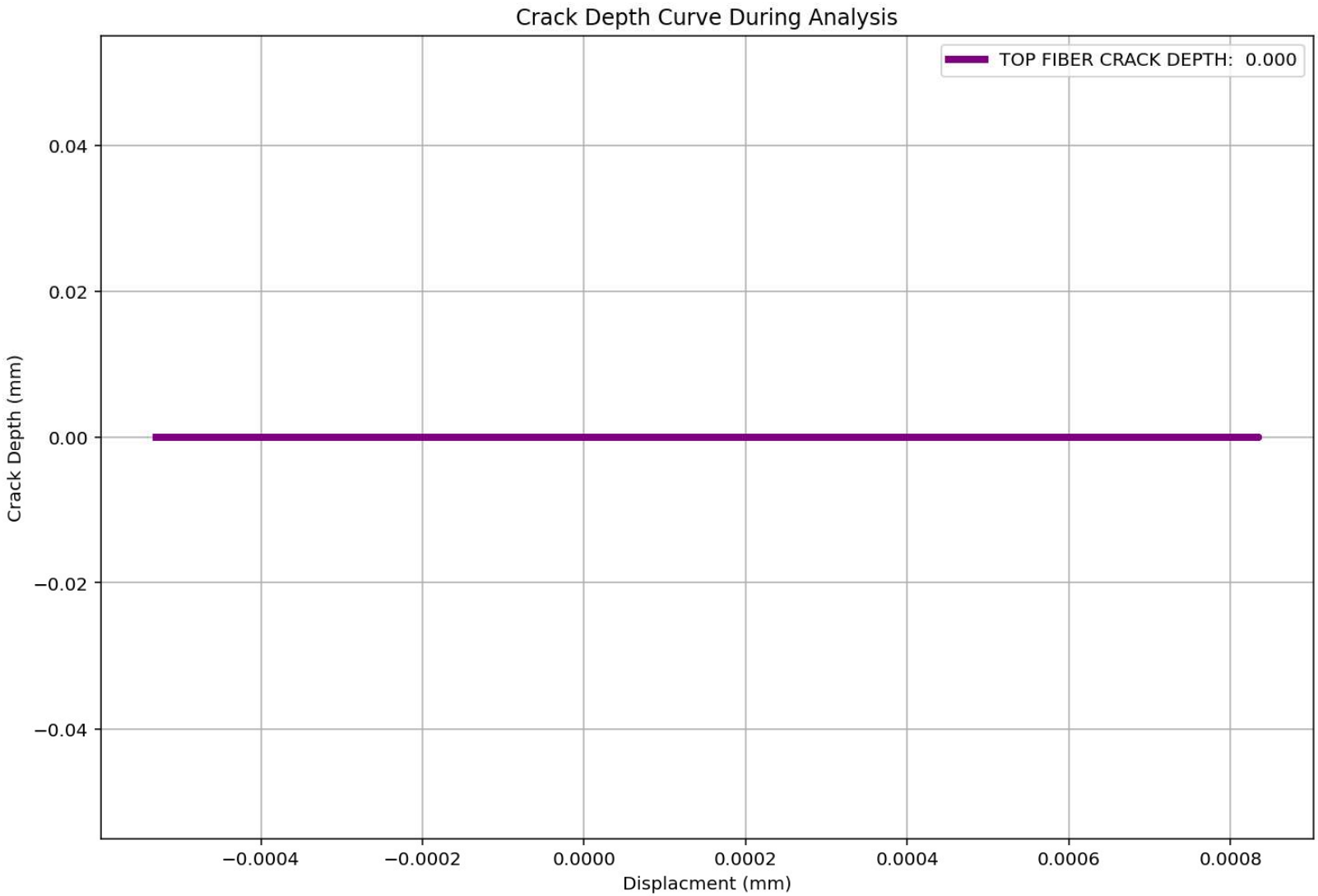

elements force in Z Dir. vs Time for Beam Element During Seismic Analysis
Element Moment Force [N.mm]
Time [s]
100000
80000
60000
40000
20000
0
−20000
−40000
−60000
0
1
2
3
4
5
6
7
8
Ele. 1 - MAX. ABS. : 4.8113e+04
Ele. 2 - MAX. ABS. : 3.2121e+04
Ele. 3 - MAX. ABS. : 4.2548e+04
Ele. 4 - MAX. ABS. : 6.5389e+03
Ele. 5 - MAX. ABS. : 3.3992e+04
Ele. 6 - MAX. ABS. : 3.2218e+04
Ele. 7 - MAX. ABS. : 8.6477e+04
Ele. 8 - MAX. ABS. : 4.7810e+04
Ele. 9 - MAX. ABS. : 3.6205e+04
Ele. 10 - MAX. ABS. : 3.5454e+04
Ele. 11 - MAX. ABS. : 3.0352e+04
Ele. 12 - MAX. ABS. : 2.5283e+04
Ele. 13 - MAX. ABS. : 9.3006e+04
Ele. 14 - MAX. ABS. : 5.0948e+04
Ele. 15 - MAX. ABS. : 3.5992e+04
Ele. 16 - MAX. ABS. : 3.2069e+04
Ele. 17 - MAX. ABS. : 3.7300e+04
Ele. 18 - MAX. ABS. : 3.5597e+04

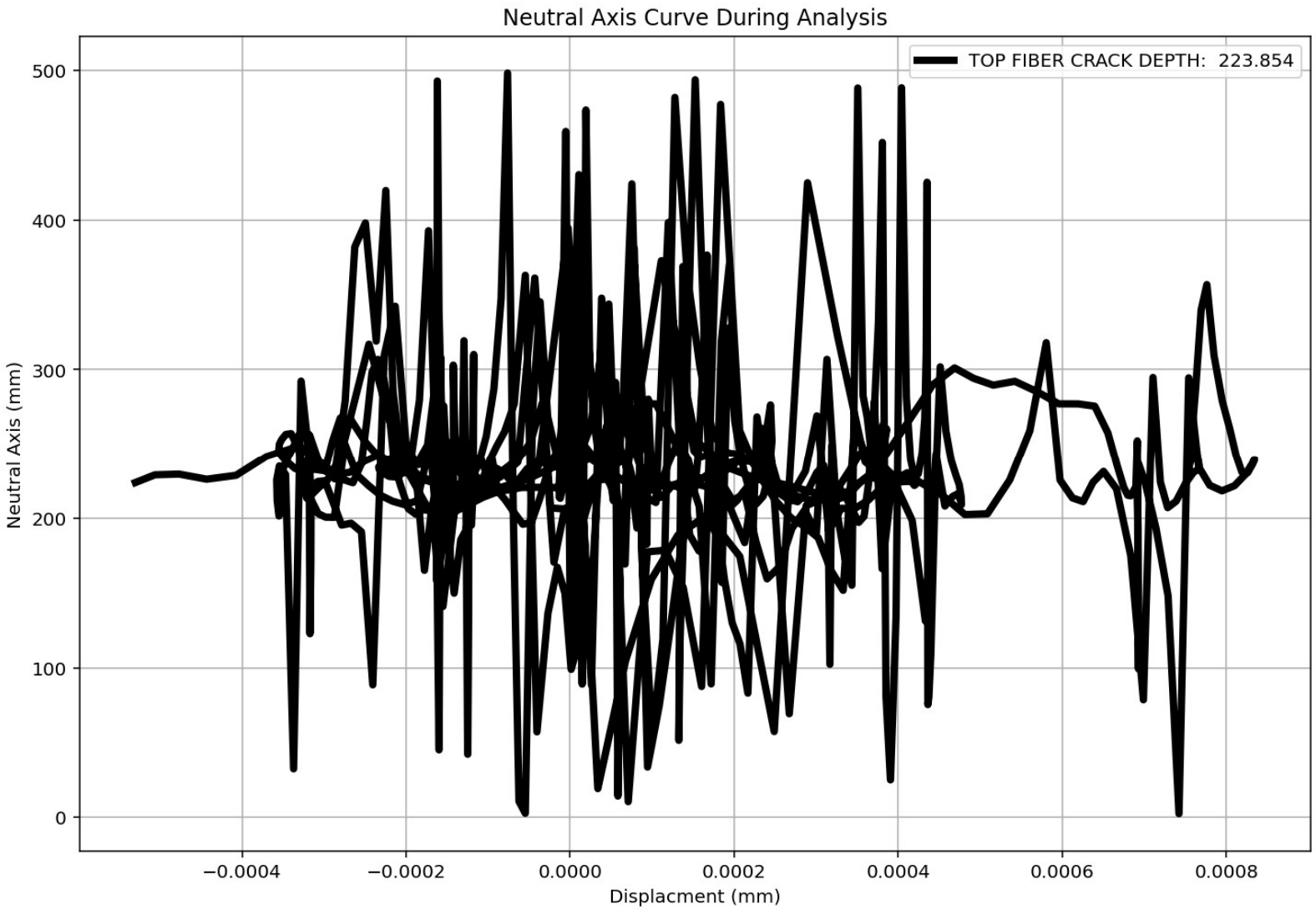

Node Displacements in X Dir. vs Time for Beam Element During Seismic Analysis
Displacement [mm]
Time [s]
0.0008
0.0006
0.0004
0.0002
0.0000
−0.0002
−0.0004
−0.0006
0
1
2
3
4
5
6
7
8
Node 1 - MAX. ABS. : 8.3448e-04
Node 2 - MAX. ABS. : 8.3859e-04
Node 3 - MAX. ABS. : 8.4140e-04
Node 4 - MAX. ABS. : 8.4214e-04
Node 5 - MAX. ABS. : 8.3884e-04
Node 6 - MAX. ABS. : 8.3349e-04
Node 7 - MAX. ABS. : 8.2682e-04
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 7.2001e-05
Node 10 - MAX. ABS. : 2.3861e-04
Node 11 - MAX. ABS. : 4.3170e-04
Node 12 - MAX. ABS. : 5.9750e-04
Node 13 - MAX. ABS. : 7.1306e-04
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 7.7334e-05
Node 16 - MAX. ABS. : 2.5545e-04
Node 17 - MAX. ABS. : 4.5964e-04
Node 18 - MAX. ABS. : 6.2947e-04
Node 19 - MAX. ABS. : 7.3528e-04

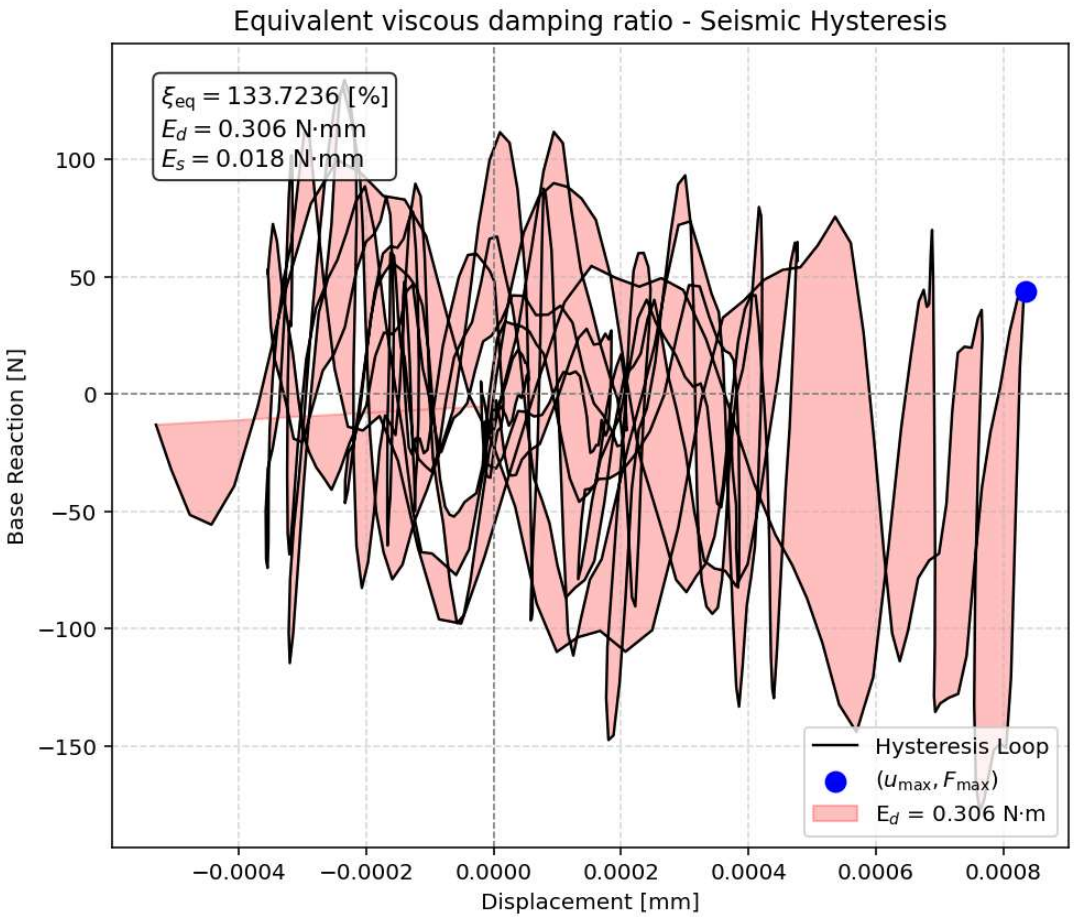

Node Displacements in Y Dir. vs Time for Beam Element During Seismic Analysis
Displacement [mm]
Time [s]
0.0002
0.0001
0.0000
−0.0001
−0.0002
−0.0003
−0.0004
0
1
2
3
4
5
6
7
8
Node 1 - MAX. ABS. : 2.9127e-05
Node 2 - MAX. ABS. : 3.6660e-04
Node 3 - MAX. ABS. : 4.4921e-04
Node 4 - MAX. ABS. : 2.0033e-04
Node 5 - MAX. ABS. : 3.5002e-04
Node 6 - MAX. ABS. : 2.7371e-04
Node 7 - MAX. ABS. : 2.3431e-05
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 5.4052e-06
Node 10 - MAX. ABS. : 1.0697e-05
Node 11 - MAX. ABS. : 1.5795e-05
Node 12 - MAX. ABS. : 2.0625e-05
Node 13 - MAX. ABS. : 2.5098e-05
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 4.5191e-06
Node 16 - MAX. ABS. : 8.9067e-06
Node 17 - MAX. ABS. : 1.3042e-05
Node 18 - MAX. ABS. : 1.6866e-05
Node 19 - MAX. ABS. : 2.0327e-05

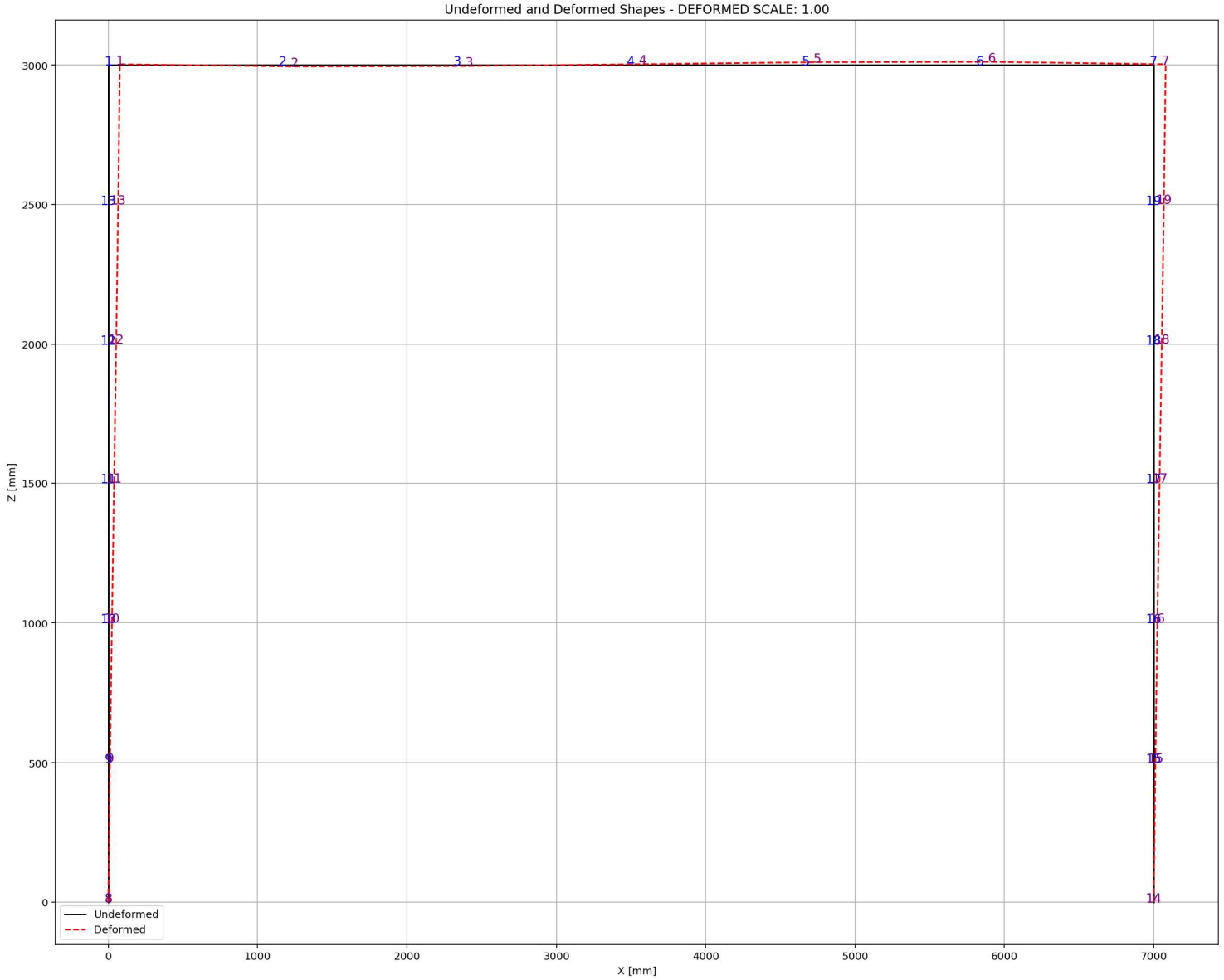

Period of Structure During Seismic Analysis
Structural Period [s]
Time [s]
PERIOD - MIN VALUES: Min: 0.411 (s) - Mean: 0.411 (s) - Max: 0.411 (s)
PERIOD - MAX VALUES: Min: 1.697 (s) - Mean: 1.697 (s) - Max: 1.697 (s)
0.4
0.6
0.8
1.0
1.2
1.4
1.6
0
1
2
3
4
5
6
7
8

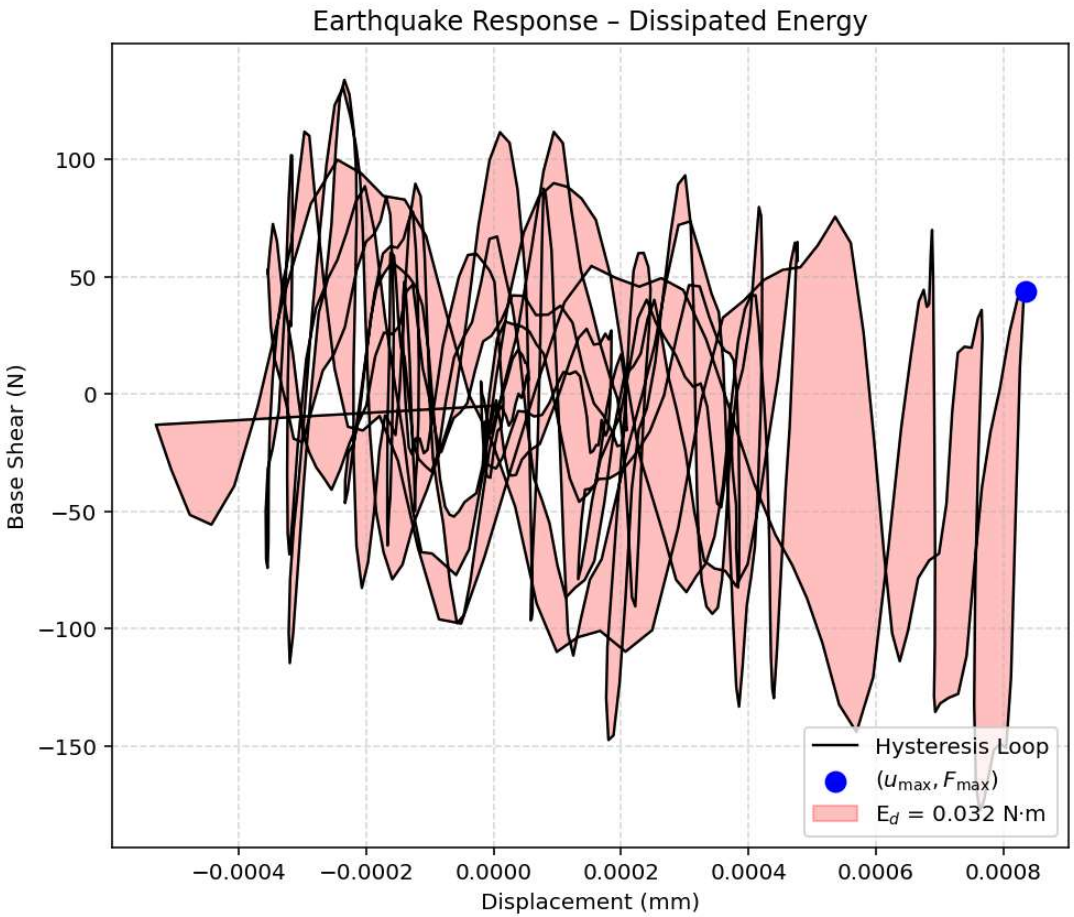

Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.003
TOP FIBER - Stress: 0.002
0.002
0.001
0.000
−0.001
−0.002
Stress (N/mm^2)
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1e−7
Strain (mm/mm)

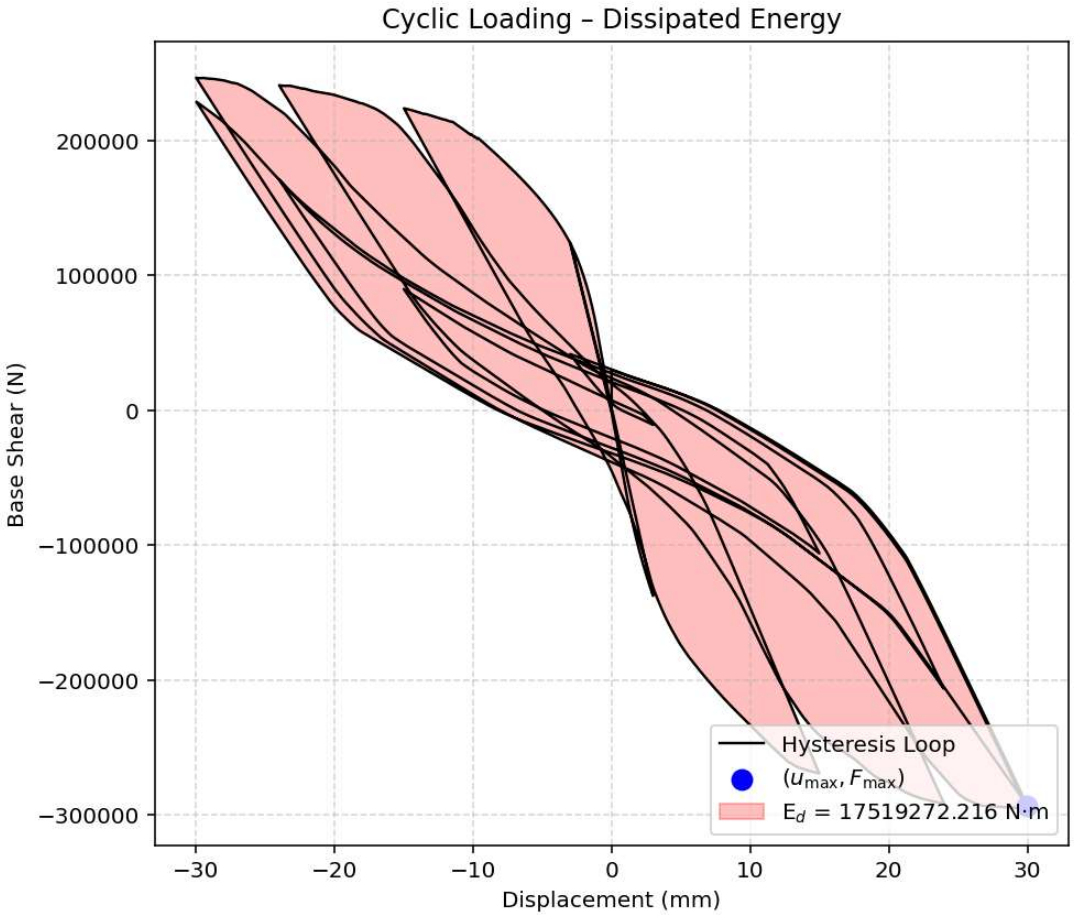

Crack Depth Curve During Analysis
TOP FIBER CRACK DEPTH: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−0.0004
−0.0002
0.0000
0.0002
0.0004
0.0006
0.0008
Displacment (mm)


Neutral Axis Curve During Analysis
TOP FIBER CRACK DEPTH: 223.854
Neutral Axis (mm)
500
400
300
200
100
0
−0.0004
−0.0002
0.0000
0.0002
0.0004
0.0006
0.0008
Displacment (mm)


Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 133.7236$ [%]
$E_d = 0.306$ N·mm
$E_s = 0.018$ N·mm
Base Reaction [N]
Displacement [mm]
100
50
0
−50
−100
−150
−0.0004
−0.0002
0.0000
0.0002
0.0004
0.0006
0.0008
Hysteresis Loop
(u_{max}, F_{max})
E_d = 0.306 N·m

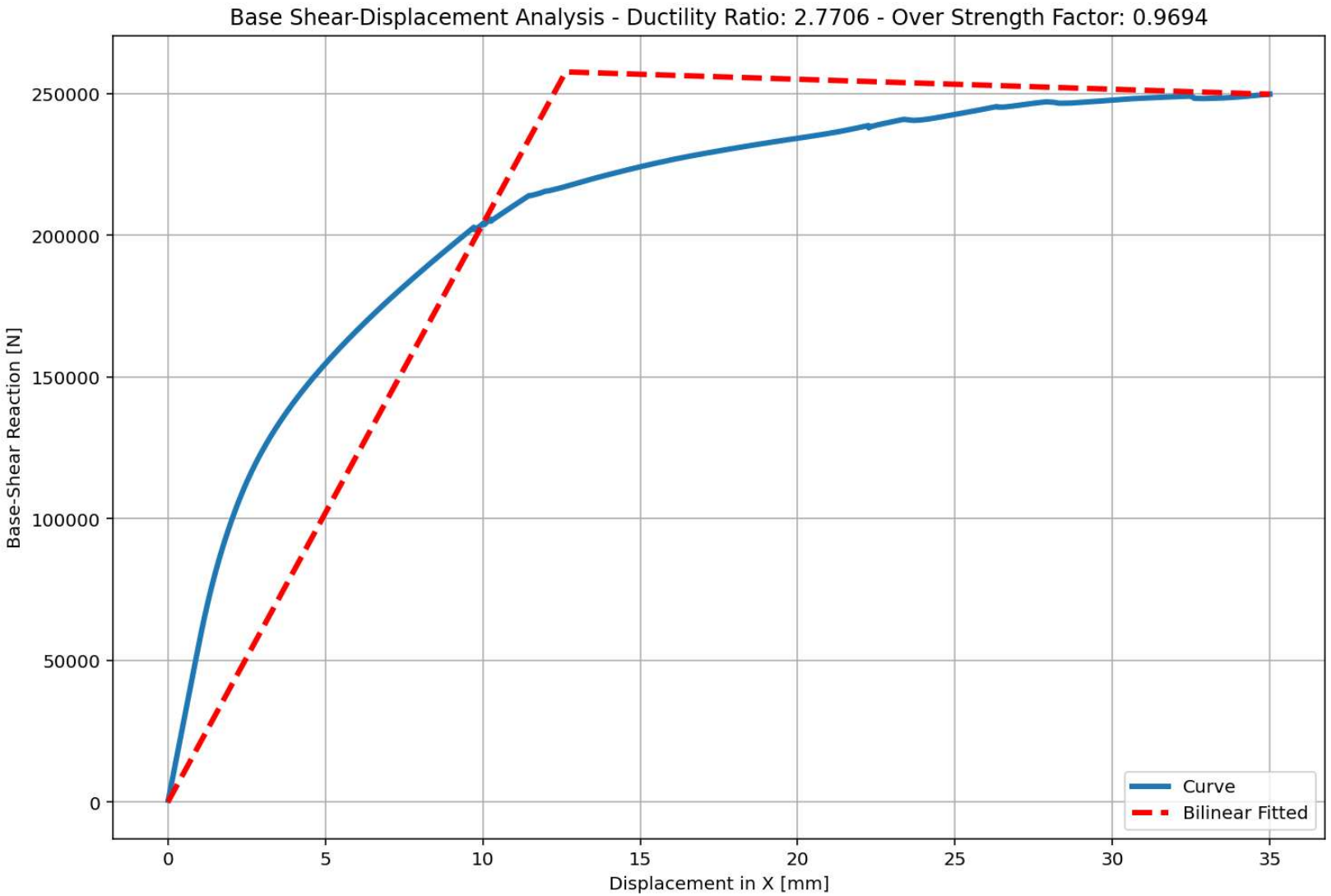

Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000
Undeformed
Deformed


Earthquake Response – Dissipated Energy
Base Shear (N)
Displacement (mm)
100
50
0
−50
−100
−150
−0.0004
−0.0002
0.0000
0.0002
0.0004
0.0006
0.0008
Hysteresis Loop
(u_{max}, F_{max})
E_d = 0.032 N·m


Cyclic Loading – Dissipated Energy
Base Shear (N)
Displacement (mm)
200000
100000
0
−100000
−200000
−300000
−30
−20
−10
0
10
20
30
Hysteresis Loop
(u_{max}, F_{max})
E_d = 17519272.216 N·m

Dissipated Energy from Earthquake= 0.03 N·mm
Dissipated Energy from Cyclic Displacement= 17519272.22 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 0.000 [%]
ZONE 1: VERY MINOR DAMAGE


Base Shear-Displacement Analysis - Ductility Ratio: 2.7706 - Over Strength Factor: 0.9694
Base-Shear Reaction [N]
250000
200000
150000
100000
50000
0
0
5
10
15
20
25
30
35
Displacement in X [mm]
Curve
Bilinear Fitted

```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
--------------------------
[[0.00000000e+00 0.00000000e+00]
 [1.26327517e+01 2.57617793e+05]
 [3.50000000e+01 2.49742938e+05]]
+==========================+
+-------------------------------------------------+
 Structure Elastic Stiffness :     20392.85
 Structure Plastic Stiffness :     7135.51
 Structure Tangent Stiffness :     -352.07
 Structure Ductility Ratio :       2.77
 Structure Over Strength Factor:   0.97
+-------------------------------------------------+
Over Strength Coefficient (Ω0):     0.9694
Displacement Ductility Ratio (μ):    2.7706
Ductility Coefficient (Rμ):         2.7706
Structural Behavior Coefficient (R): 2.6859
Structural Ductility Damage Index in X Direction: -56.4751 (%)
```